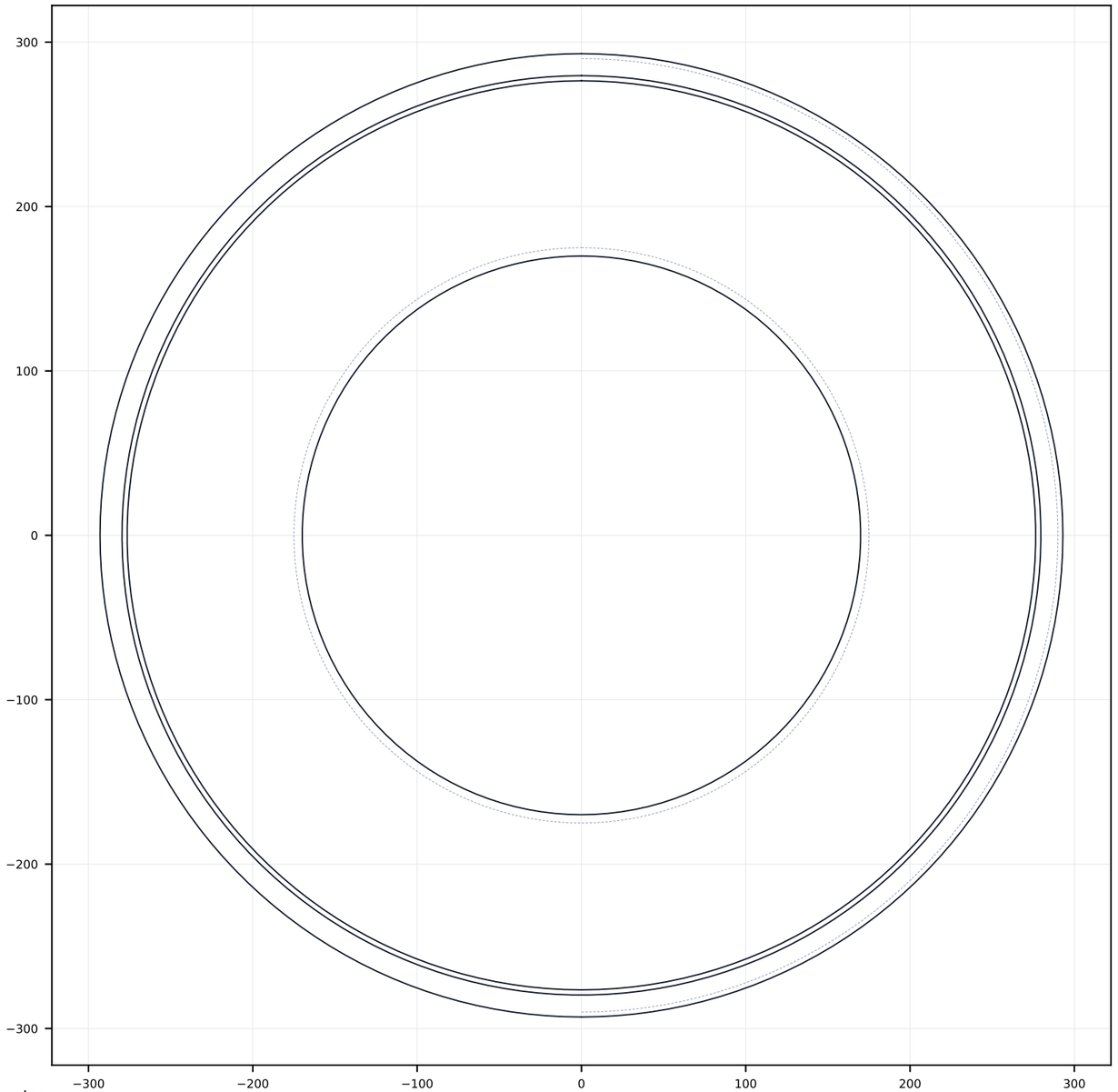
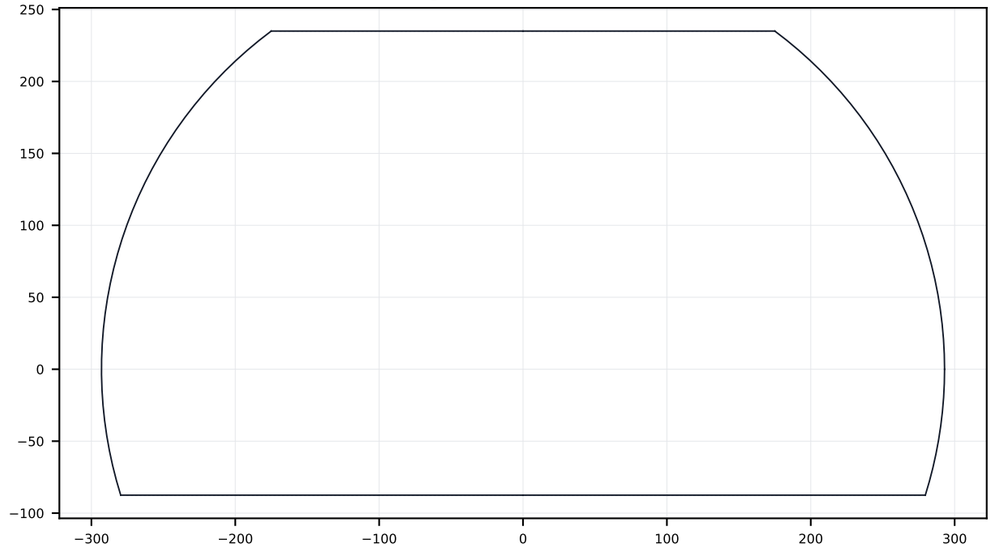
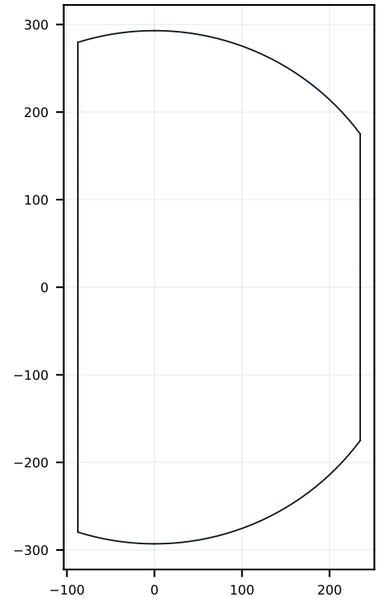
front (-y) (axes in mm from the view origin)



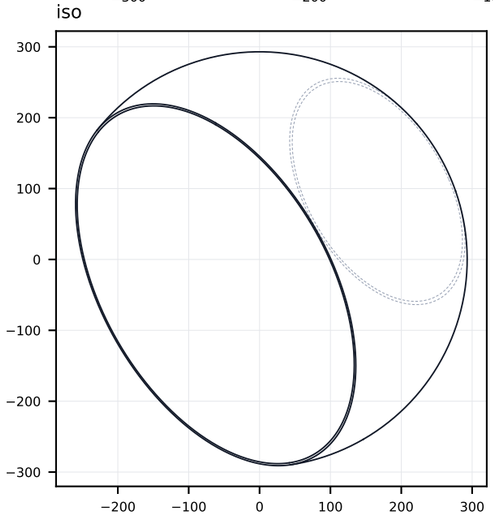
top (+z) (axes in mm from the view origin)



side (+x) (axes in mm from the view origin)

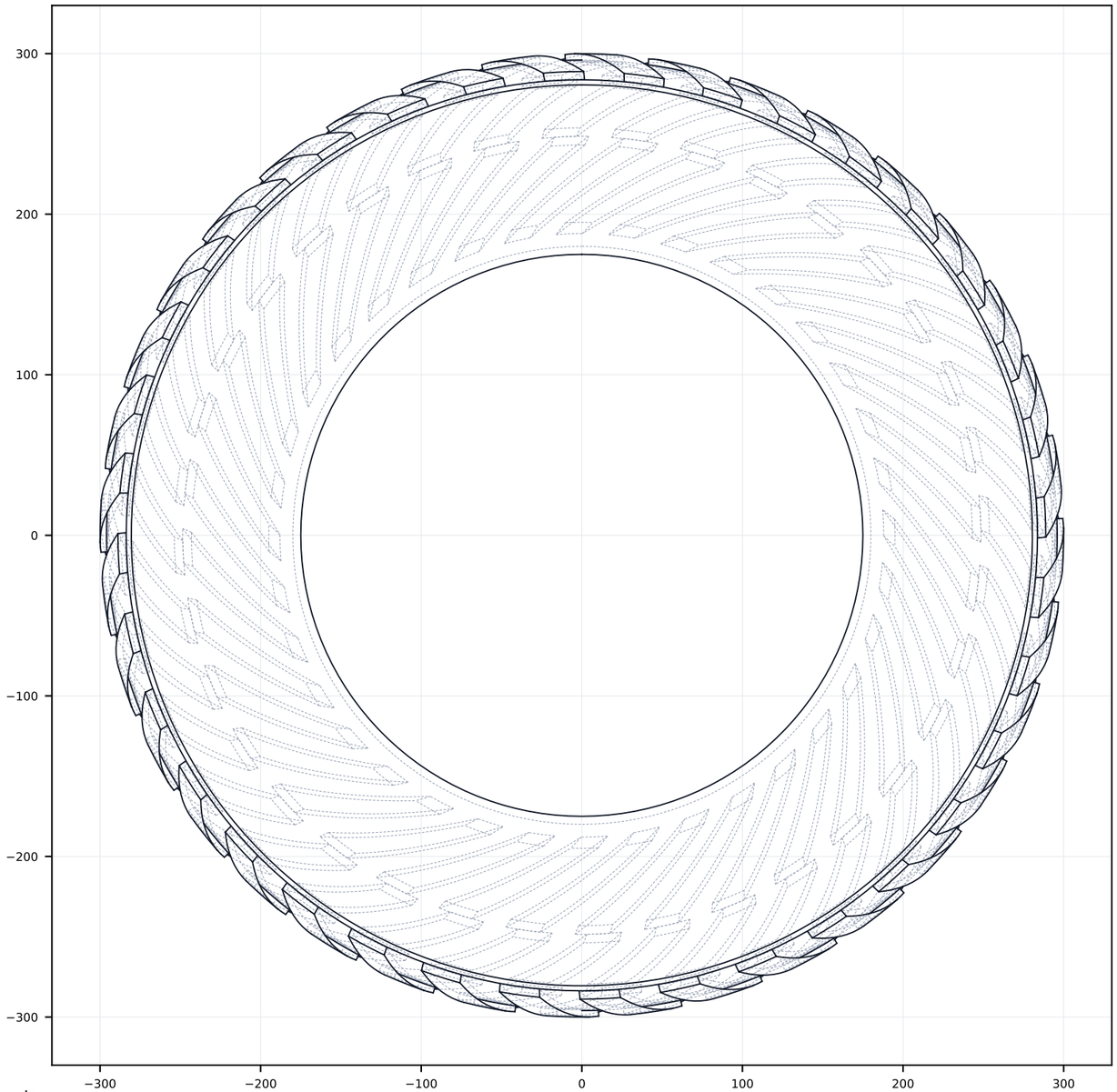


iso

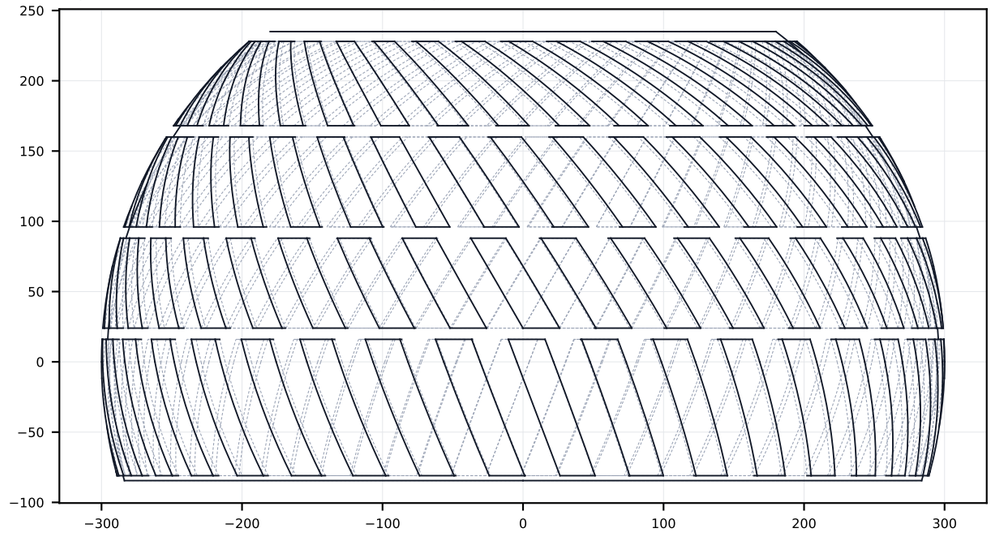


DRAWING drum
qty per robot: 2 (R-hand part = this one mirrored about XZ)
material: E-glass / epoxy, quasi-isotropic, 3.0 mm
process: vacuum infusion, 2-piece female mould split at the crown
finish: peel-ply inside at bond zones
mass: 3278.2 g overall: 586.0 x 586.0 x 586.0 mm (x, y, z)
note:
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/drum.step
view origin = bounding-box centre = robot (0.00, 90.00, 0.00)

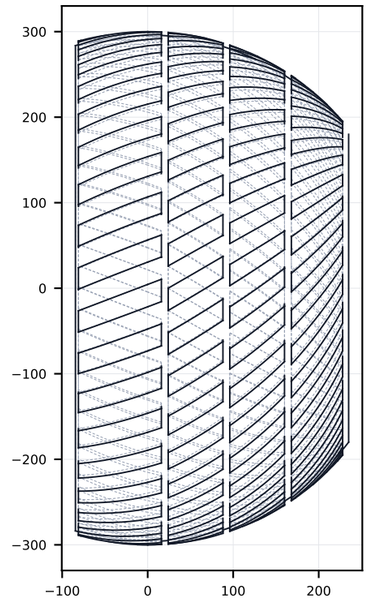
front (-y) (axes in mm from the view origin)



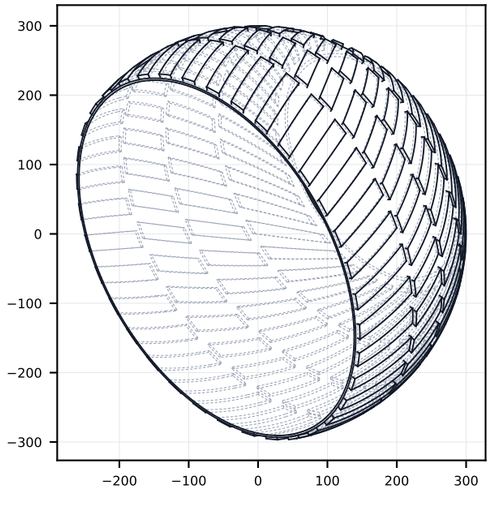
top (+z) (axes in mm from the view origin)



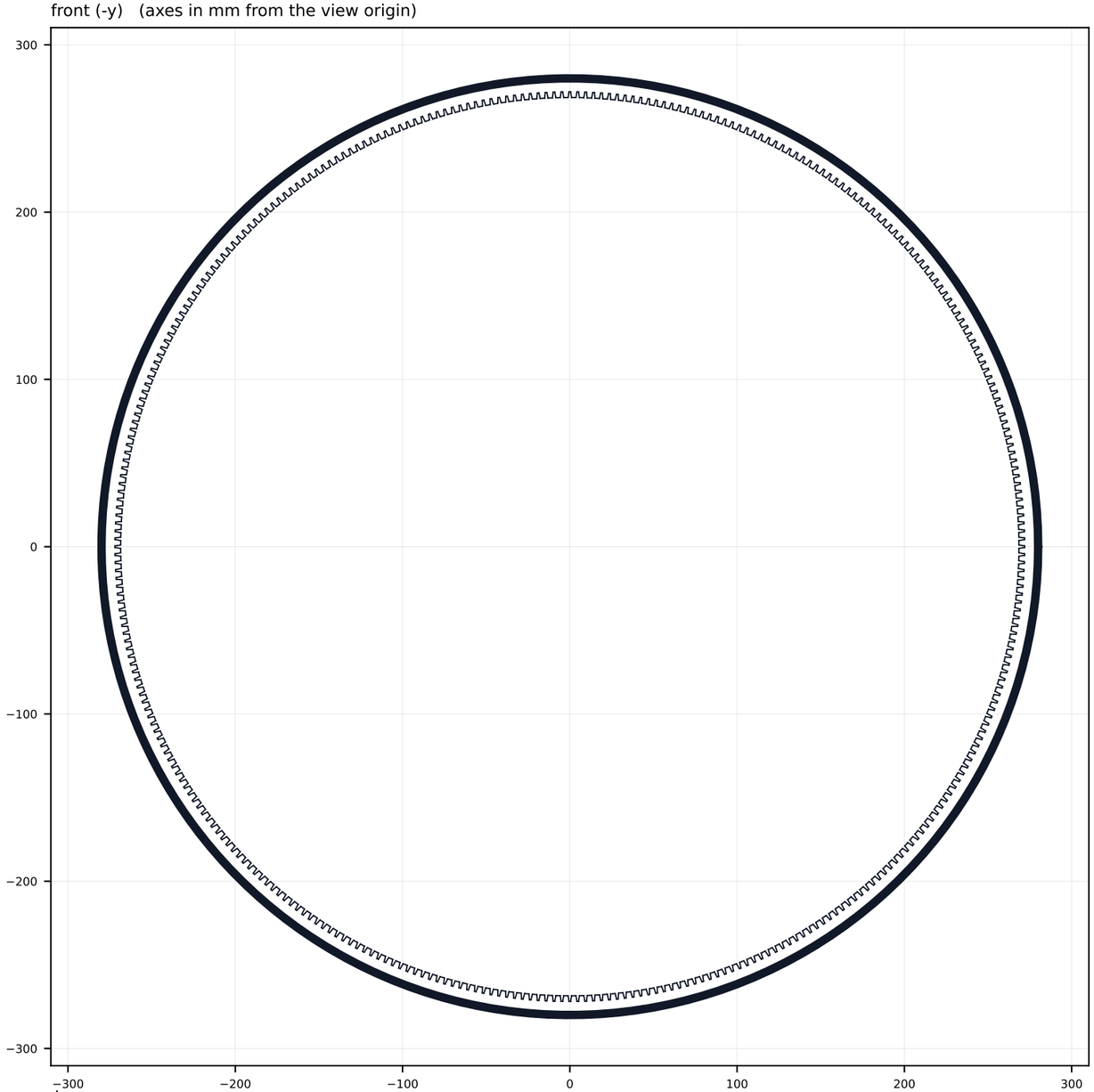
side (+x) (axes in mm from the view origin)

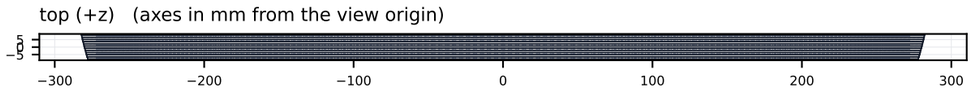


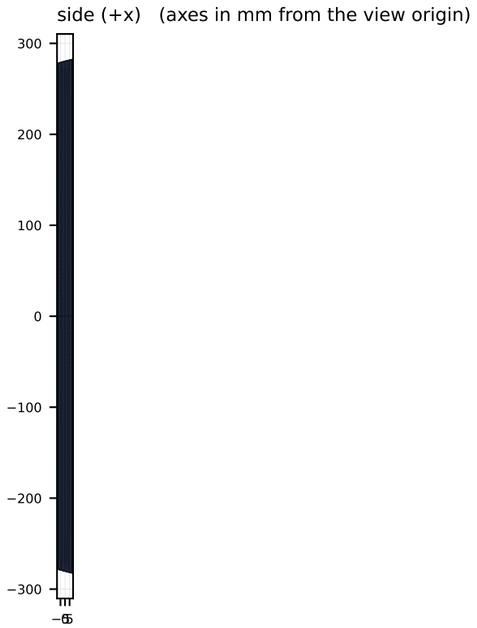
iso

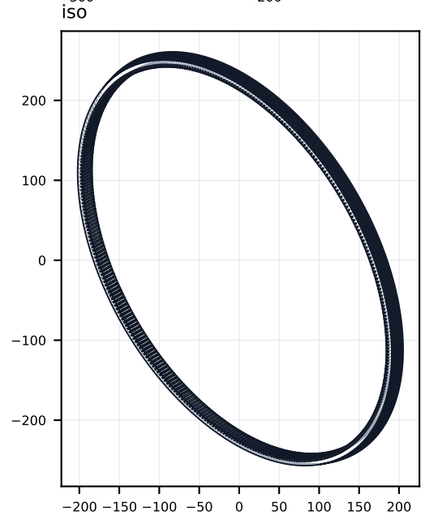


DRAWING tread
qty per robot: 2 (R-hand part = this one mirrored about XZ)
material: PU elastomer 70 Shore A (e.g. Smooth-On VytaFlex 70 class)
process: cast in place in a closed 2-piece mould
finish: as cast
mass: 3419.3 g overall: 600.0 x 592.0 x 600.0 mm (x, y, z)
note:
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/tread.step
view origin = bounding-box centre = robot (0.00, 90.00, 0.00)



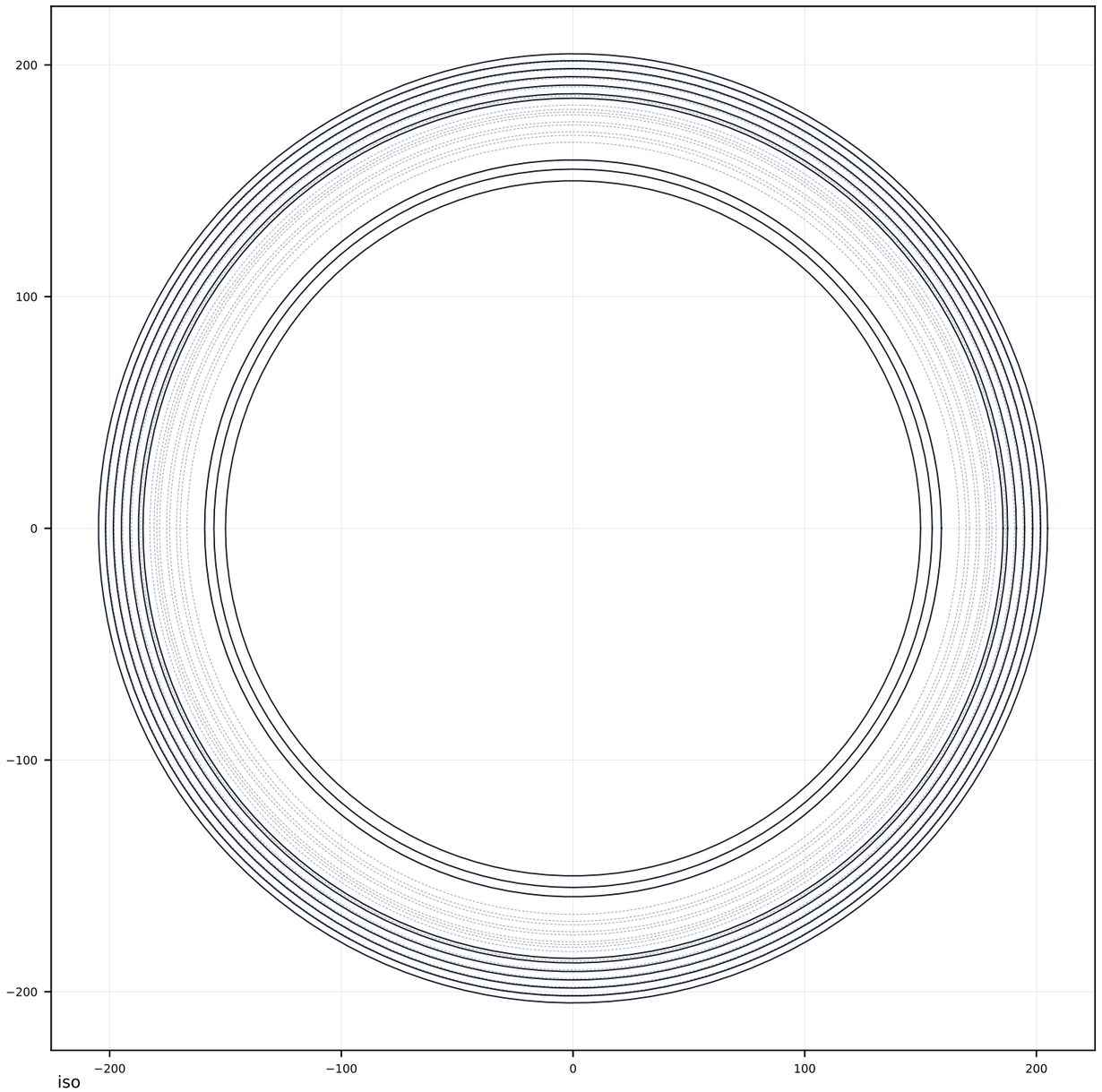




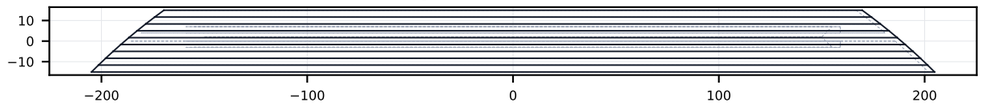


DRAWING ring_gear
qty per robot: 2 (R-hand part = this one mirrored about XZ)
material: 7075-T6 ring blank
process: CNC turn back to drum sphere, cut teeth by gear shaping or wire EDM
finish: hard anodise 25 um
mass: 813.5 g overall: 564.2 x 16.0 x 564.2 mm (x, y, z)
note: internal m1.5 z360 alpha 20 deg x=0, ISO 1328 grade 8
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/ring_gear.step
view origin = bounding-box centre = robot (0.00, 16.00, 0.00)

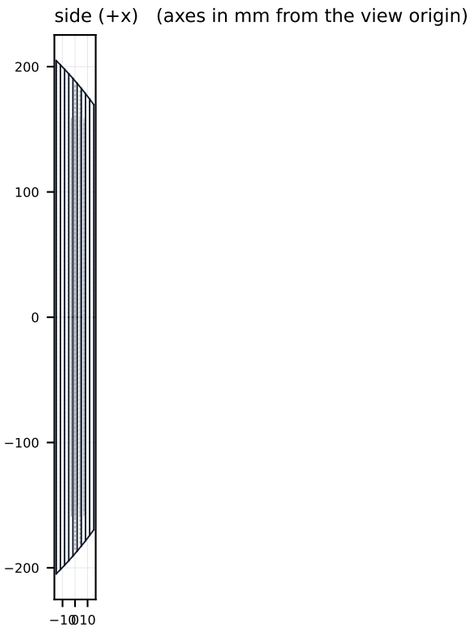
front (-y) (axes in mm from the view origin)



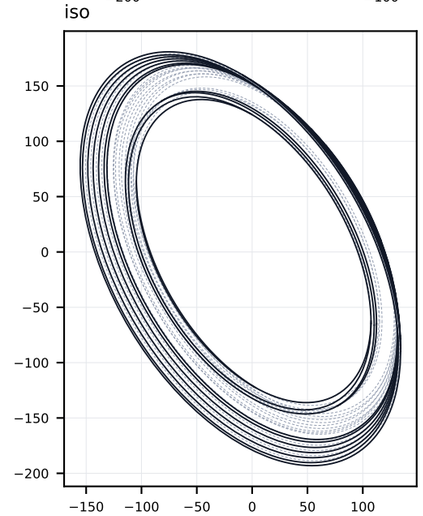
top (+z) (axes in mm from the view origin)



side (+x) (axes in mm from the view origin)

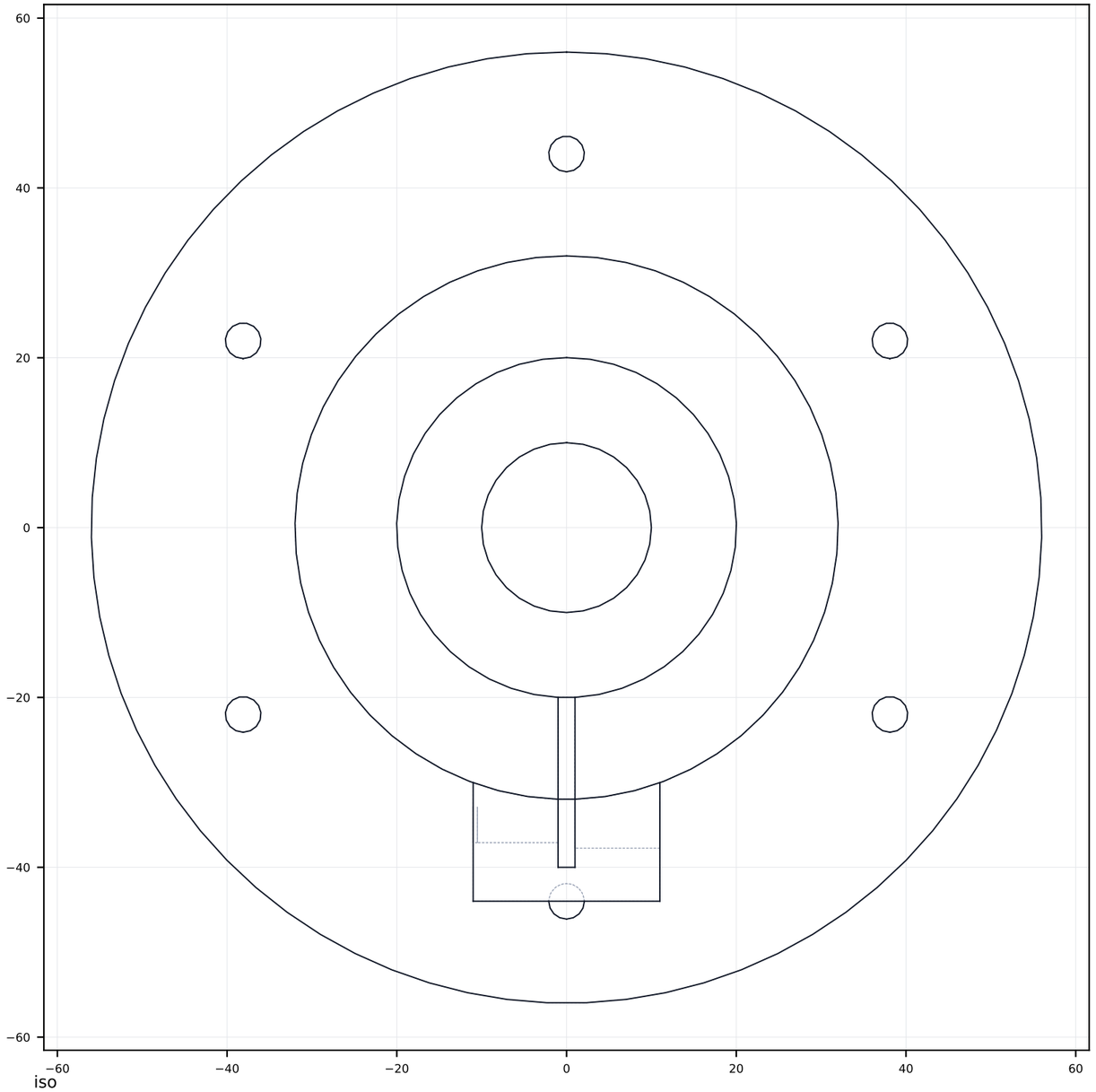


iso

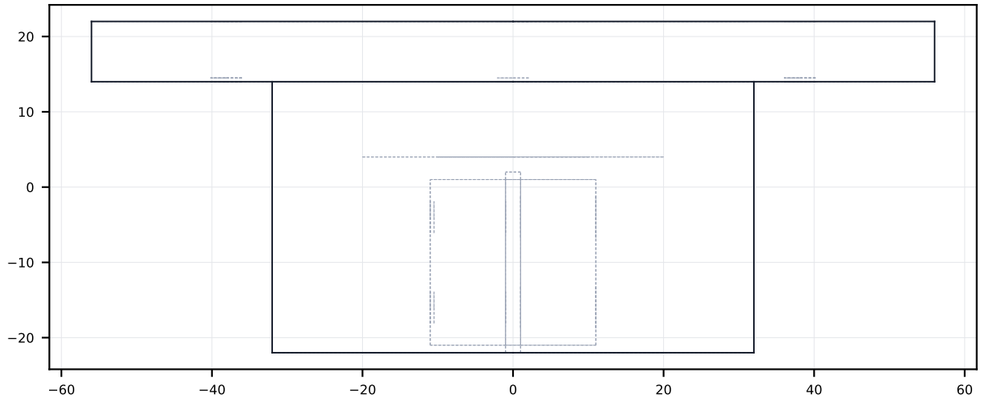


DRAWING rim
qty per robot: 2 (R-hand part = this one mirrored about XZ)
material: 6061-T6
process: CNC turn
finish: clear anodise; track flanks hard anodise
mass: 738.5 g overall: 409.6 x 30.0 x 409.6 mm (x, y, z)
note:
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/rim.step
view origin = bounding-box centre = robot (0.00, 310.00, 0.00)

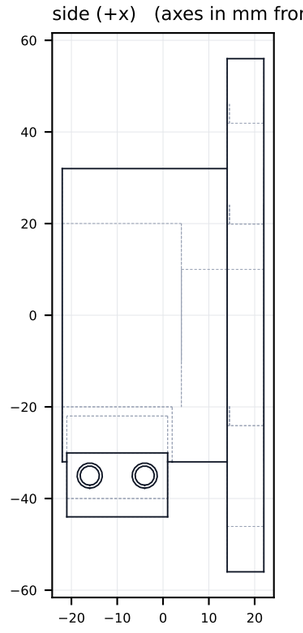
front (-y) (axes in mm from the view origin)



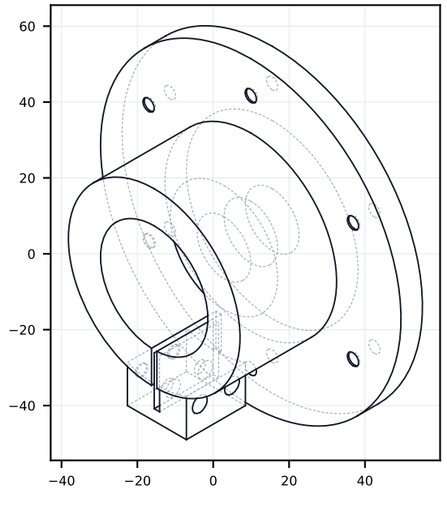
top (+z) (axes in mm from the view origin)



side (+x) (axes in mm from the view origin)



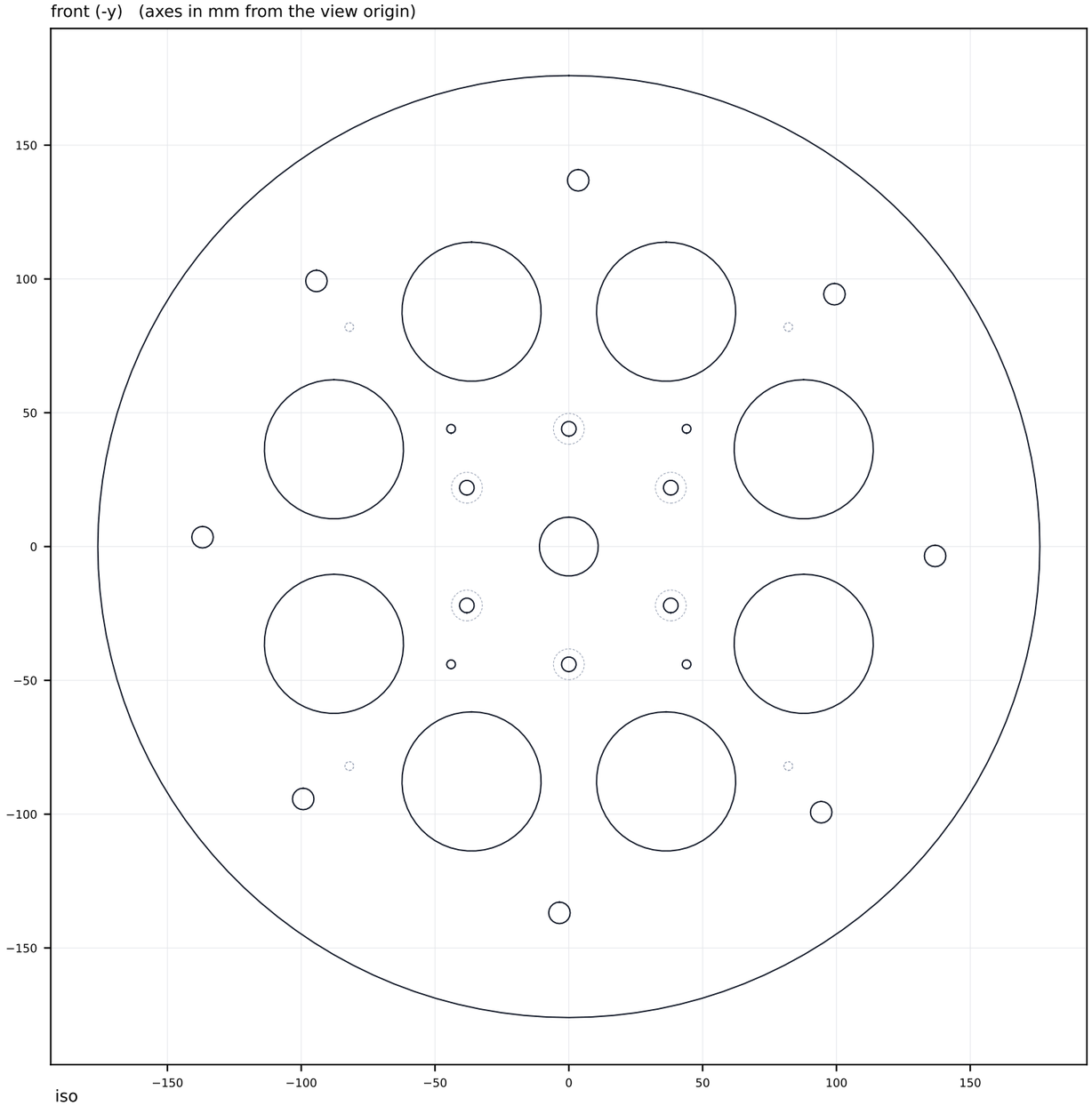
iso

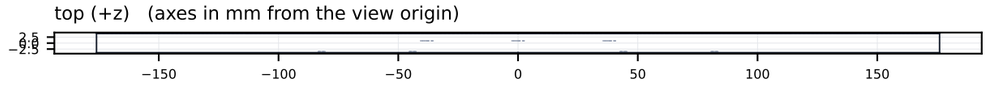


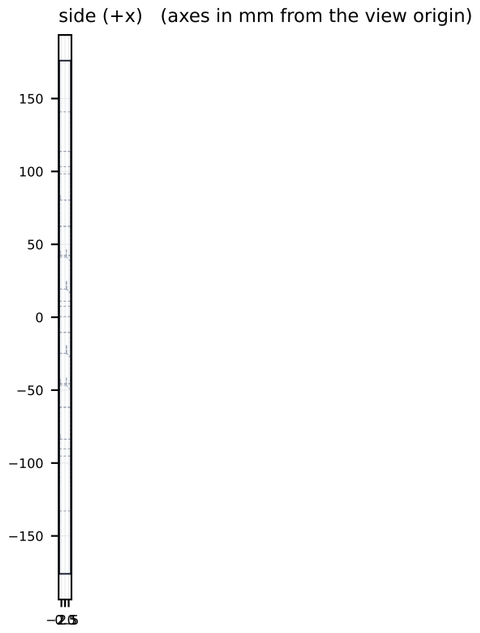
DRAWING pod_hub
qty per robot: 2 (R-hand part = this one mirrored about XZ)
material: 6061-T6
process: CNC mill-turn
finish: clear anodise
mass: 432.0 g overall: 112.0 x 44.0 x 112.0 mm (x, y, z)
note: bore 40 H7, slit clamp
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/pod_hub.step
view origin = bounding-box centre = robot (0.00, 308.00, 0.00)

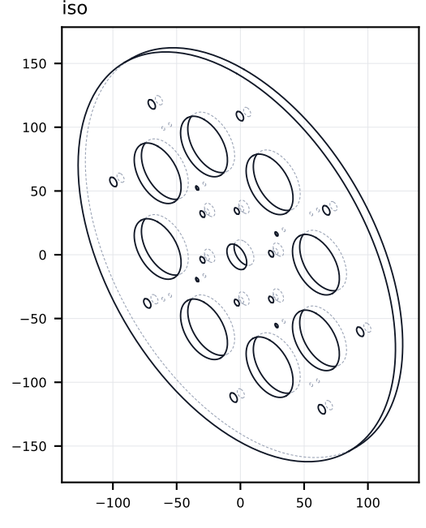
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
M5 tapped 10.0 deep (drill 4.2)	11.00	-16.00	-35.00	-x
M5 tapped 10.0 deep (drill 4.2)	11.00	-4.00	-35.00	-x
M5 tapped 10.6 deep (drill 4.2)	-0.00	30.00	-44.00	-y
M5 tapped 8.0 deep (drill 4.2)	-38.10	30.00	-22.00	-y
M5 tapped 8.0 deep (drill 4.2)	-38.10	30.00	22.00	-y
M5 tapped 8.0 deep (drill 4.2)	0.00	30.00	44.00	-y
M5 tapped 8.0 deep (drill 4.2)	38.10	30.00	-22.00	-y
M5 tapped 8.0 deep (drill 4.2)	38.10	30.00	22.00	-y
d5.5 through	11.00	-16.00	-35.00	-x
d5.5 through	11.00	-4.00	-35.00	-x





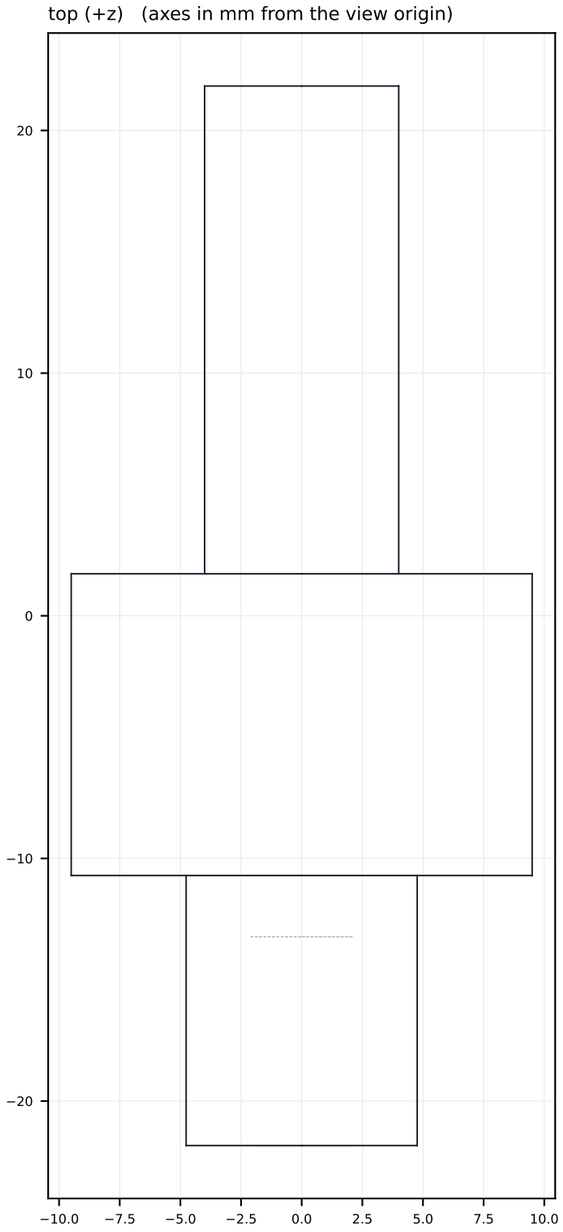


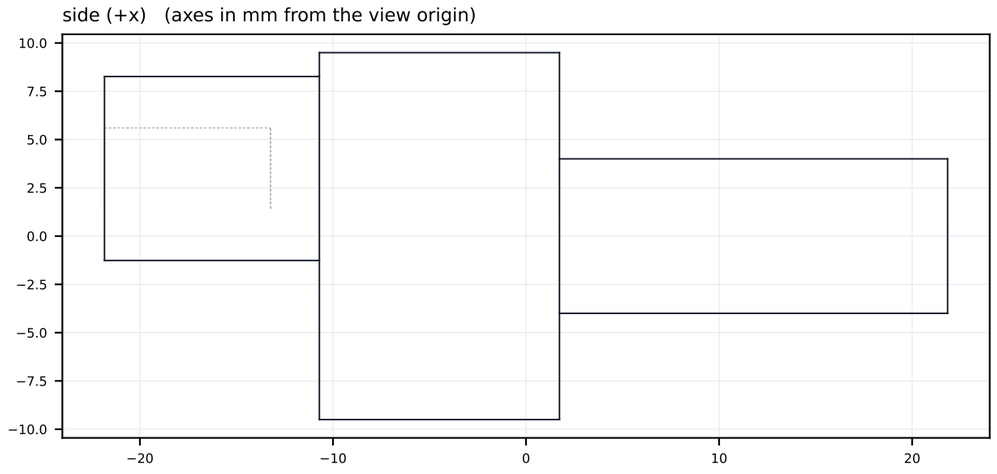


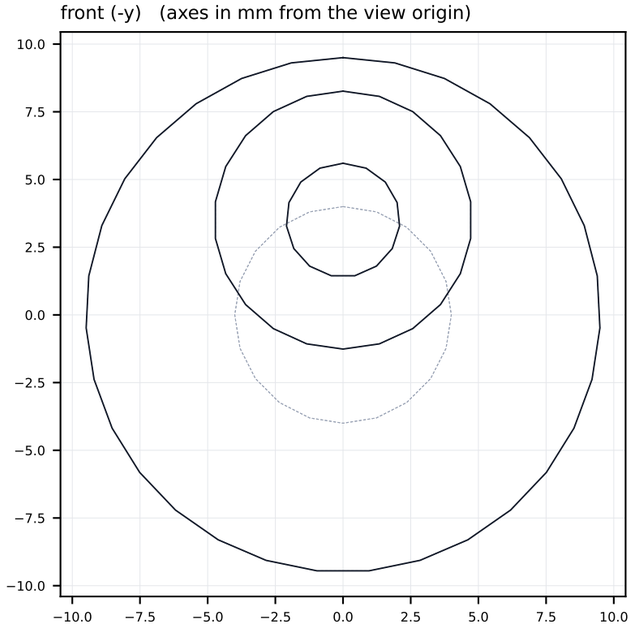
DRAWING pod_plate
qty per robot: 2 (R-hand part = this one mirrored about XZ)
material: 6061-T6 plate 8 mm
process: waterjet + CNC finish (stud holes reamed 8 H7)
finish: black anodise
mass: 1711.7 g overall: 352.0 x 8.0 x 352.0 mm (x, y, z)
note:
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/pod_plate.step
view origin = bounding-box centre = robot (0.00, 334.00, 0.00)

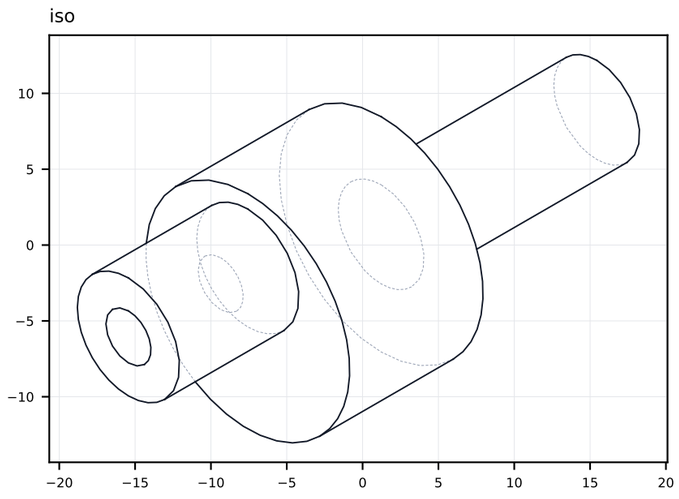
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
M4 tapped 8.0 deep (drill 3.3)	-44.00	9.00	-44.00	-y
M4 tapped 8.0 deep (drill 3.3)	-44.00	9.00	44.00	-y
M4 tapped 8.0 deep (drill 3.3)	44.00	9.00	-44.00	-y
M4 tapped 8.0 deep (drill 3.3)	44.00	9.00	44.00	-y
d5.5 through, csk 90 deg	-38.10	4.00	-22.00	-y
d5.5 through, csk 90 deg	-38.10	4.00	22.00	-y
d5.5 through, csk 90 deg	-0.00	4.00	-44.00	-y
d5.5 through, csk 90 deg	0.00	4.00	44.00	-y
d5.5 through, csk 90 deg	38.10	4.00	-22.00	-y
d5.5 through, csk 90 deg	38.10	4.00	22.00	-y







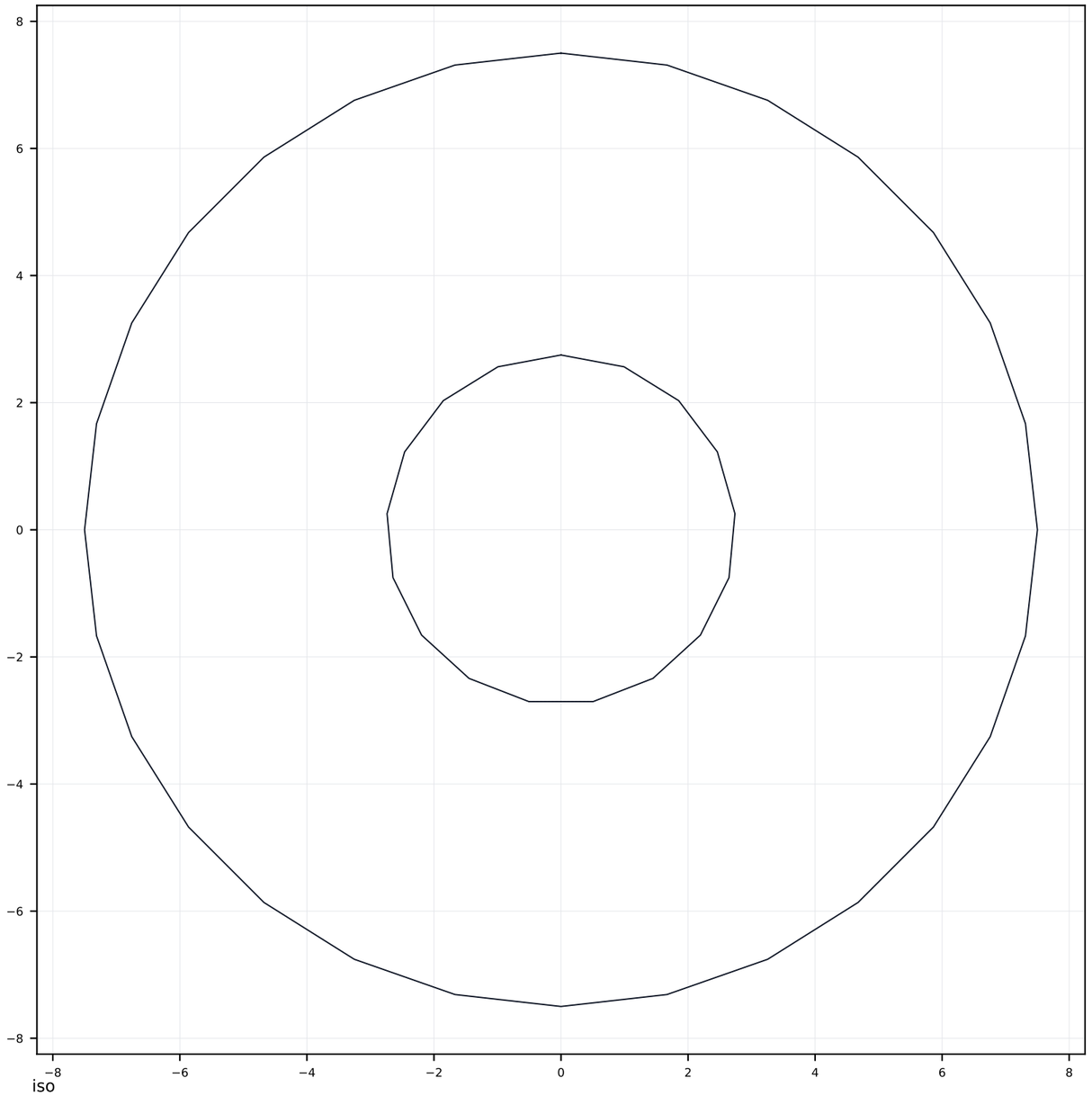


DRAWING wheel_stud
qty per robot: 16 (R-hand part = this one mirrored about XZ)
material: 303 stainless
process: CNC turn
finish:
mass: 41.4 g overall: 19.0 x 43.7 x 19.0 mm (x, y, z)
note: seat g6, M8x1.25 stem, axial M5 tapped end for the wheel retainer
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/wheel_stud.step
view origin = bounding-box centre = robot (136.88, 328.27, -3.50)

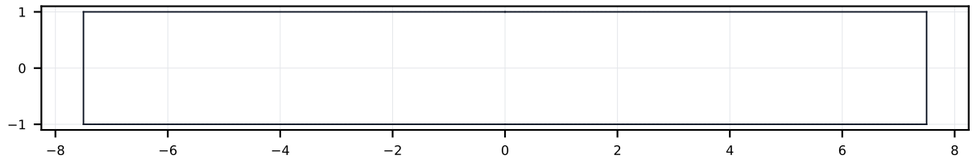
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
M5 tapped 8.6 deep (drill 4.2)	0.00	-23.83	3.50	+y

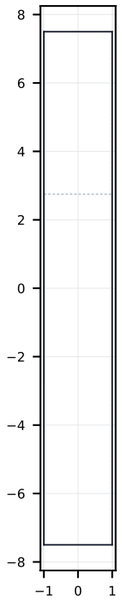
front (-y) (axes in mm from the view origin)



top (+z) (axes in mm from the view origin)



side (+x) (axes in mm from the view origin)



iso

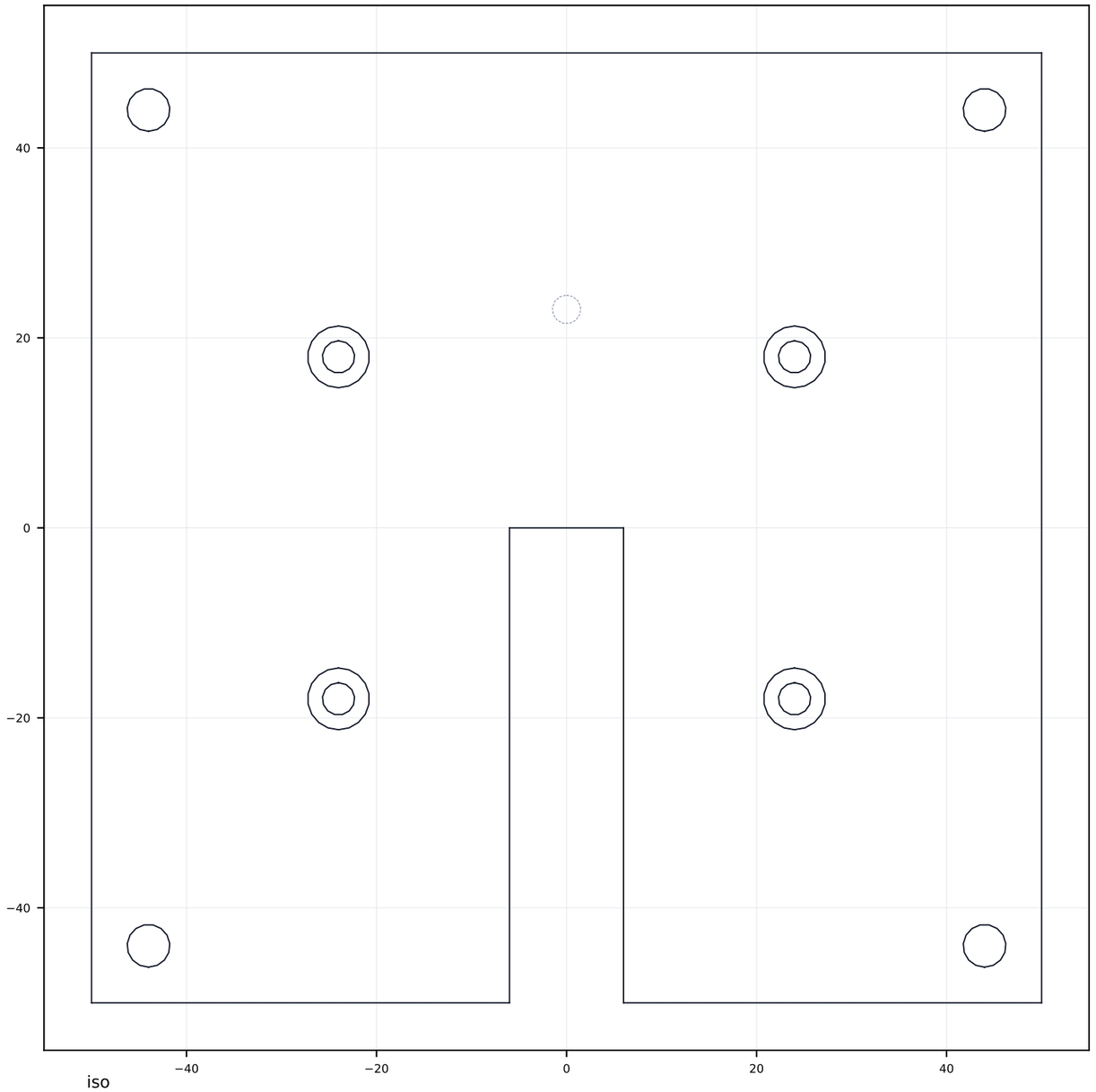


DRAWING wheel_retainer
qty per robot: 16 (R-hand part = this one mirrored about XZ)
material: 303 stainless
process: turned washer 15 x 2 (clamps the inner race)
finish:
mass: 2.4 g overall: 15.0 x 2.0 x 15.0 mm (x, y, z)
note:
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/wheel_retainer.step
view origin = bounding-box centre = robot (136.88, 305.44, 0.00)

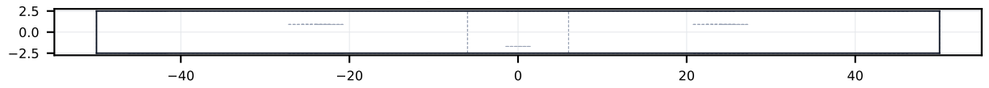
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
d5.5 through	0.00	-1.00	0.00	+y

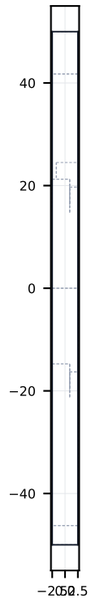
front (-y) (axes in mm from the view origin)



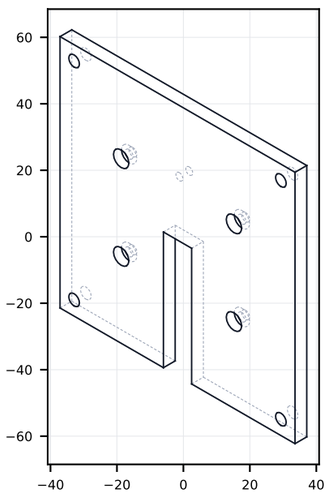
top (+z) (axes in mm from the view origin)



side (+x) (axes in mm from the view origin)



iso

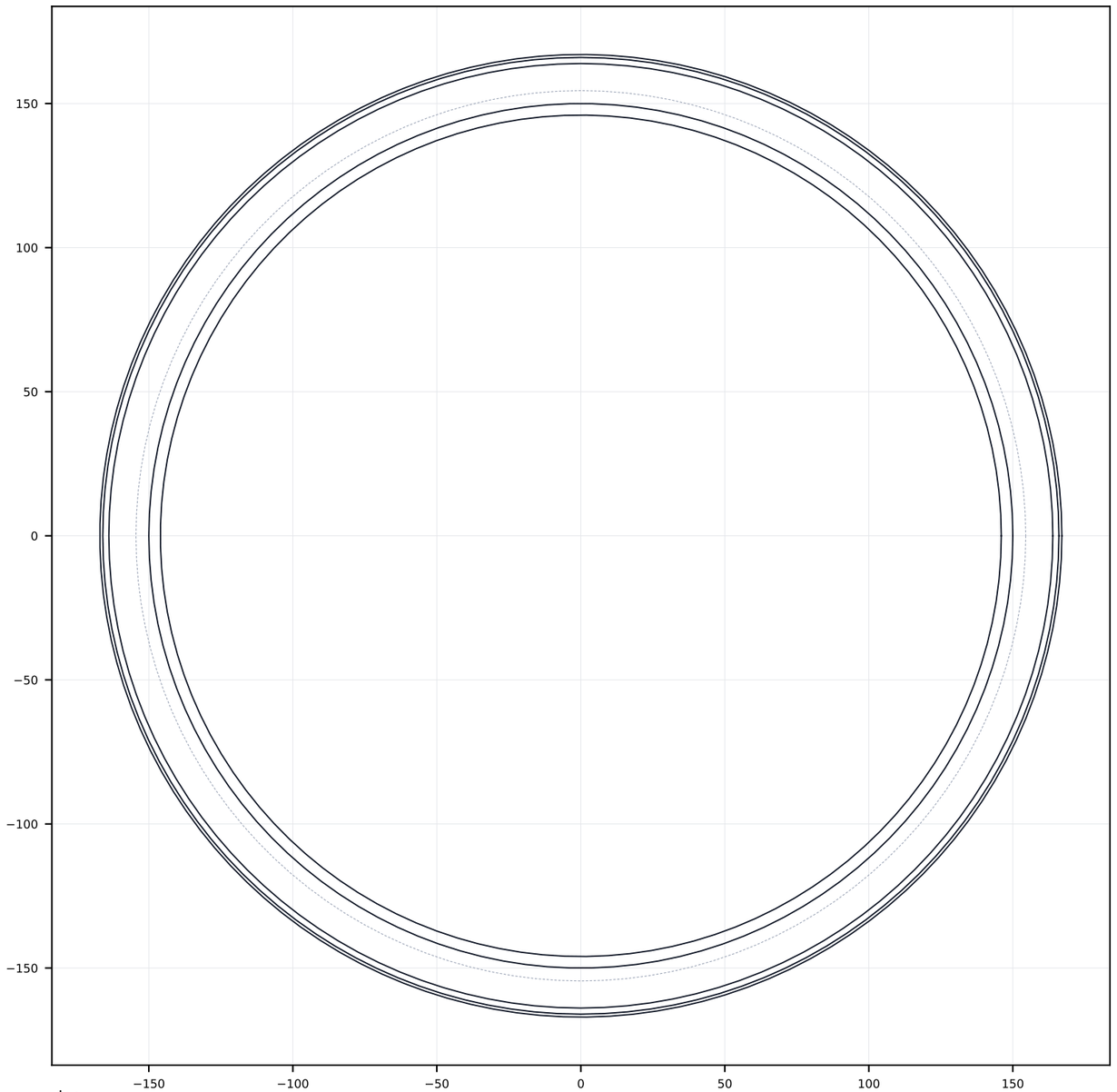


DRAWING lidar_riser
qty per robot: 2 (R-hand part = this one mirrored about XZ)
material: 6061-T6
process: CNC
finish: black anodise
mass: 124.6 g overall: 100.0 x 5.0 x 100.0 mm (x, y, z)
note: 2 x d3 m6 dowels to the LiDAR
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/lidar_riser.step
view origin = bounding-box centre = robot (0.00, 340.50, 0.00)

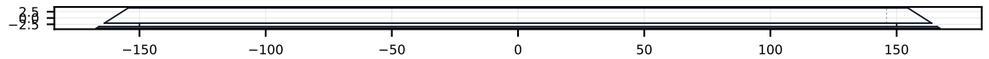
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
d3.4 through	-24.00	0.90	-18.00	+y
d3.4 through	-24.00	0.90	18.00	+y
d3.4 through	24.00	0.90	-18.00	+y
d3.4 through	24.00	0.90	18.00	+y
d4.5 through	-44.00	2.50	-44.00	-y
d4.5 through	-44.00	2.50	44.00	-y
d4.5 through	44.00	2.50	-44.00	-y
d4.5 through	44.00	2.50	44.00	-y

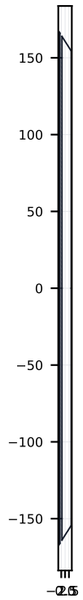
front (-y) (axes in mm from the view origin)



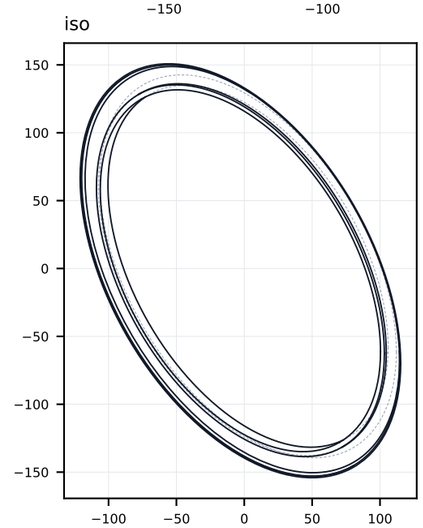
top (+z) (axes in mm from the view origin)



side (+x) (axes in mm from the view origin)

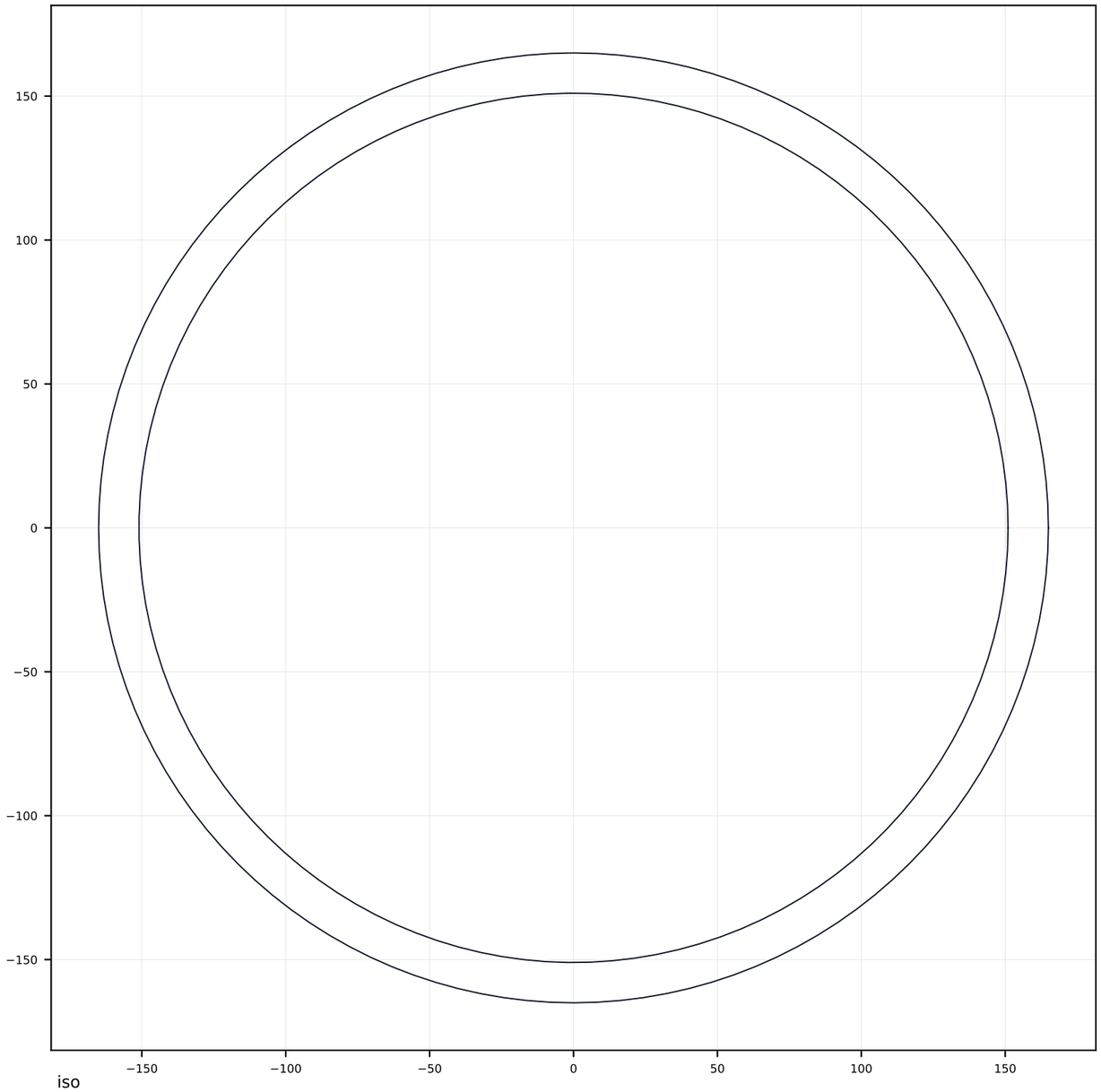


iso

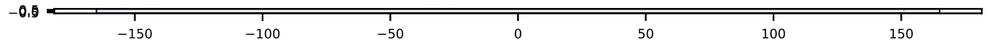


DRAWING light_ring
qty per robot: 2 (R-hand part = this one mirrored about XZ)
material: PC clear (Makrolon 2405)
process: CNC turn + vapour polish
finish:
mass: 100.5 g overall: 334.0 x 8.0 x 334.0 mm (x, y, z)
note: diffuses the status LEDs
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/light_ring.step
view origin = bounding-box centre = robot (0.00, 342.04, 0.00)

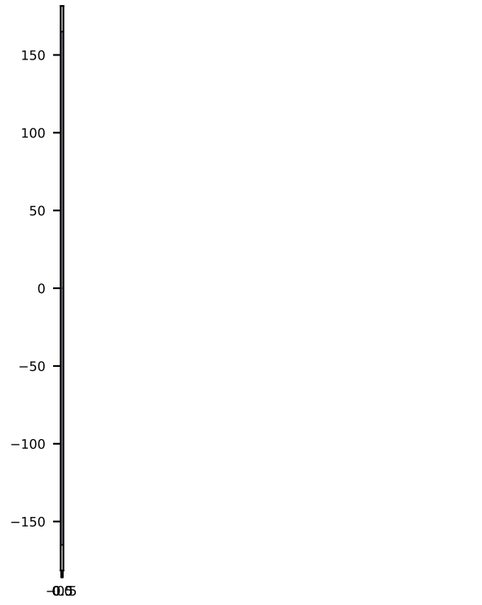
front (-y) (axes in mm from the view origin)



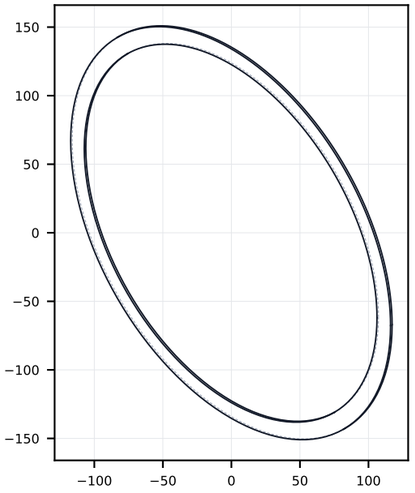
top (+z) (axes in mm from the view origin)



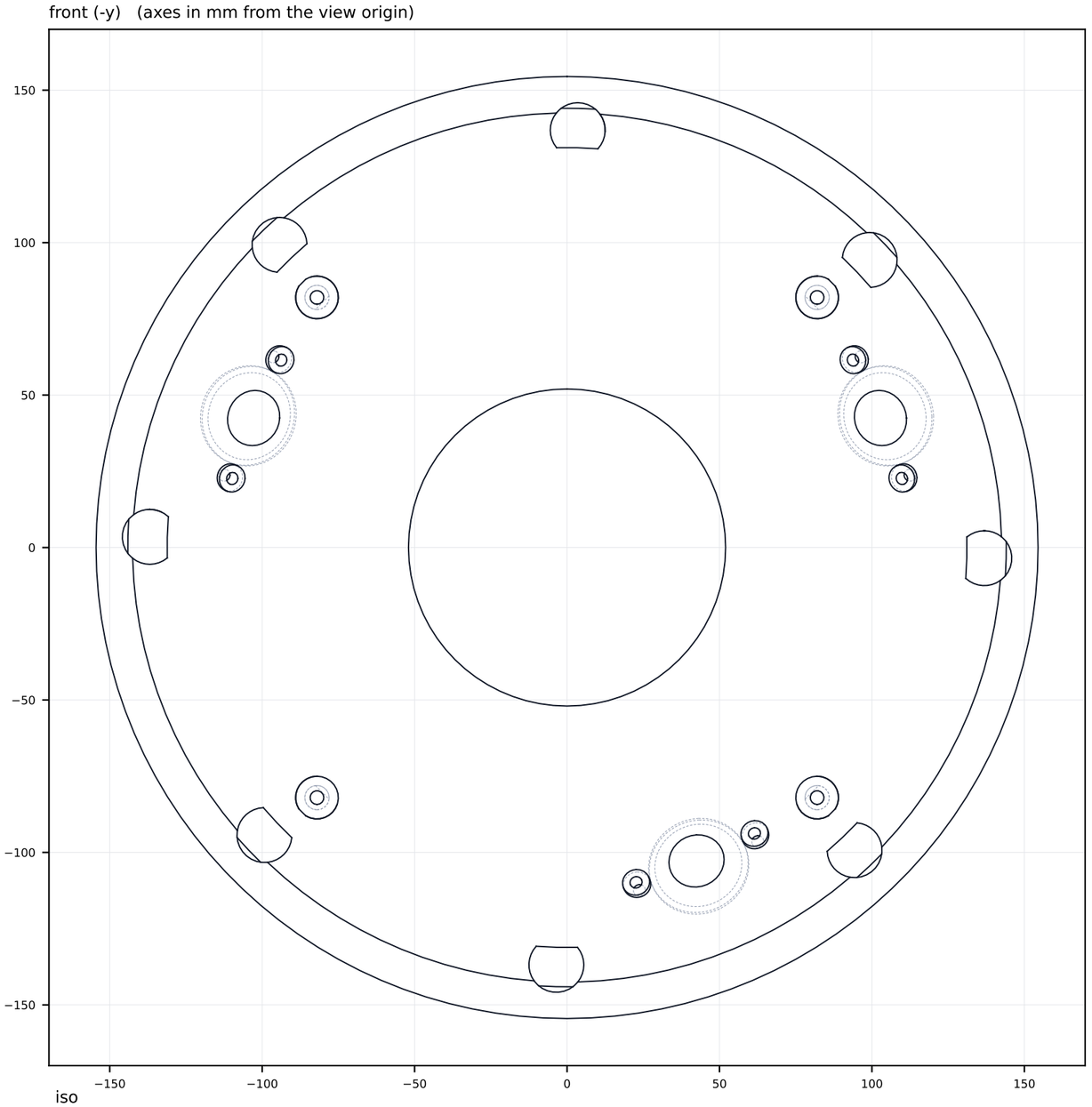
side (+x) (axes in mm from the view origin)

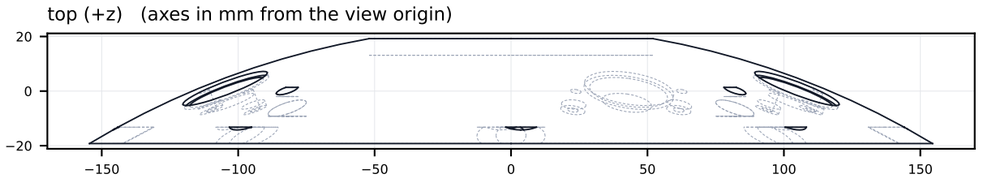


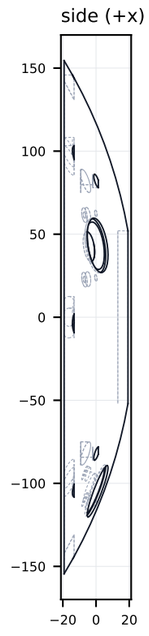
iso

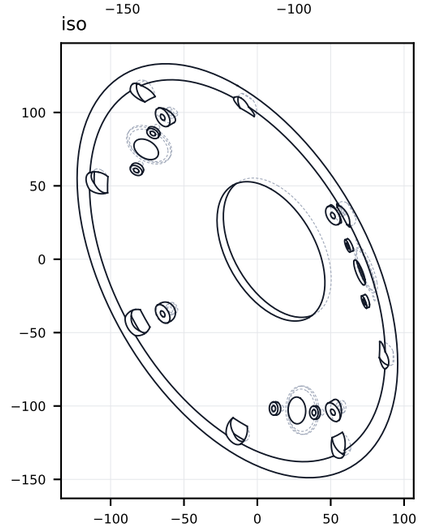


DRAWING led_pcb
qty per robot: 2 (R-hand part = this one mirrored about XZ)
material: FR4 1.6 mm
process: PCBA: 24 x WS2812B-2020 RGB LEDs, on-board 12 V -> 5 V 1 A buck (TPS62933
    class), JST-GH 3-pin (12 V, data, GND)
finish:
mass: 30.0 g overall: 330.0 x 1.6 x 330.0 mm (x, y, z)
note:
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/led_pcb.step
view origin = bounding-box centre = robot (0.00, 338.80, 0.00)







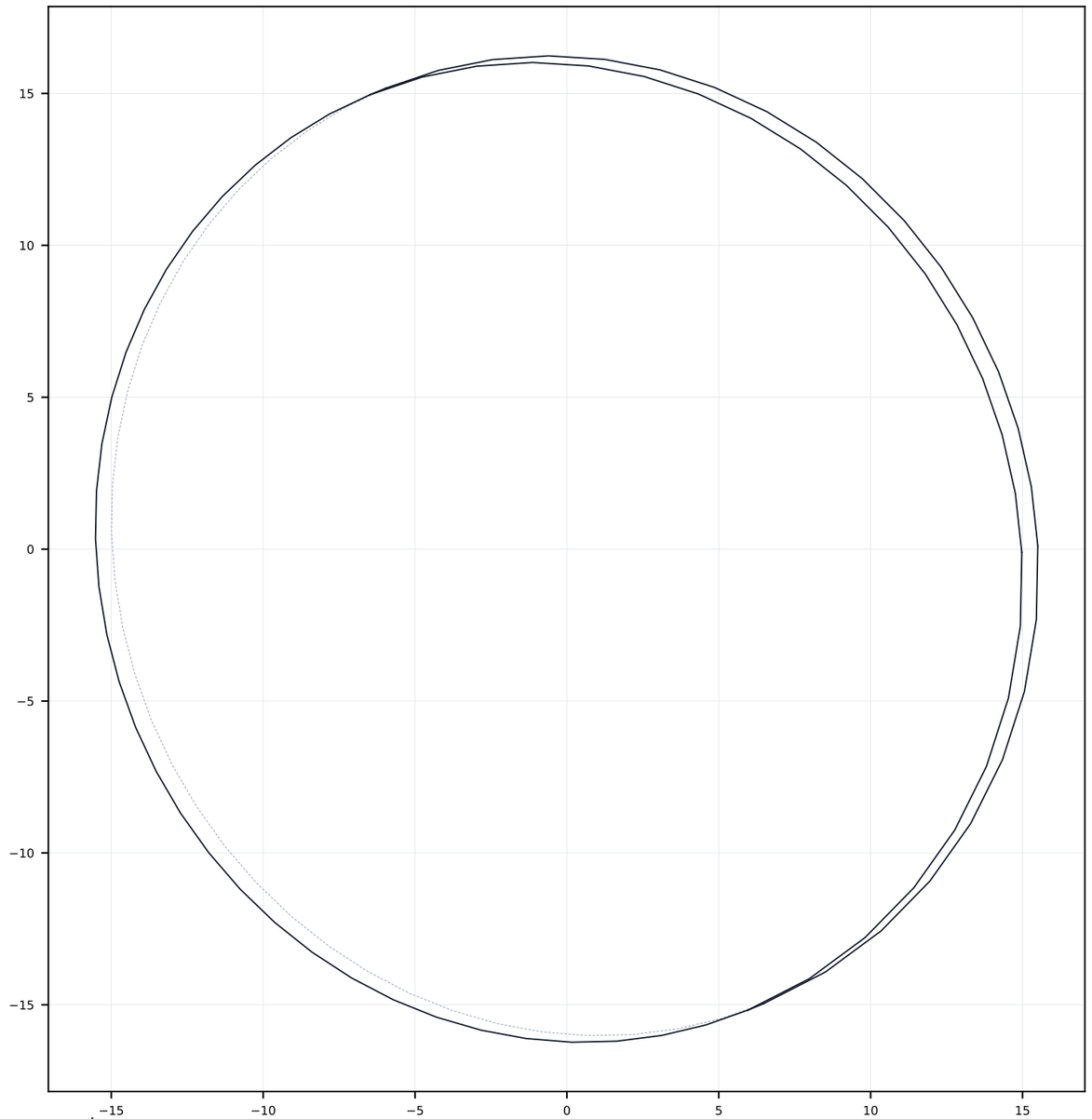


DRAWING pod_shield
qty per robot: 2 (R-hand part = this one mirrored about XZ)
material: PA12 (HP MJF), dyed black
process: MJF print
finish: bead blast + clear coat
mass: 384.0 g overall: 309.0 x 38.4 x 309.0 mm (x, y, z)
note: 4 x M4 button heads on spot faces into the standoffs; serviceable from outside
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/pod_shield.step
view origin = bounding-box centre = robot (-0.00, 365.22, 0.00)

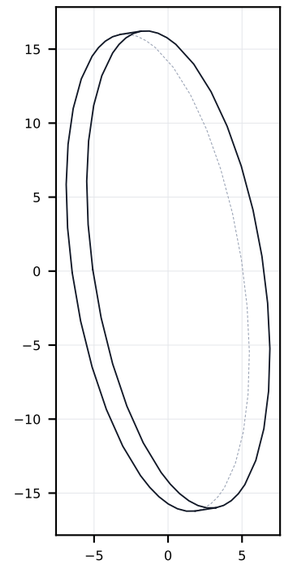
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
d4.0 x 7.4 for M3 heat-set insert	-108.06	-11.58	21.92	+y
d4.0 x 7.4 for M3 heat-set insert	-92.02	-11.58	60.74	+y
d4.0 x 7.4 for M3 heat-set insert	21.92	-11.58	-108.06	+y
d4.0 x 7.4 for M3 heat-set insert	60.74	-11.58	-92.02	+y
d4.0 x 7.4 for M3 heat-set insert	92.02	-11.58	60.74	+y
d4.0 x 7.4 for M3 heat-set insert	108.06	-11.58	21.92	+y
d4.5 through	-82.02	-2.00	-82.02	-y
d4.5 through	-82.02	-2.00	82.02	-y
d4.5 through	82.02	-2.00	-82.02	-y
d4.5 through	82.02	-2.00	82.02	-y

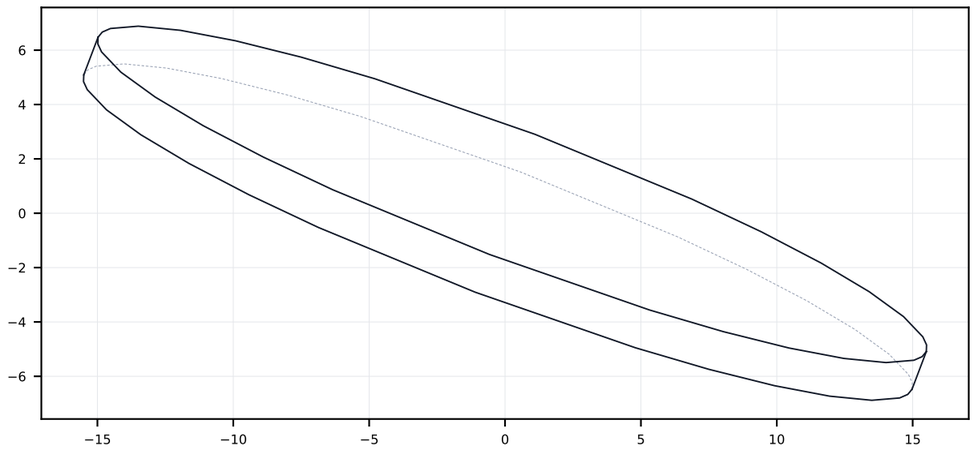
front (-y) (axes in mm from the view origin)



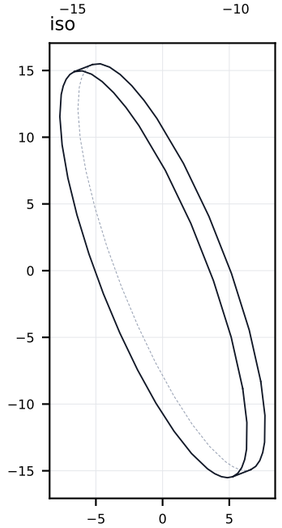
side (+x) (axes in mm from the view origin)



top (+z) (axes in mm from the view origin)

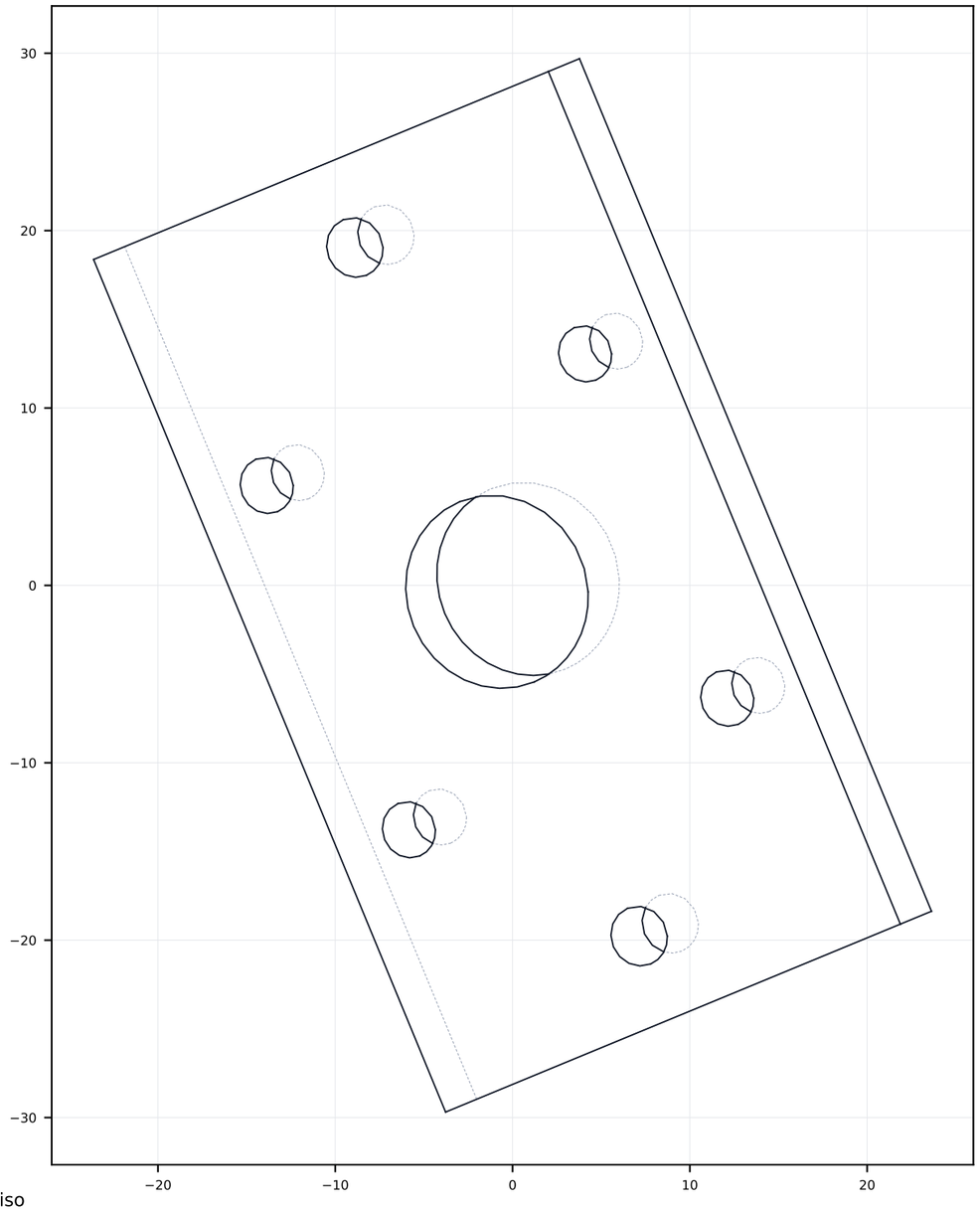


iso

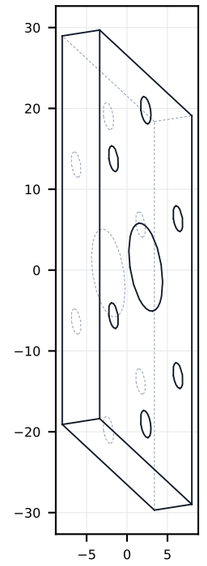


DRAWING cam_window
qty per robot: 6 (R-hand part = this one mirrored about XZ)
material: PC 1.5 mm, AR coated
process: laser cut disc d32.6
finish:
mass: 1.5 g overall: 31.1 x 13.8 x 32.5 mm (x, y, z)
note:
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/cam_window.step
view origin = bounding-box centre = robot (104.52, 365.44, 43.18)

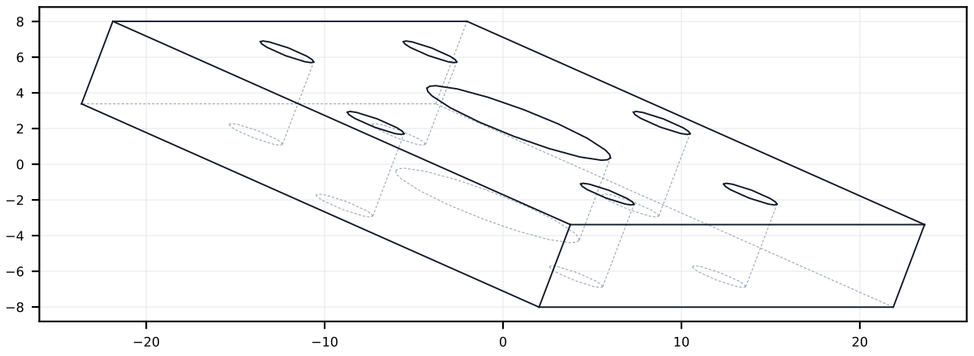
front (-y) (axes in mm from the view origin)



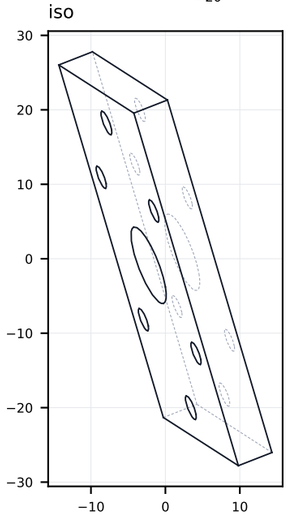
side (+x) (axes in mm from the view origin)



top (+z) (axes in mm from the view origin)



iso

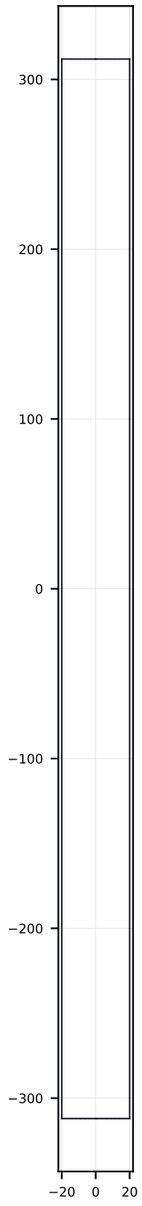


DRAWING cam_holder
qty per robot: 6 (R-hand part = this one mirrored about XZ)
material: PA12
process: MJF print
finish:
mass: 7.1 g overall: 47.3 x 16.0 x 59.4 mm (x, y, z)
note: 2 x M3 into shield inserts; camera on 4 x M2 into M2 inserts
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/cam_holder.step
view origin = bounding-box centre = robot (100.92, 355.96, 41.69)

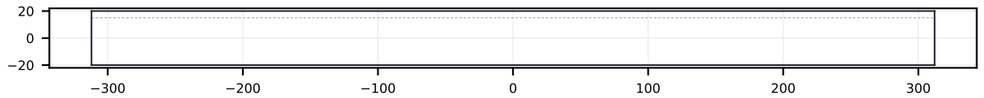
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
d3.2 x 5.8 for M2 heat-set insert	-16.01	-3.97	4.75	+y
d3.2 x 5.8 for M2 heat-set insert	-7.99	-3.97	-14.66	+y
d3.2 x 5.8 for M2 heat-set insert	1.95	-11.94	12.17	+y
d3.2 x 5.8 for M2 heat-set insert	9.97	-11.94	-7.24	+y
d3.4 through	-8.90	-2.31	19.05	+y
d3.4 through	7.14	-2.31	-19.77	+y

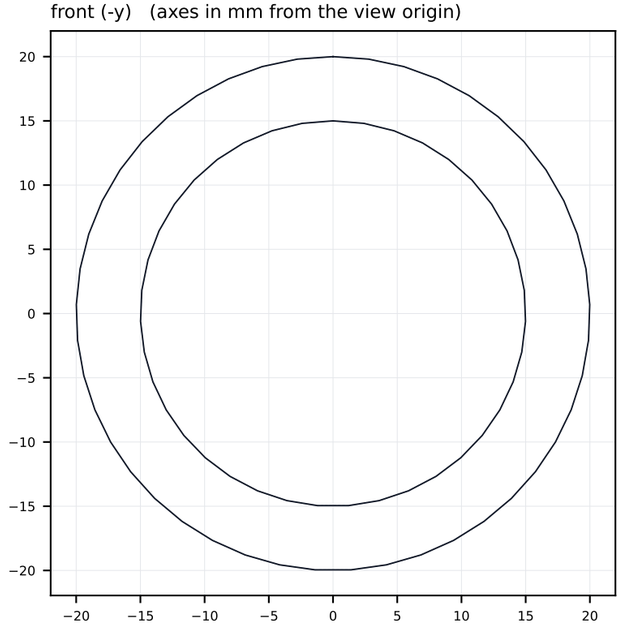
top (+z) (axes in mm from the view origin)



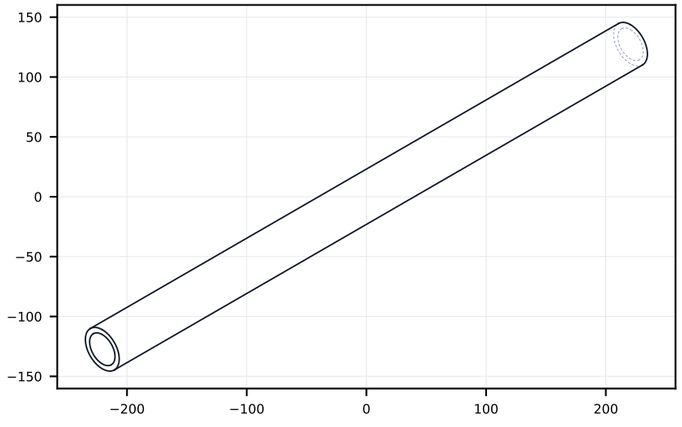
side (+x) (axes in mm from the view origin)



front (-y) (axes in mm from the view origin)

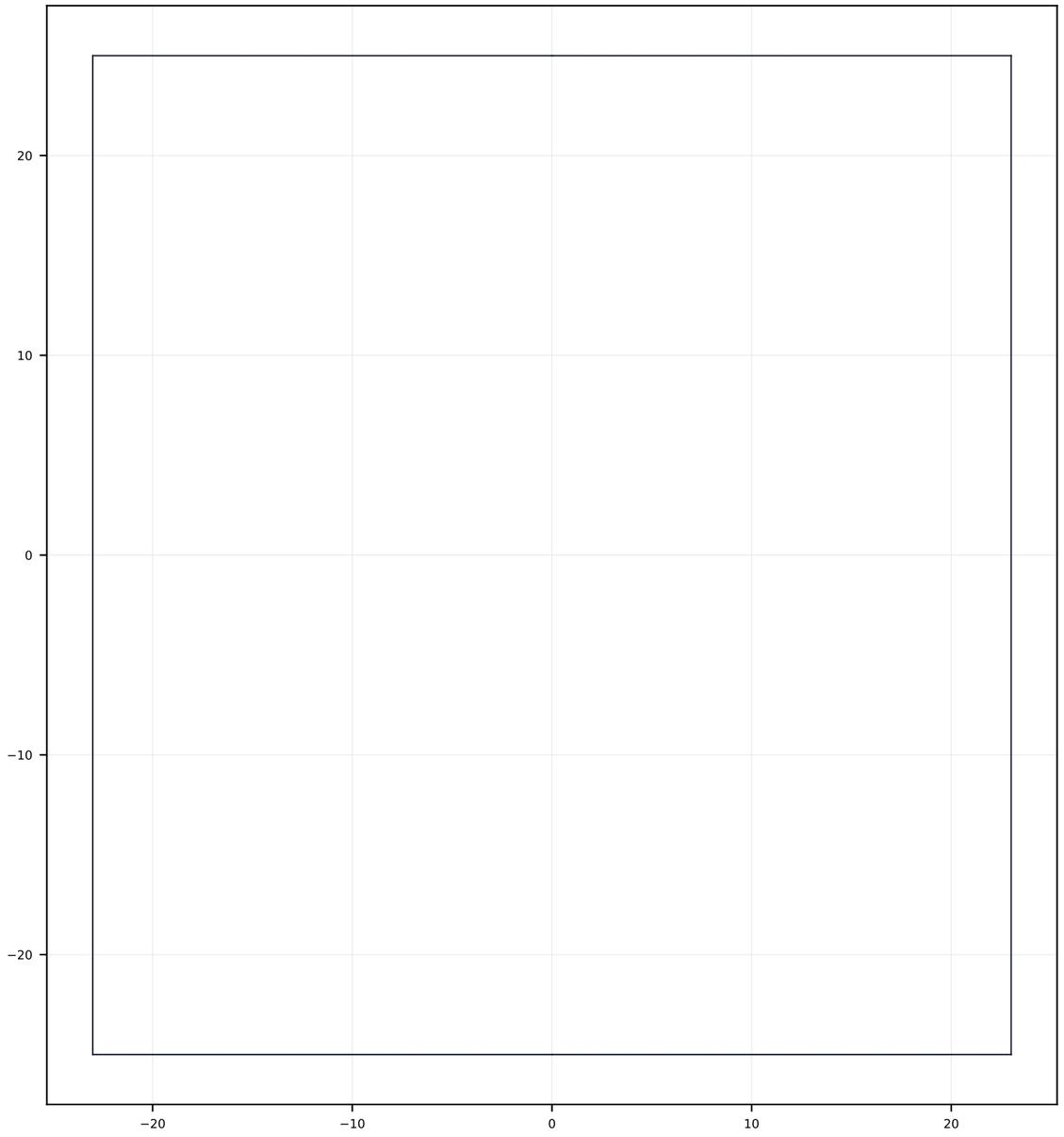


iso

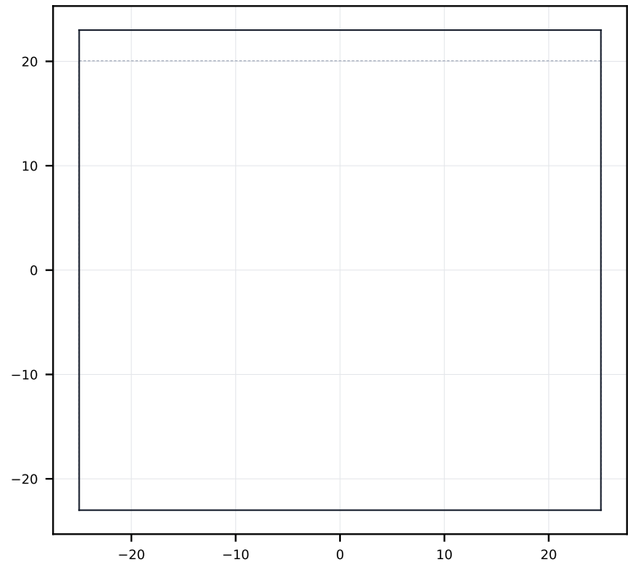


DRAWING axle
qty per robot: 1
material: roll-wrapped CFRP tube 40 x 30, high-modulus fibre (E_axial >= 150 GPa),
    mostly 0 deg + +-45
process: cut to 624 mm, ends faced
finish:
mass: 531.7 g overall: 40.0 x 624.0 x 40.0 mm (x, y, z)
note: cable slots 10 x 30 at y +-170 and +-296 (deburred, edge-sealed)
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/axle.step
view origin = bounding-box centre = robot (0.00, 0.00, 0.00)

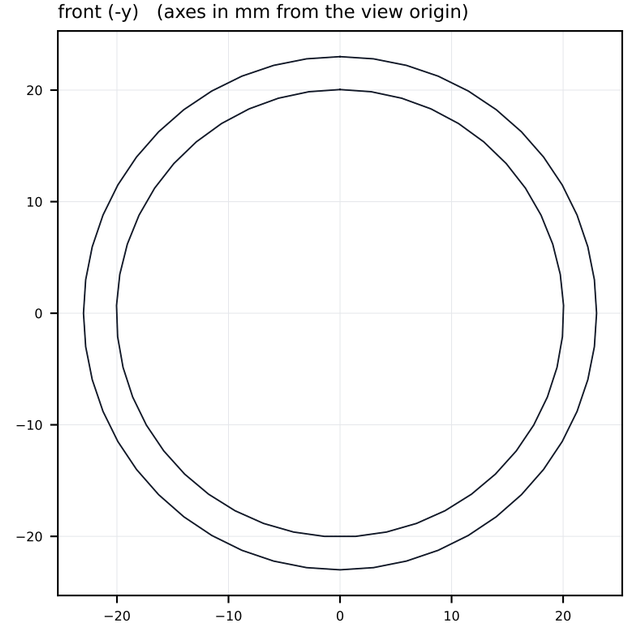
top (+z) (axes in mm from the view origin)



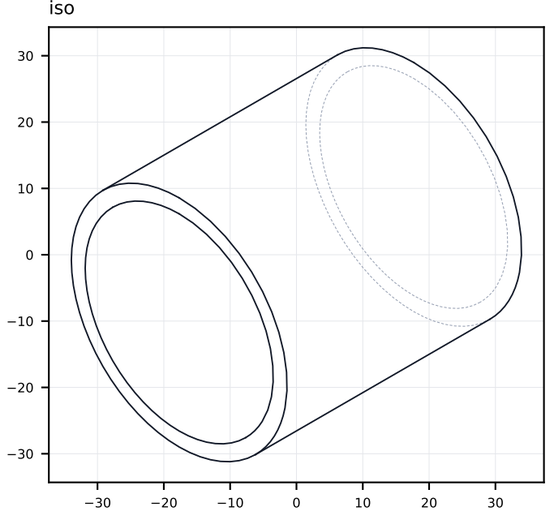
side (+x) (axes in mm from the view origin)



front (-y) (axes in mm from the view origin)

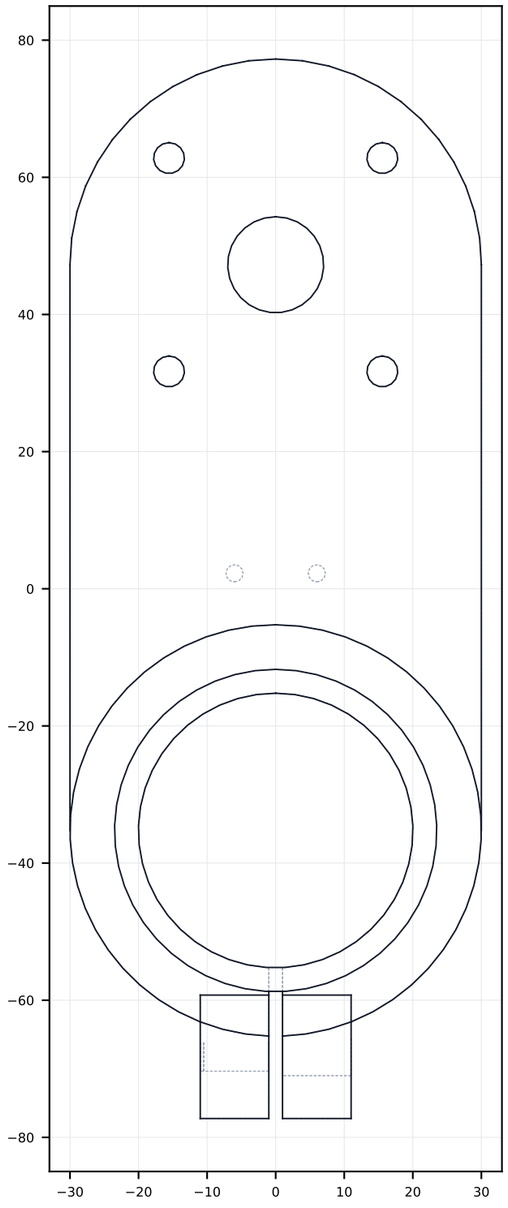


iso

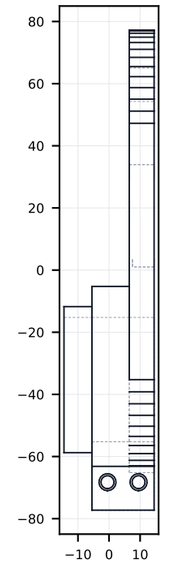


DRAWING hub_spacer
qty per robot: 1
material: 6061-T6 tube
process: CNC turn, faces parallel 0.02
finish:
mass: 53.9 g overall: 46.0 x 50.0 x 46.0 mm (x, y, z)
note: clamps the two inner races
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/hub_spacer.step
view origin = bounding-box centre = robot (0.00, 0.00, 0.00)

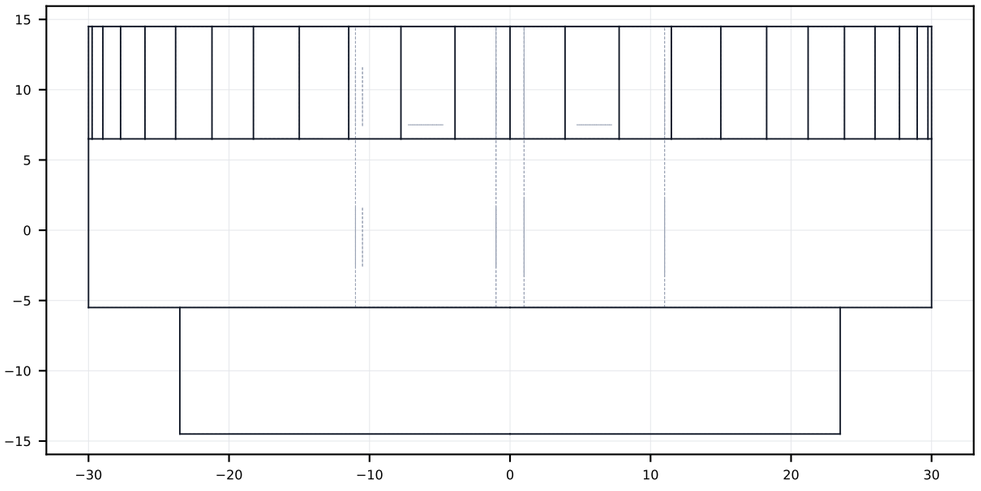
front (-y) (axes in mm from the view origin)



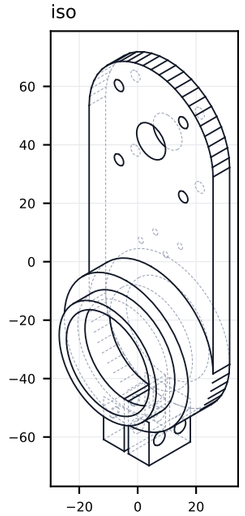
side (+x) (axes in mm from the view origin)



top (+z) (axes in mm from the view origin)



iso

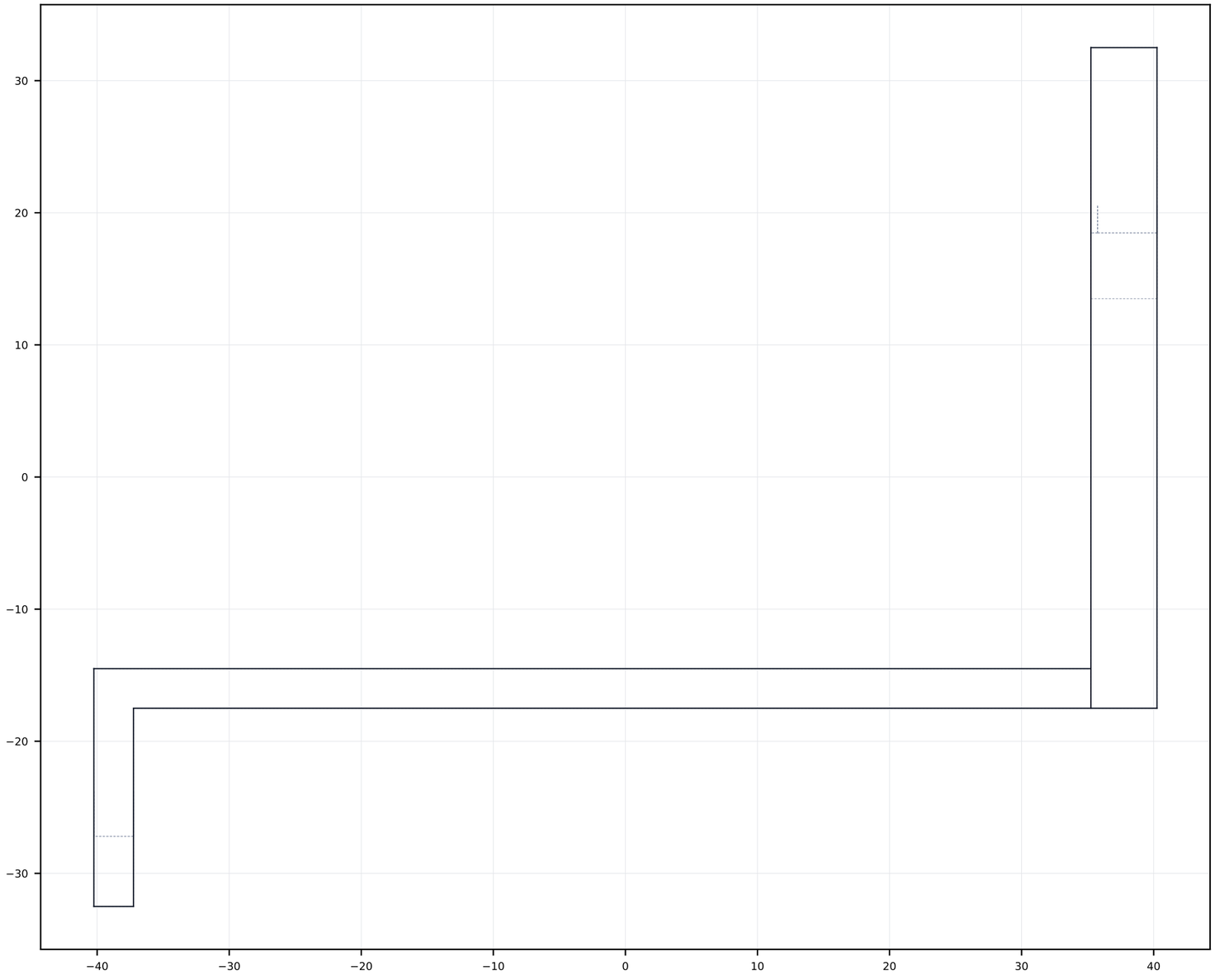


DRAWING level_bracket
qty per robot: 1
material: 6061-T6
process: CNC
finish: clear anodise
mass: 208.9 g overall: 60.0 x 29.0 x 154.5 mm (x, y, z)
note: split clamp locates the left inner race; arm carries the levelling motor
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/level_bracket.step
view origin = bounding-box centre = robot (0.00, 51.50, 35.25)

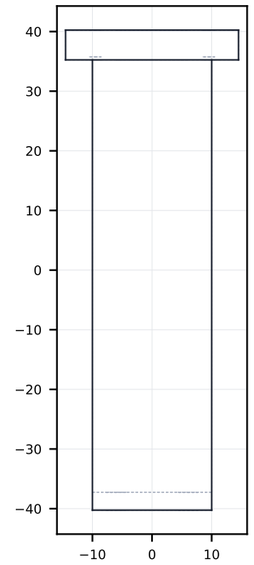
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
M3 tapped 7.0 deep (drill 2.5)	-6.00	17.50	2.25	-y
M3 tapped 7.0 deep (drill 2.5)	6.00	17.50	2.25	-y
M5 tapped 10.0 deep (drill 4.2)	11.00	-0.50	-68.25	-x
M5 tapped 10.0 deep (drill 4.2)	11.00	9.50	-68.25	-x
d4.5 through	-15.56	6.50	31.69	+y
d4.5 through	-15.56	6.50	62.81	+y
d4.5 through	15.56	6.50	31.69	+y
d4.5 through	15.56	6.50	62.81	+y
d5.5 through	11.00	-0.50	-68.25	-x
d5.5 through	11.00	9.50	-68.25	-x

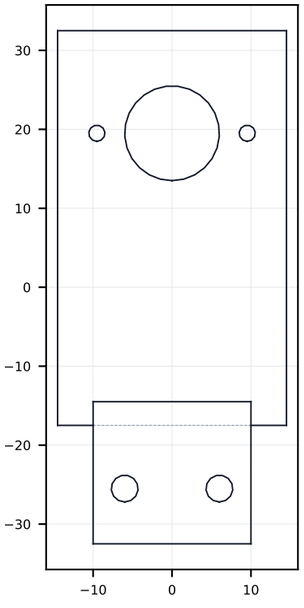
side (+x) (axes in mm from the view origin)



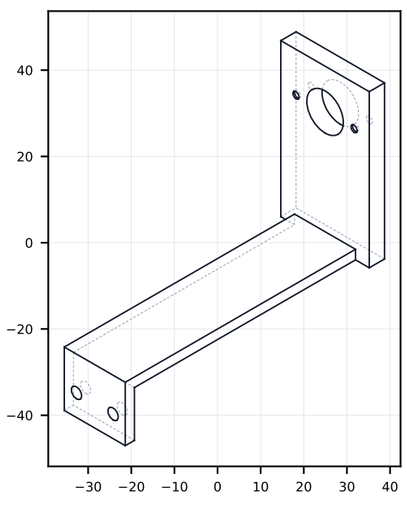
top (+z) (axes in mm from the view origin)



front (-y) (axes in mm from the view origin)



iso

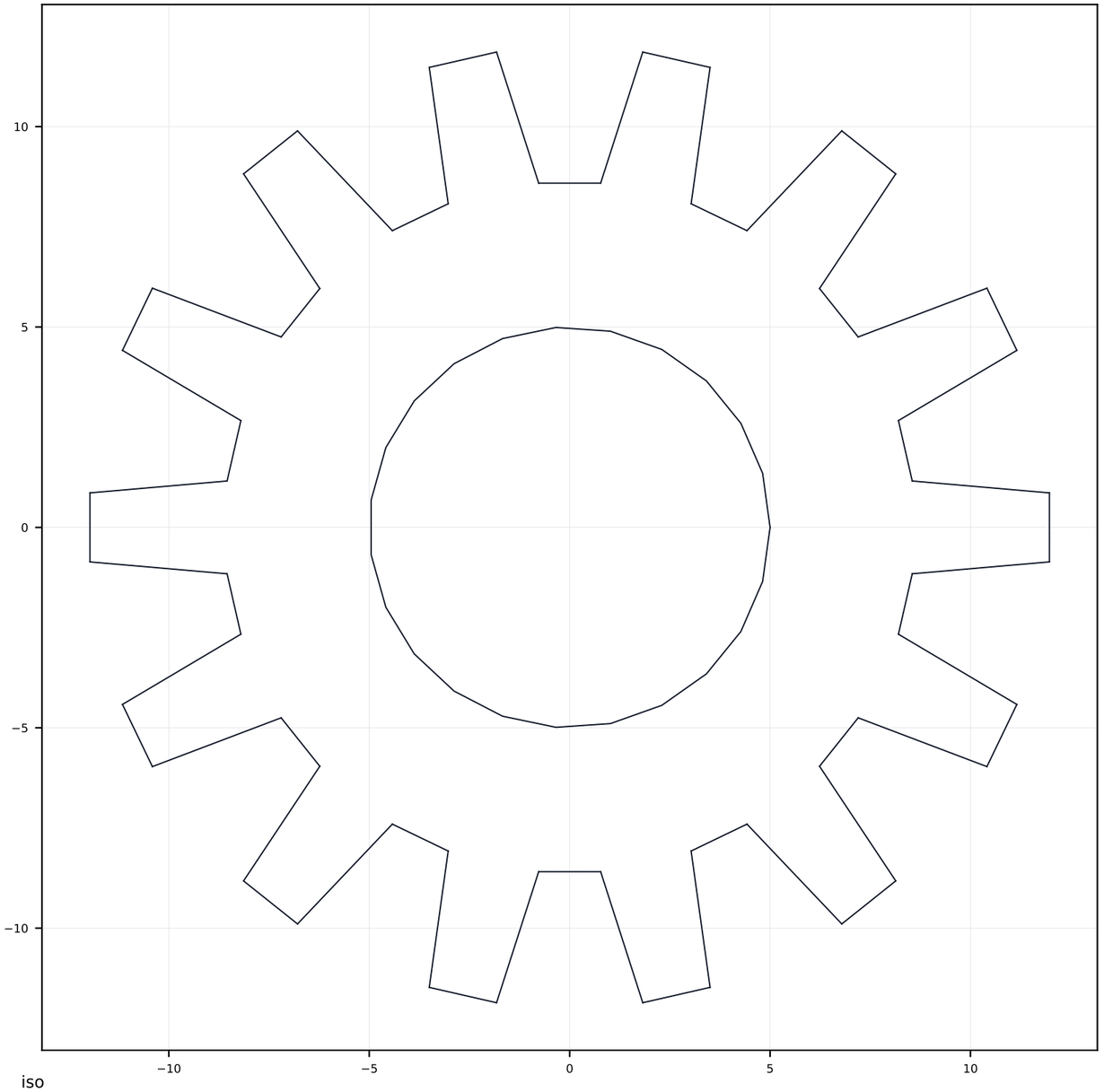


DRAWING level_enc_bracket
qty per robot: 1
material: 6061-T6
process: CNC from 6 mm plate (or 3 mm bent L + 5 mm bridge)
finish:
mass: 32.5 g overall: 29.0 x 80.5 x 65.0 mm (x, y, z)
note: M2.5 tapped through in the bridge; foot holes d3.4
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/level_enc_bracket.step
view origin = bounding-box centre = robot (0.00, 106.25, 63.00)

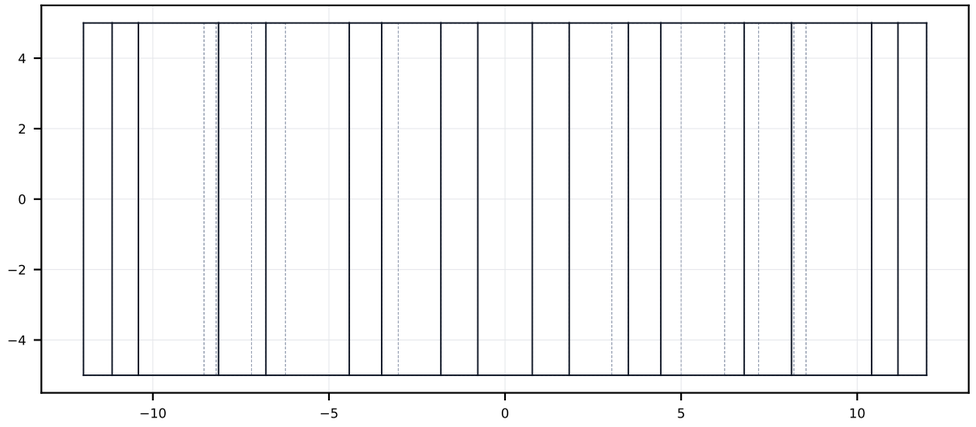
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
M2.5 tapped 5.0 deep (drill 2.05)	-9.53	41.75	19.50	-y
M2.5 tapped 5.0 deep (drill 2.05)	9.53	41.75	19.50	-y
d3.4 through	-6.00	-37.25	-25.50	-y
d3.4 through	6.00	-37.25	-25.50	-y

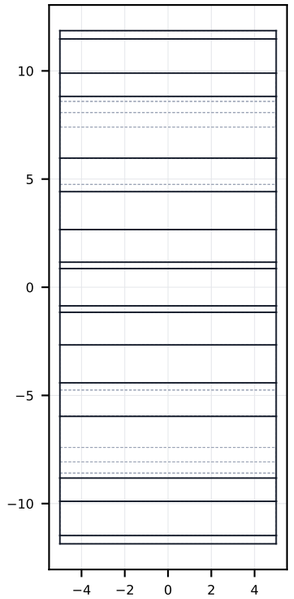
front (-y) (axes in mm from the view origin)



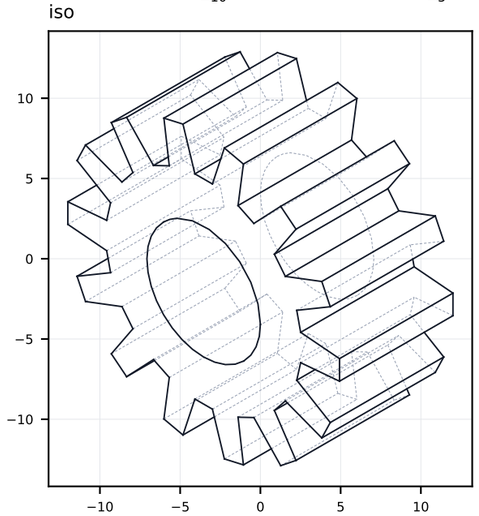
top (+z) (axes in mm from the view origin)



side (+x) (axes in mm from the view origin)

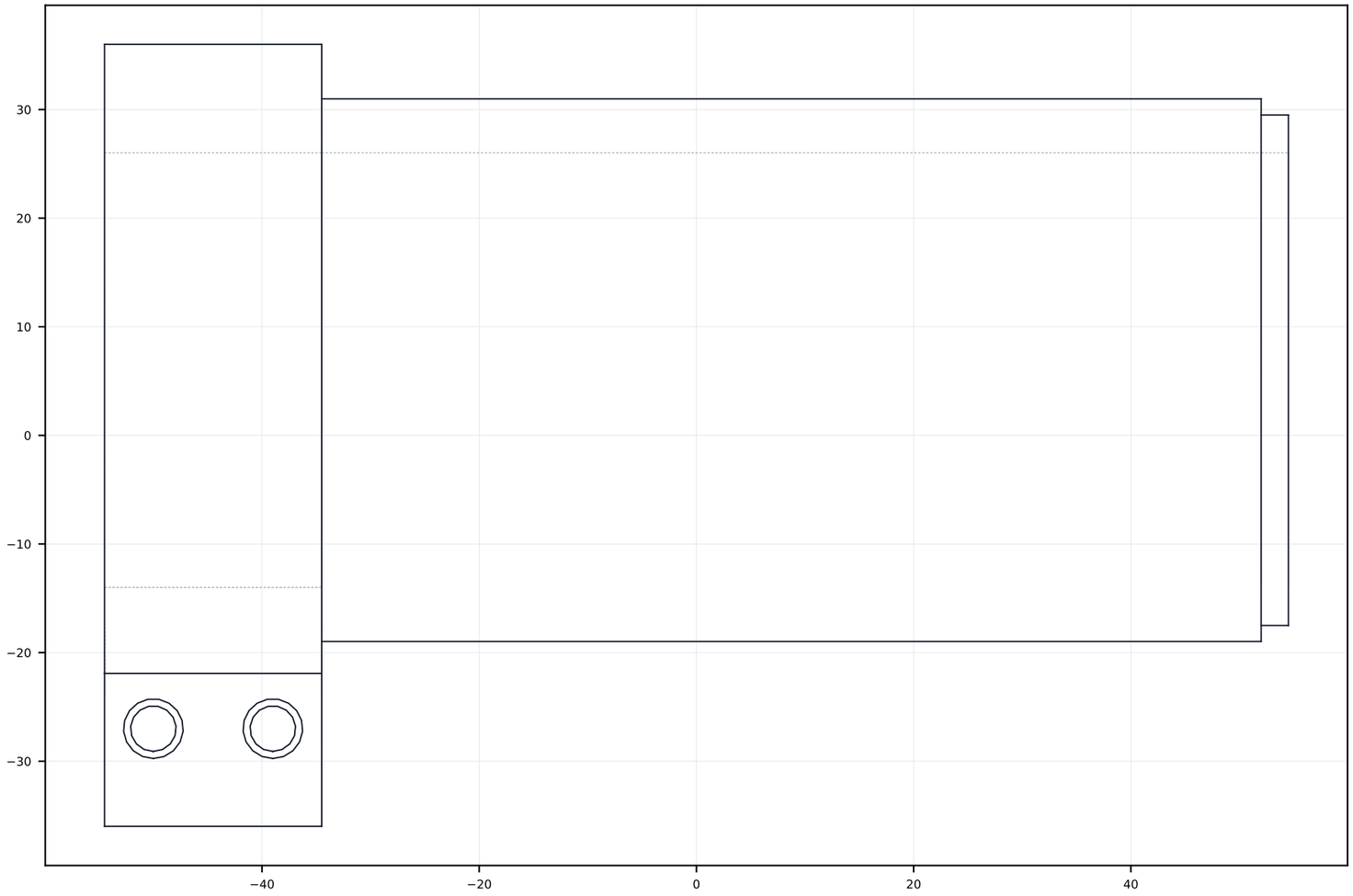


iso

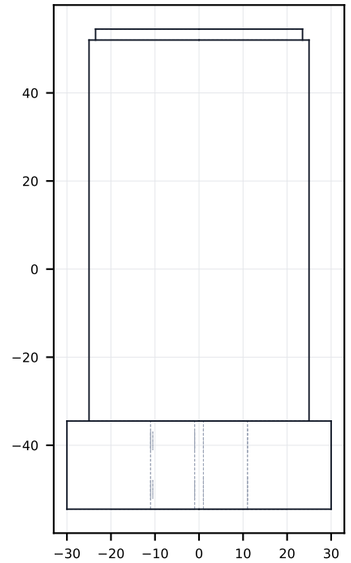


DRAWING level_pinion
qty per robot: 1
material: 42CrMo4 QT
process: hob + bore 10H7, 3 mm keyway, M4 set screw
finish:
mass: 19.6 g overall: 23.9 x 10.0 x 23.7 mm (x, y, z)
note: m1.5 z14 x=+0.3 (no undercut), face 10
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/level_pinion.step
view origin = bounding-box centre = robot (0.00, 41.00, 82.50)

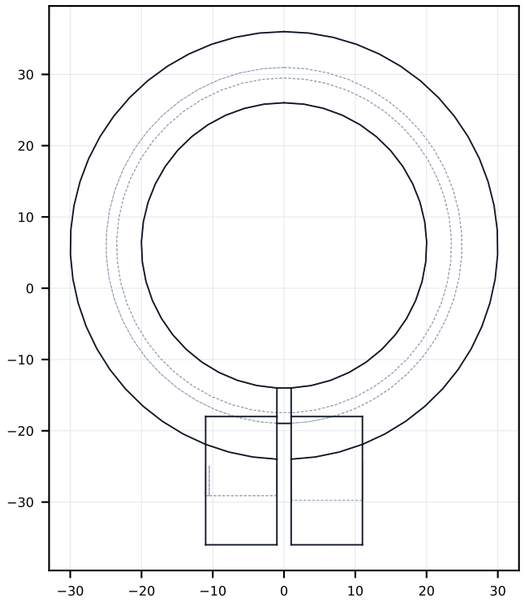
side (+x) (axes in mm from the view origin)



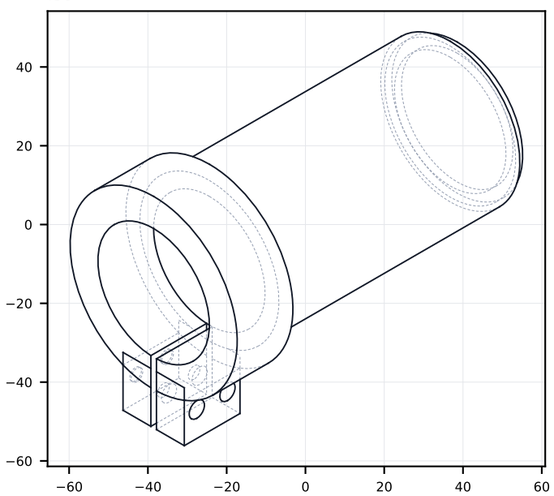
top (+z) (axes in mm from the view origin)



front (-y) (axes in mm from the view origin)



iso

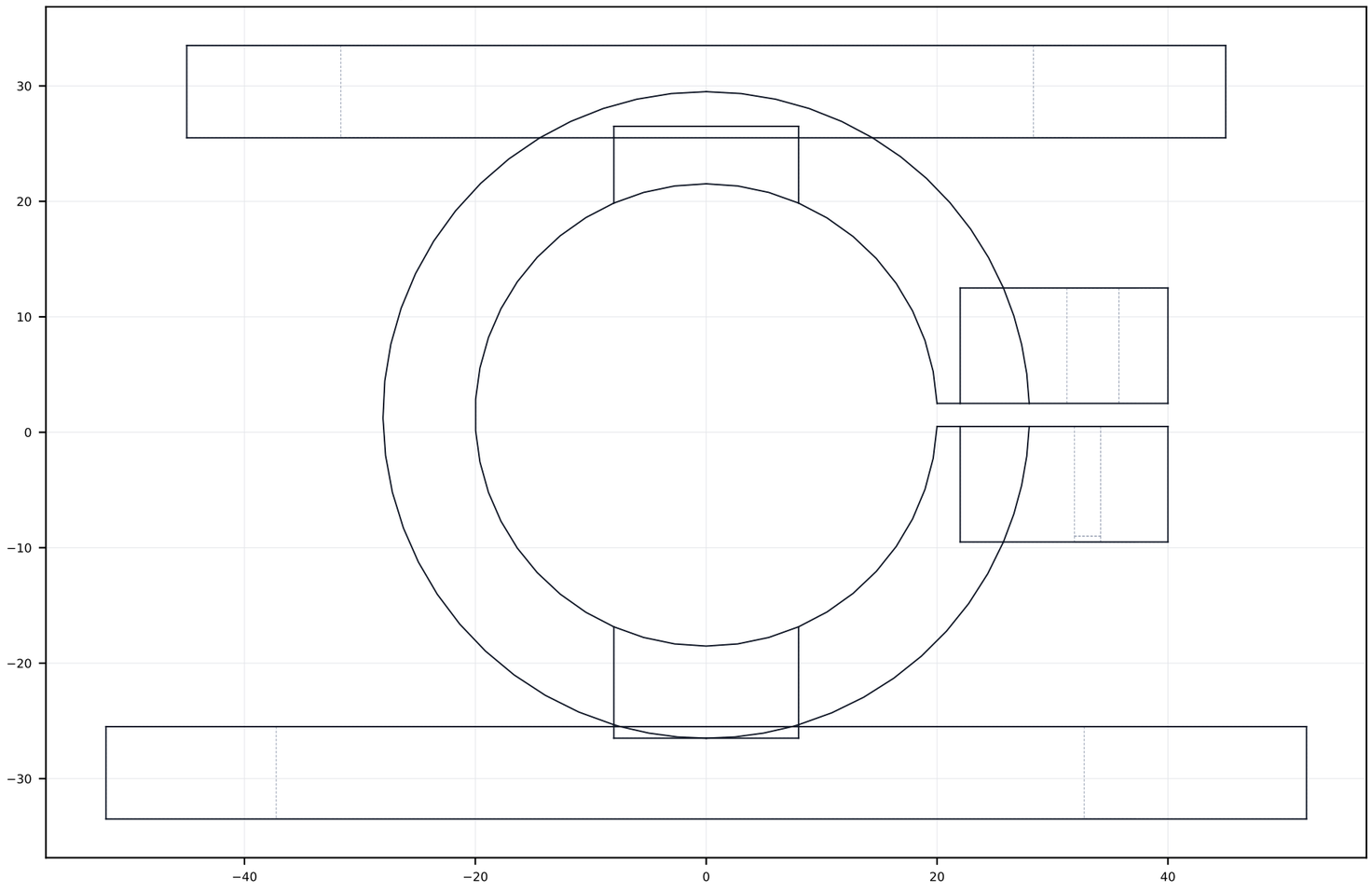


DRAWING slip_sleeve
qty per robot: 1
material: 6061-T6
process: CNC turn + slit
finish:
mass: 262.3 g overall: 60.0 x 109.0 x 72.0 mm (x, y, z)
note: locates the right inner race; slip-ring rotor collar set screws bear on it
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/slip_sleeve.step
view origin = bounding-box centre = robot (0.00, -91.50, -6.00)

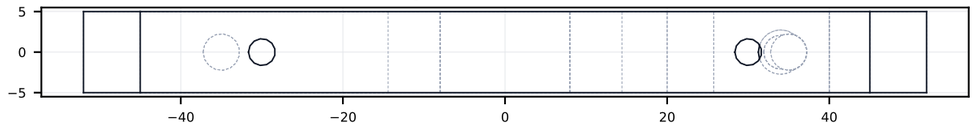
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
M5 tapped 10.0 deep (drill 4.2)	11.00	-50.00	-27.00	-x
M5 tapped 10.0 deep (drill 4.2)	11.00	-39.00	-27.00	-x
d5.5 through	11.00	-50.00	-27.00	-x
d5.5 through	11.00	-39.00	-27.00	-x

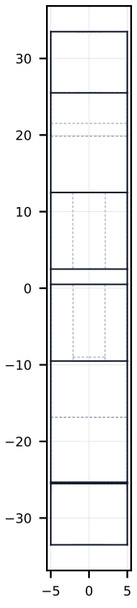
front (-y) (axes in mm from the view origin)



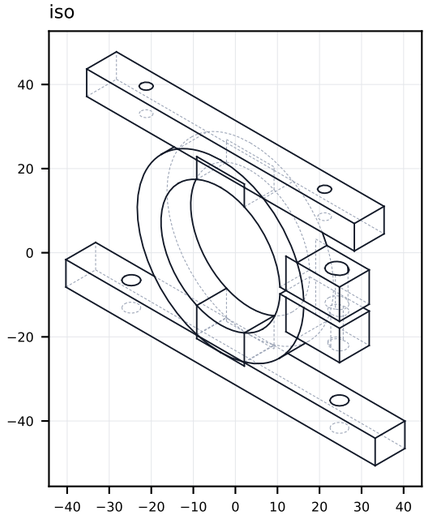
top (+z) (axes in mm from the view origin)



side (+x) (axes in mm from the view origin)



iso

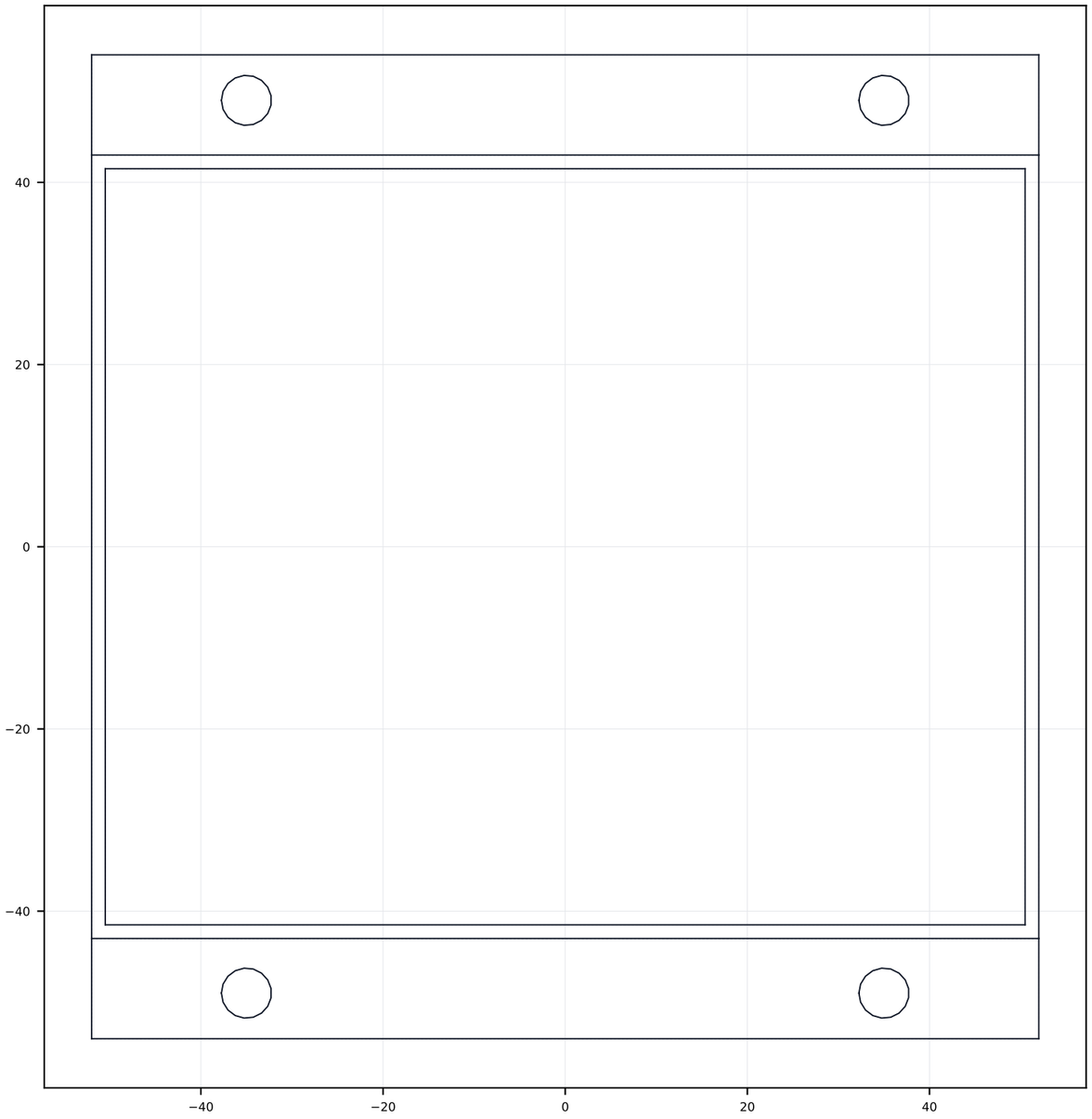


DRAWING bay_clamp
qty per robot: 4 (R-hand part = this one mirrored about XZ)
material: 6061-T6
process: CNC
finish: clear anodise
mass: 76.3 g overall: 104.0 x 10.0 x 67.0 mm (x, y, z)
note: bore 40 H7 slit clamp on the CFRP axle
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/bay_clamp.step
view origin = bounding-box centre = robot (0.00, 173.00, -1.50)

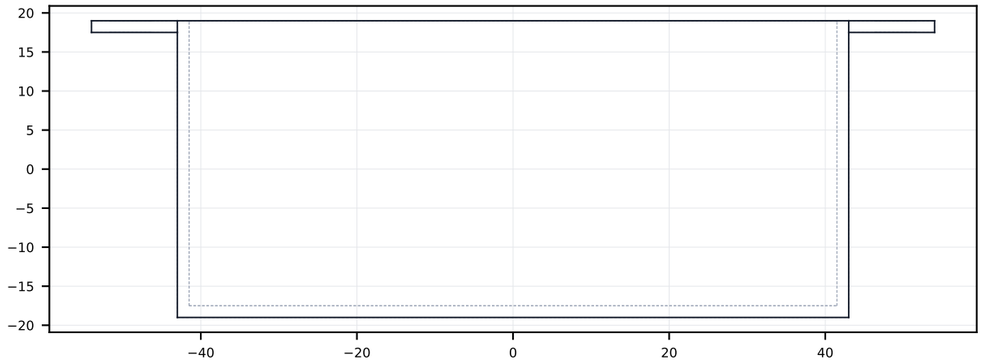
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
M4 tapped 9.4 deep (drill 3.3)	-30.00	0.00	36.50	-z
M4 tapped 9.4 deep (drill 3.3)	30.00	0.00	36.50	-z
M5 tapped 10.0 deep (drill 4.2)	34.00	0.00	12.50	-z
d4.5 through	-35.00	0.00	-25.50	-z
d4.5 through	35.00	0.00	-25.50	-z
d5.5 through	34.00	0.00	12.50	-z

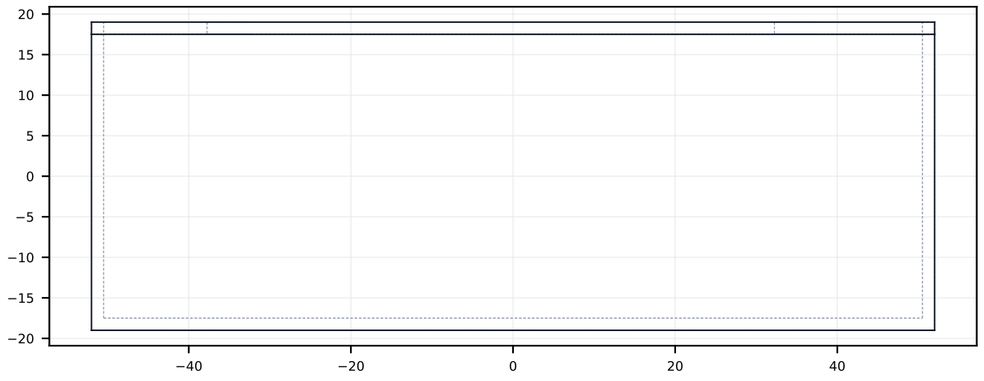
top (+z) (axes in mm from the view origin)



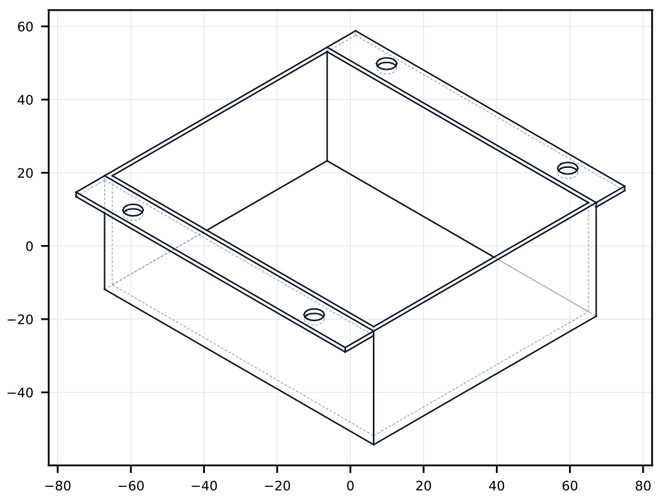
side (+x) (axes in mm from the view origin)



front (-y) (axes in mm from the view origin)



iso

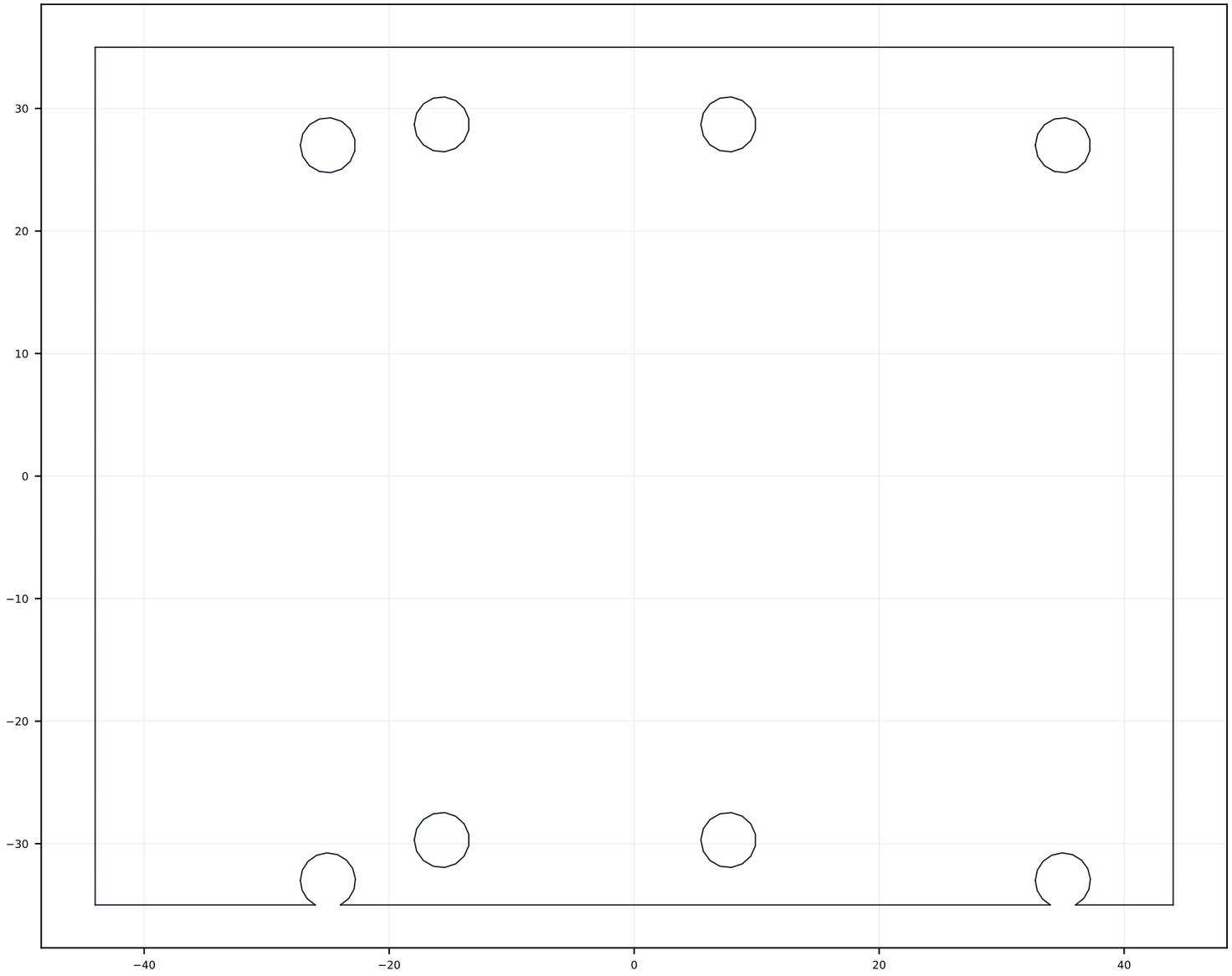


DRAWING bay_tray
qty per robot: 2 (R-hand part = this one mirrored about XZ)
material: 5052-H32 sheet 1.5 mm
process: laser cut + bend, PEM nuts
finish: black powder coat
mass: 100.4 g overall: 104.0 x 108.0 x 38.0 mm (x, y, z)
note: lid = the clamp tabs; cable entry through the axle slots
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/bay_tray.step
view origin = bounding-box centre = robot (0.00, 222.00, -54.00)

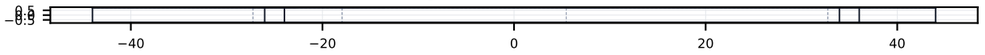
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
d5.5 for M4 PEM nut	-35.00	-49.00	27.00	-z
d5.5 for M4 PEM nut	-35.00	49.00	27.00	-z
d5.5 for M4 PEM nut	35.00	-49.00	27.00	-z
d5.5 for M4 PEM nut	35.00	49.00	27.00	-z

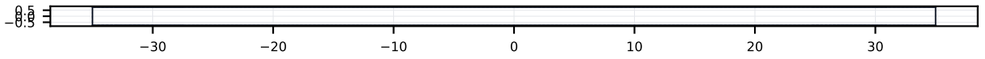
top (+z) (axes in mm from the view origin)



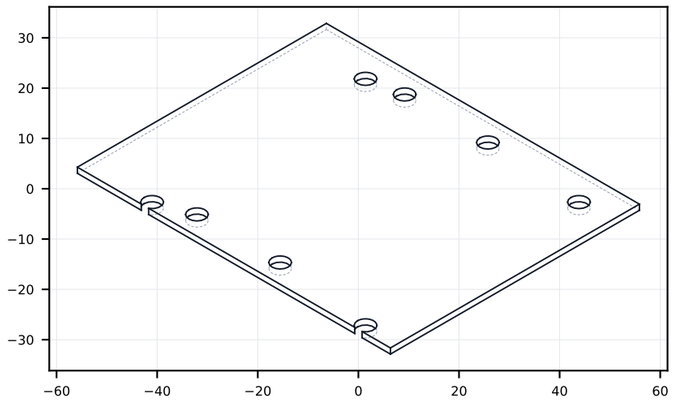
front (-y) (axes in mm from the view origin)



side (+x) (axes in mm from the view origin)



iso

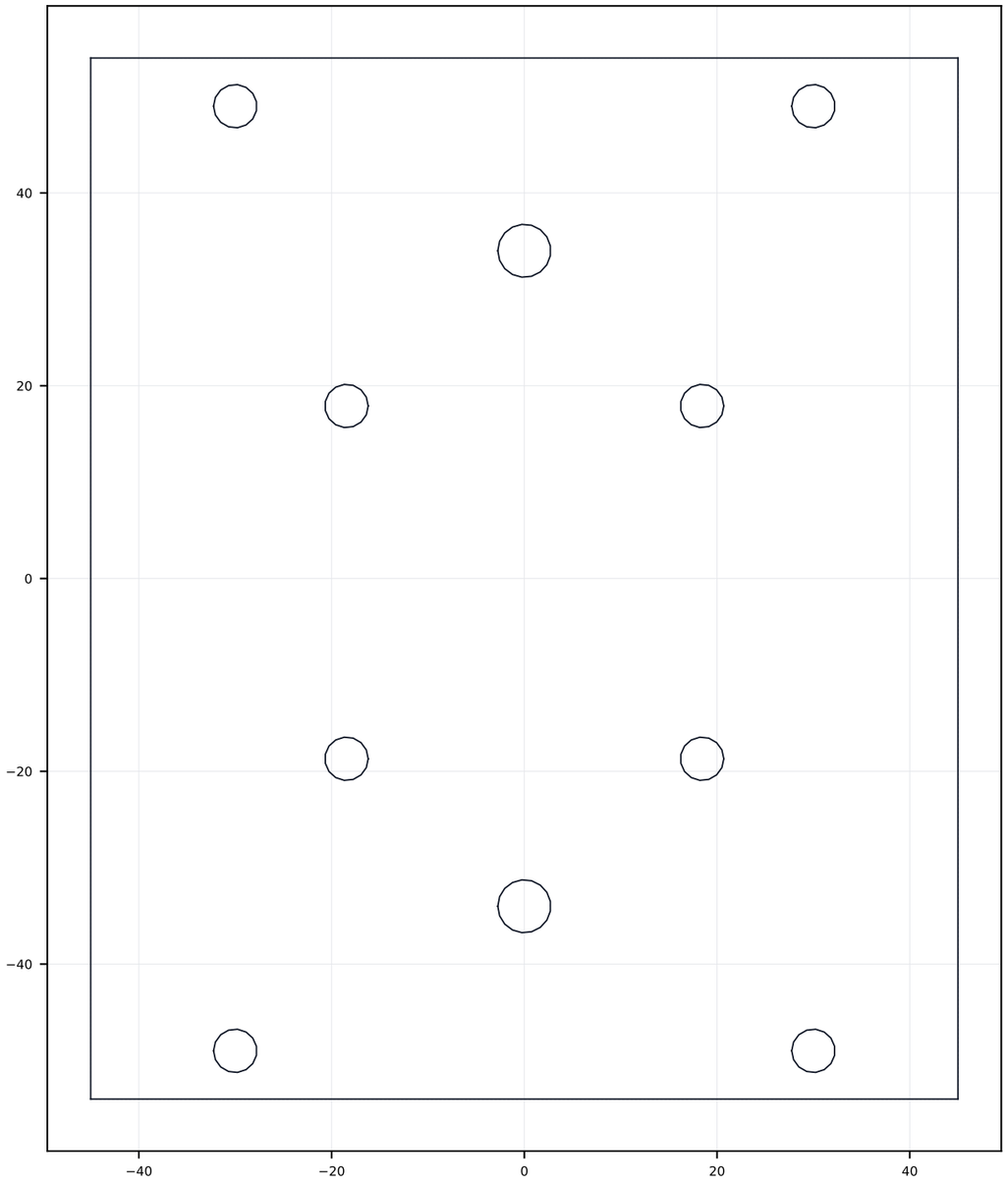


DRAWING bay_mezzanine
qty per robot: 1
material: 5052-H32 1.5 mm
process: laser cut, PEM nuts (M3)
finish:
mass: 24.4 g overall: 88.0 x 70.0 x 1.5 mm (x, y, z)
note: cable tie slots along the edges; USB hub on PEM nuts
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/bay_mezzanine.step
view origin = bounding-box centre = robot (-5.00, -218.00, -54.75)

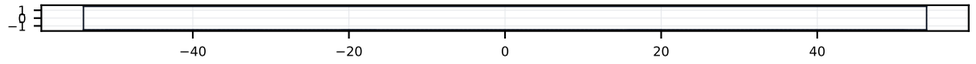
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
d4.5 for M3 PEM nut	-15.70	-29.70	7.35	-z
d4.5 for M3 PEM nut	-15.70	28.70	7.35	-z
d4.5 for M3 PEM nut	7.70	-29.70	7.35	-z
d4.5 for M3 PEM nut	7.70	28.70	7.35	-z
d4.5 through	-25.00	-33.00	0.75	-z
d4.5 through	-25.00	27.00	0.75	-z
d4.5 through	35.00	-33.00	0.75	-z
d4.5 through	35.00	27.00	0.75	-z

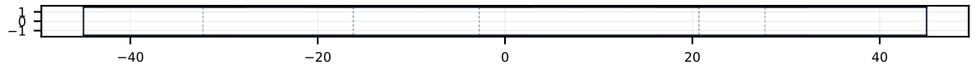
top (+z) (axes in mm from the view origin)



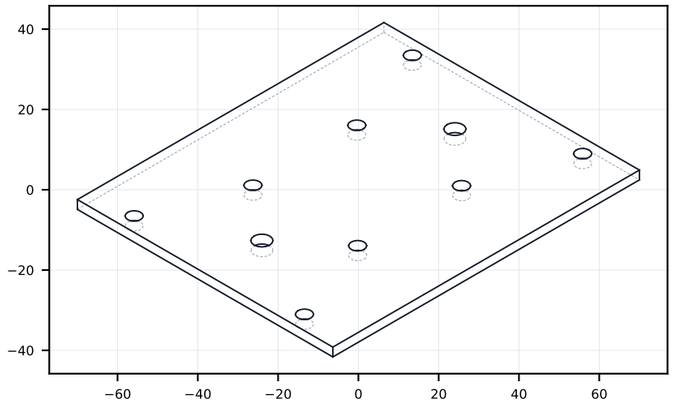
side (+x) (axes in mm from the view origin)



front (-y) (axes in mm from the view origin)



iso

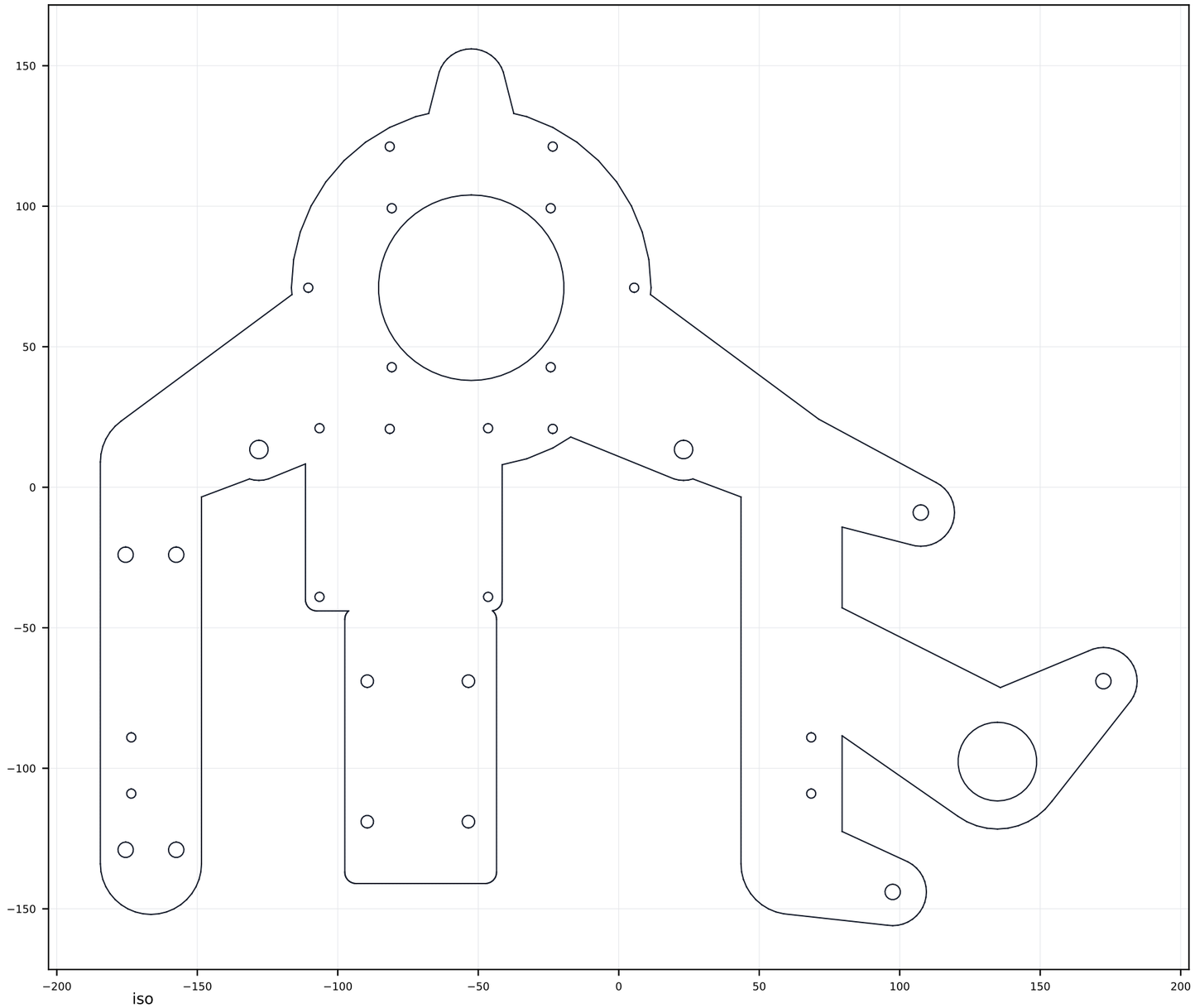


DRAWING gnss_plate
qty per robot: 1
material: 5052-H32 3 mm
process: laser cut, PEM nuts
finish:
mass: 77.3 g overall: 90.0 x 108.0 x 3.0 mm (x, y, z)
note: 120 mm ground plane for the antenna
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/gnss_plate.step
view origin = bounding-box centre = robot (0.00, 222.00, 33.50)

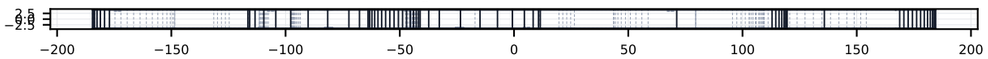
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
d4.5 for M3 PEM nut	-18.45	-18.70	-11.00	+z
d4.5 for M3 PEM nut	-18.45	17.90	-11.00	+z
d4.5 for M3 PEM nut	18.45	-18.70	-11.00	+z
d4.5 for M3 PEM nut	18.45	17.90	-11.00	+z
d4.5 through	-30.00	-49.00	1.50	-z
d4.5 through	-30.00	49.00	1.50	-z
d4.5 through	30.00	-49.00	1.50	-z
d4.5 through	30.00	49.00	1.50	-z
d5.5 for M4 PEM nut	0.00	-34.00	7.50	-z
d5.5 for M4 PEM nut	0.00	34.00	7.50	-z

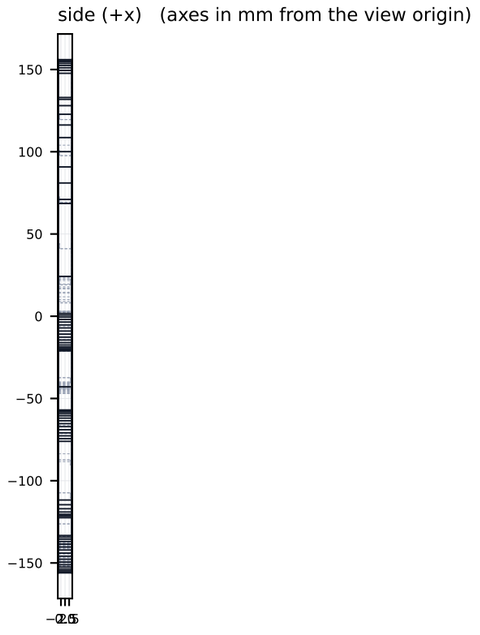
front (-y) (axes in mm from the view origin)



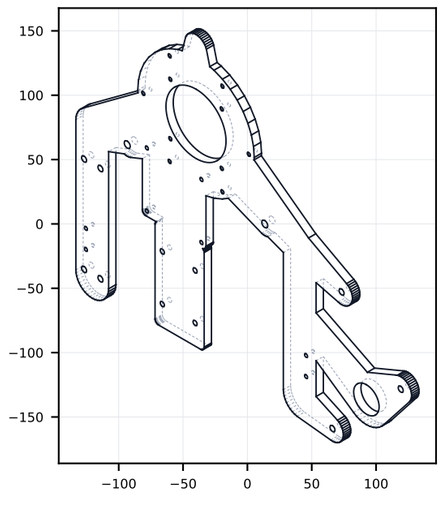
top (+z) (axes in mm from the view origin)



side (+x) (axes in mm from the view origin)



iso



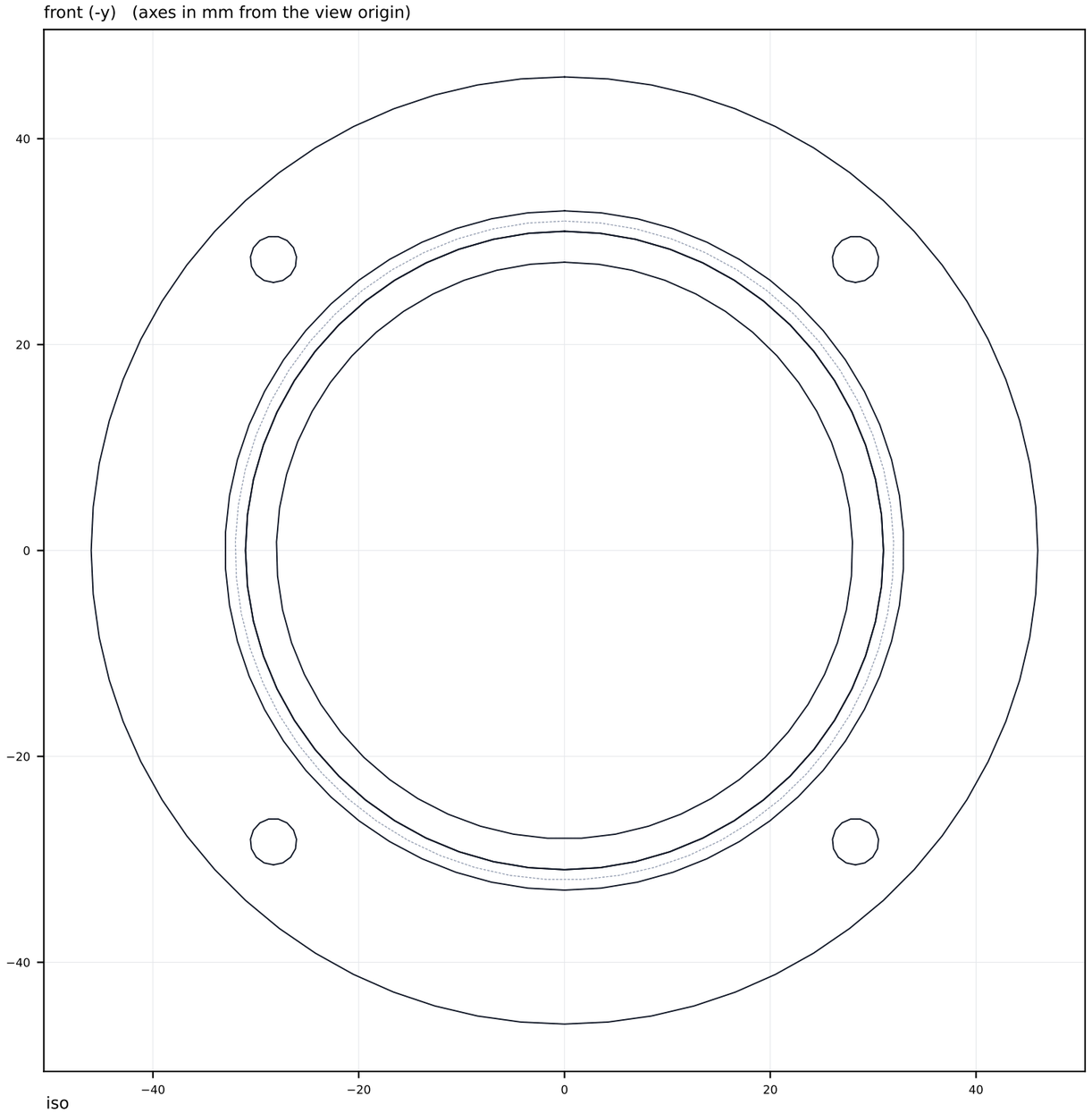
DRAWING hanger
qty per robot: 2 (R-hand part = this one mirrored about XZ)
material: 6061-T6 plate 8 mm
process: waterjet + CNC (bores, tapped holes)
finish: clear anodise
mass: 911.8 g overall: 369.0 x 8.0 x 312.0 mm (x, y, z)
note:
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/hanger.step
view origin = bounding-box centre = robot (52.50, 31.00, -71.00)

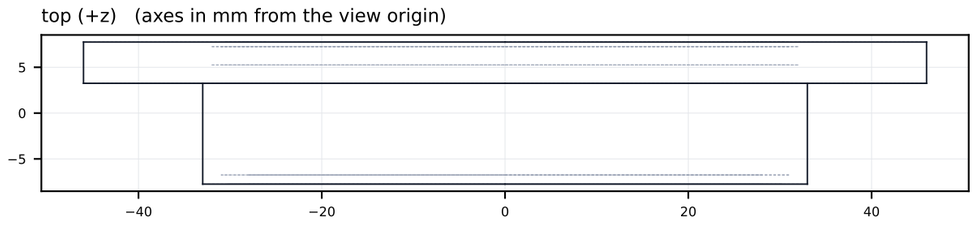
HOLE TABLE (view coords of the screw seat, mm)

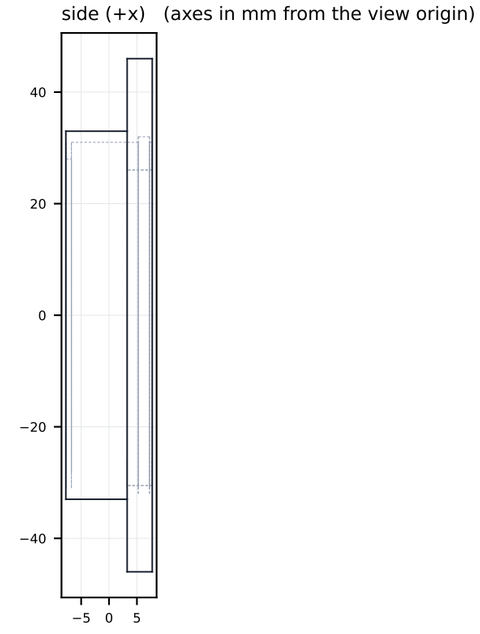
feature	x	y	z	axis
M4 tapped 8.0 deep (drill 3.3)	-173.50	-10.00	-109.00	+y
M4 tapped 8.0 deep (drill 3.3)	-173.50	-10.00	-89.00	+y
M4 tapped 8.0 deep (drill 3.3)	-110.50	14.00	71.00	-y
M4 tapped 8.0 deep (drill 3.3)	-106.50	-8.00	-39.00	+y
M4 tapped 8.0 deep (drill 3.3)	-106.50	-8.00	21.00	+y
M4 tapped 8.0 deep (drill 3.3)	-81.50	14.00	20.77	-y
M4 tapped 8.0 deep (drill 3.3)	-81.50	14.00	121.23	-y
M4 tapped 8.0 deep (drill 3.3)	-80.78	8.50	42.72	-y
M4 tapped 8.0 deep (drill 3.3)	-80.78	8.50	99.28	-y
M4 tapped 8.0 deep (drill 3.3)	-46.50	-8.00	-39.00	+y
M4 tapped 8.0 deep (drill 3.3)	-46.50	-8.00	21.00	+y
M4 tapped 8.0 deep (drill 3.3)	-24.22	8.50	42.72	-y
M4 tapped 8.0 deep (drill 3.3)	-24.22	8.50	99.28	-y
M4 tapped 8.0 deep (drill 3.3)	-23.50	14.00	20.77	-y
M4 tapped 8.0 deep (drill 3.3)	-23.50	14.00	121.23	-y
M4 tapped 8.0 deep (drill 3.3)	5.50	14.00	71.00	-y
M4 tapped 8.0 deep (drill 3.3)	68.50	-10.00	-109.00	+y
M4 tapped 8.0 deep (drill 3.3)	68.50	-10.00	-89.00	+y
d4.5 through	-89.50	4.00	-119.00	-y
d4.5 through	-89.50	4.00	-69.00	-y
d4.5 through	-53.50	4.00	-119.00	-y
d4.5 through	-53.50	4.00	-69.00	-y
d5.5 through	-175.50	-4.00	-129.00	+y
d5.5 through	-175.50	-4.00	-24.00	+y
d5.5 through	-157.50	-4.00	-129.00	+y
d5.5 through	-157.50	-4.00	-24.00	+y

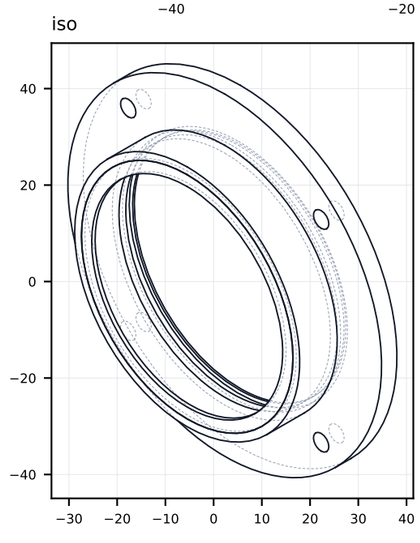
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
d5.5 through	97.50	-4.00	-144.00	+y
d5.5 through	107.50	-4.00	-9.00	+y
d5.5 through	172.50	-4.00	-69.00	+y
d6.6 through	-128.07	-4.00	13.43	+y
d6.6 through	23.07	-4.00	13.43	+y







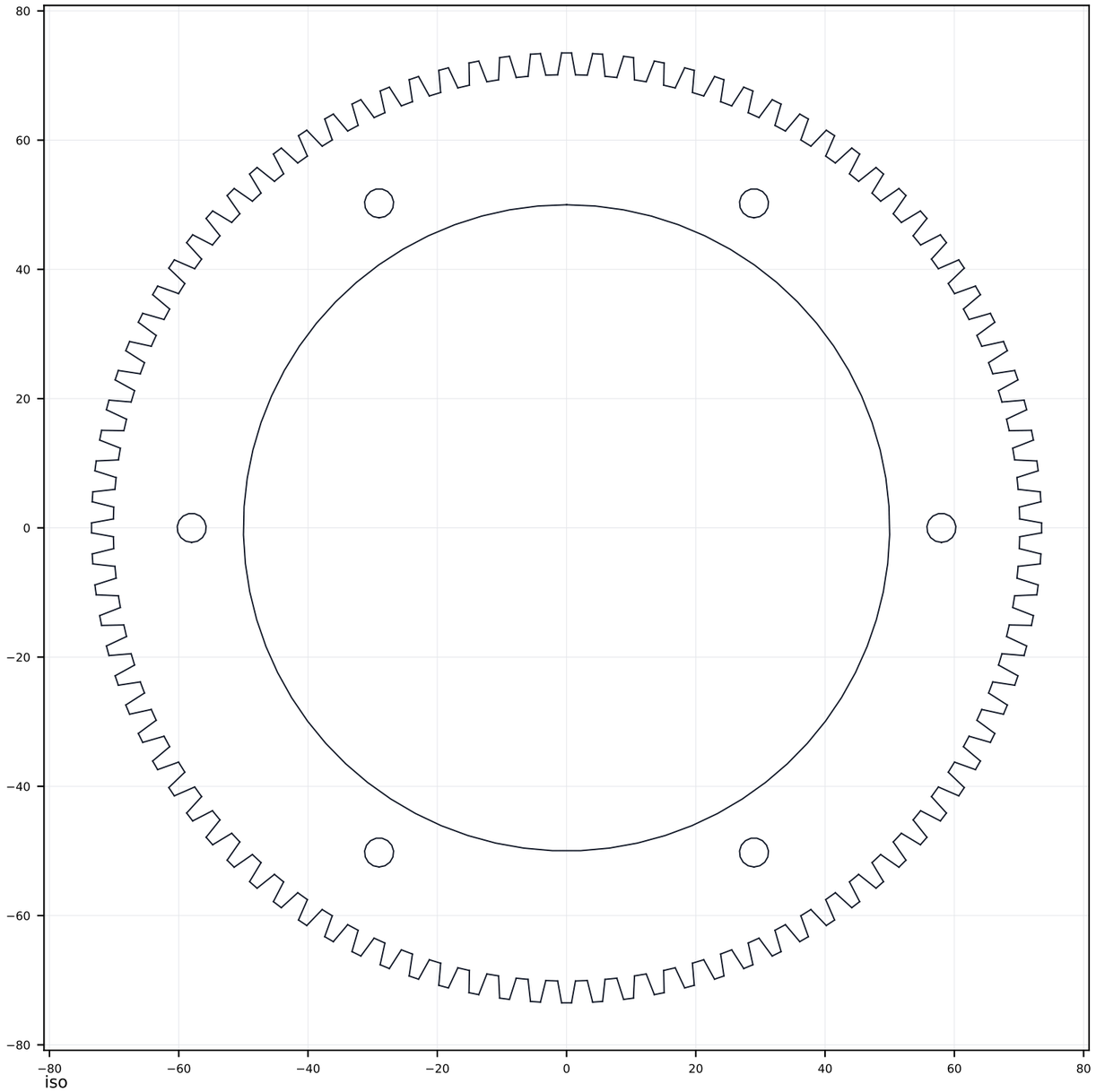


DRAWING hub_cartridge
qty per robot: 2 (R-hand part = this one mirrored about XZ)
material: 6061-T6
process: CNC turn
finish:
mass: 55.6 g overall: 92.0 x 15.5 x 92.0 mm (x, y, z)
note: spigot 66 h7 in hanger, bore 62 H7, DIN 472 groove
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/hub_cartridge.step
view origin = bounding-box centre = robot (0.00, 31.75, 0.00)

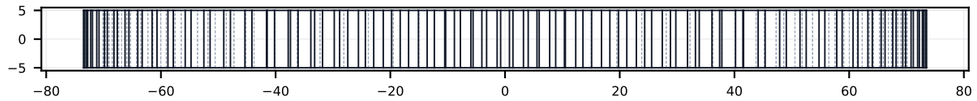
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
d4.5 through	-28.28	7.75	-28.28	-y
d4.5 through	-28.28	7.75	28.28	-y
d4.5 through	28.28	7.75	-28.28	-y
d4.5 through	28.28	7.75	28.28	-y

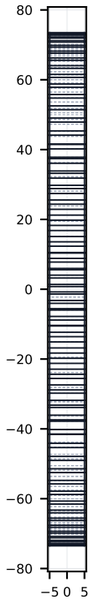
front (-y) (axes in mm from the view origin)



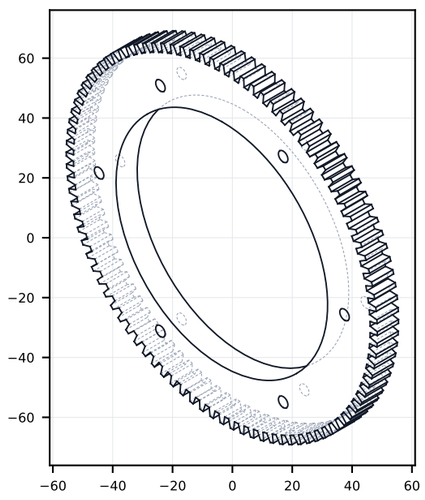
top (+z) (axes in mm from the view origin)



side (+x) (axes in mm from the view origin)



iso

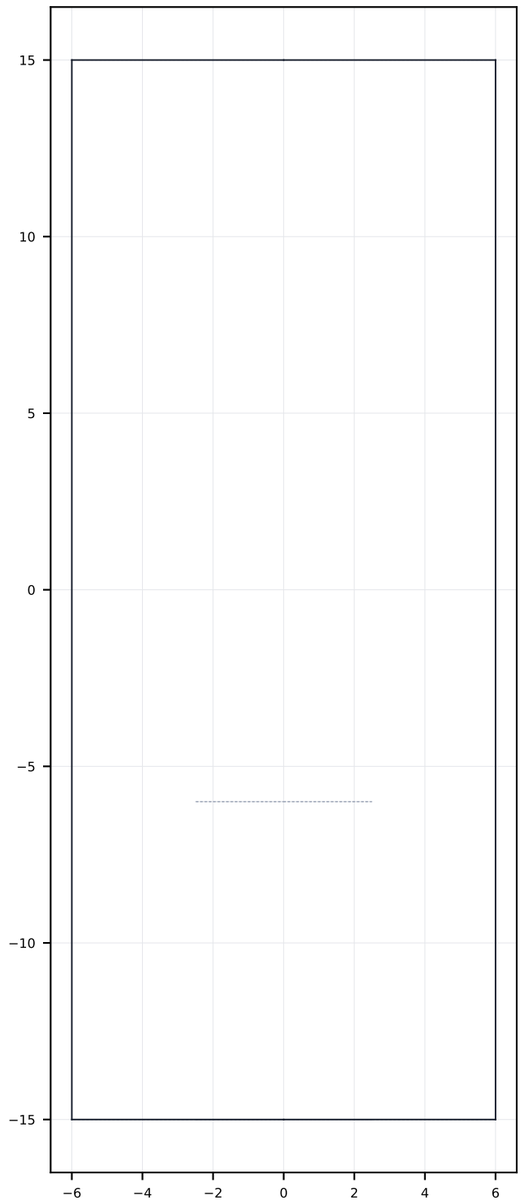


DRAWING level_gear
qty per robot: 1
material: 7075-T6
process: hob / wire EDM
finish: hard anodise
mass: 230.2 g overall: 147.0 x 10.0 x 147.0 mm (x, y, z)
note: m1.5 z96, face 10
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/level_gear.step
view origin = bounding-box centre = robot (0.00, 40.00, 0.00)

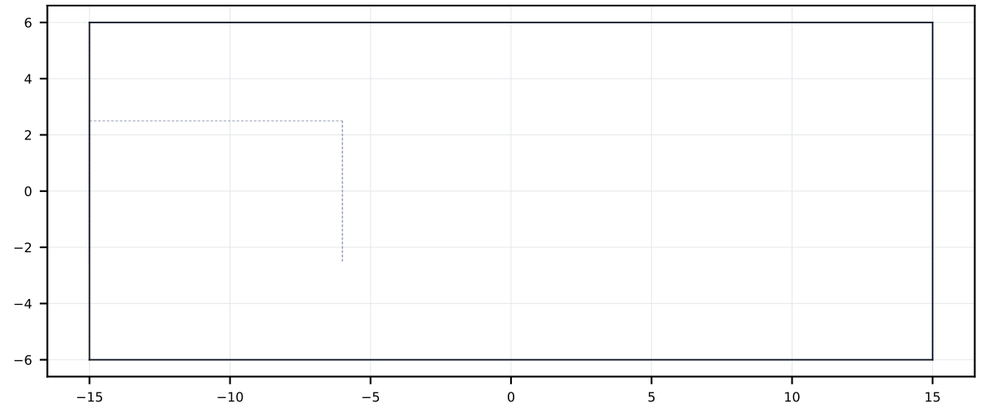
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
d4.5 through	-58.00	5.00	0.00	-y
d4.5 through	-29.00	5.00	-50.23	-y
d4.5 through	-29.00	5.00	50.23	-y
d4.5 through	29.00	5.00	-50.23	-y
d4.5 through	29.00	5.00	50.23	-y
d4.5 through	58.00	5.00	0.00	-y

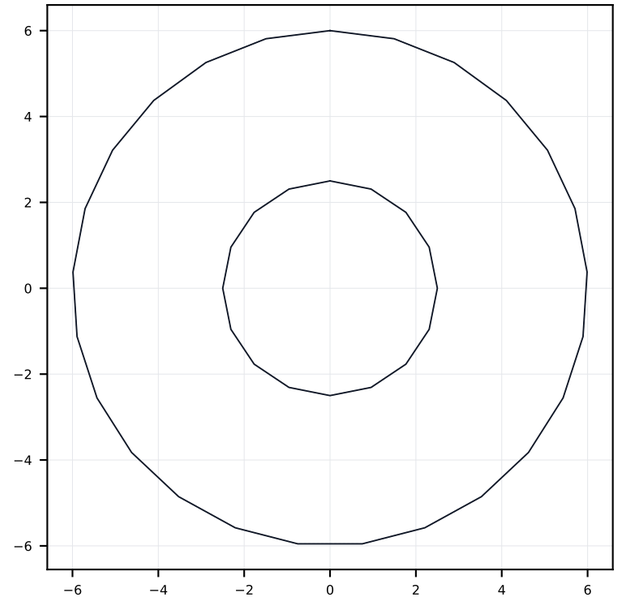
top (+z) (axes in mm from the view origin)



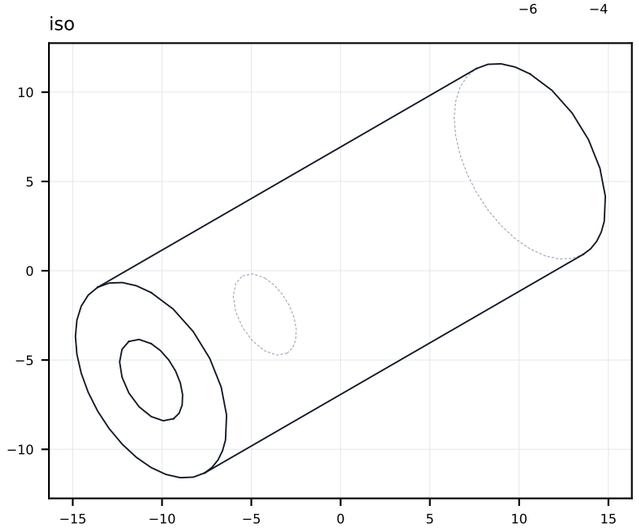
side (+x) (axes in mm from the view origin)



front (-y) (axes in mm from the view origin)



iso

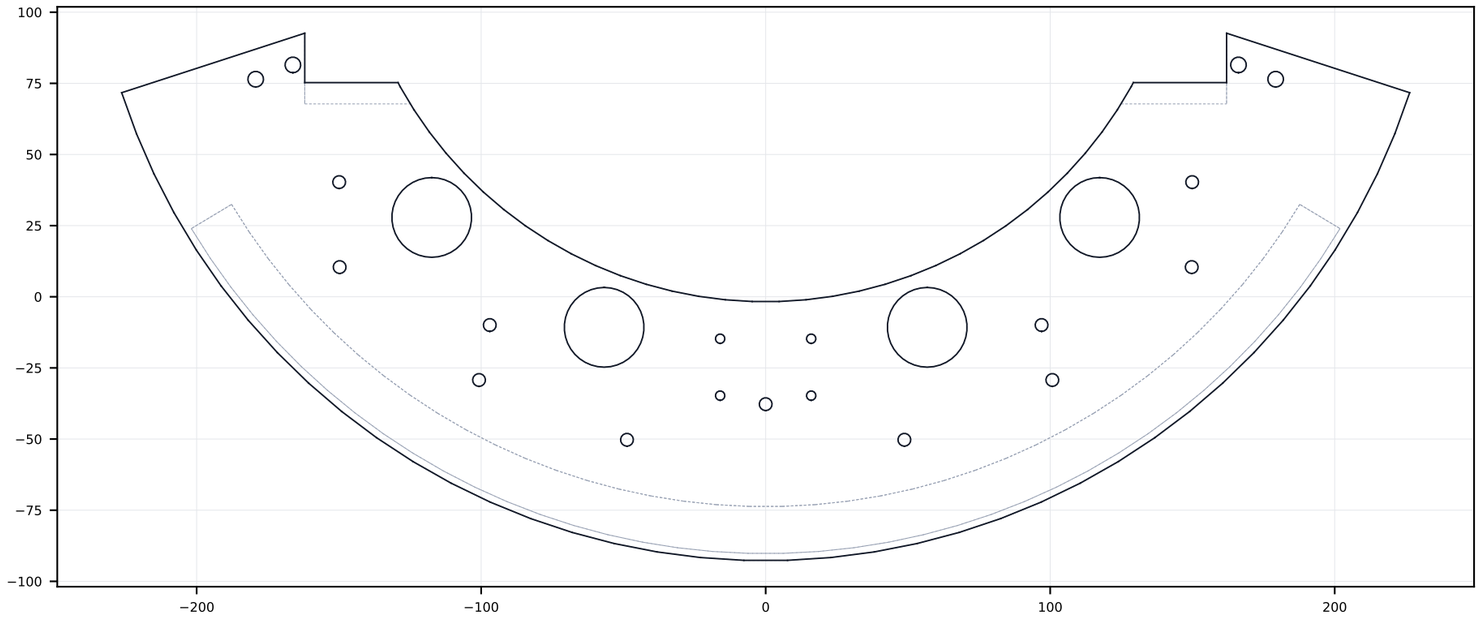


DRAWING pitch_striker
qty per robot: 2 (R-hand part = this one mirrored about XZ)
material: C45 (1045) steel d12, zinc plated
process: turned 30 long, M6 tapped 14 deep
finish:
mass: 25.2 g overall: 12.0 x 30.0 x 12.0 mm (x, y, z)
note: meets the level-bracket arm flank at 102 deg yoke-to-spine pitch
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/pitch_striker.step
view origin = bounding-box centre = robot (75.57, 50.00, -57.57)

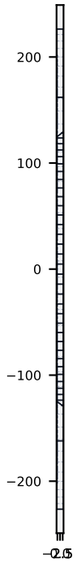
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
M6 tapped 9.0 deep (drill 5.0)	0.00	-23.00	-0.00	+y

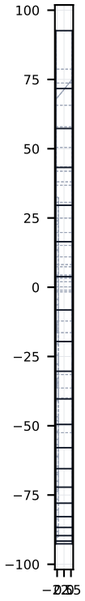
side (+x) (axes in mm from the view origin)



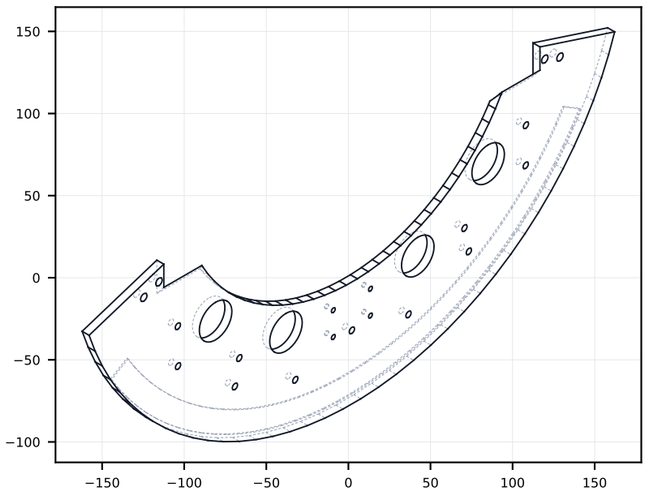
top (+z) (axes in mm from the view origin)



front (-y) (axes in mm from the view origin)



iso

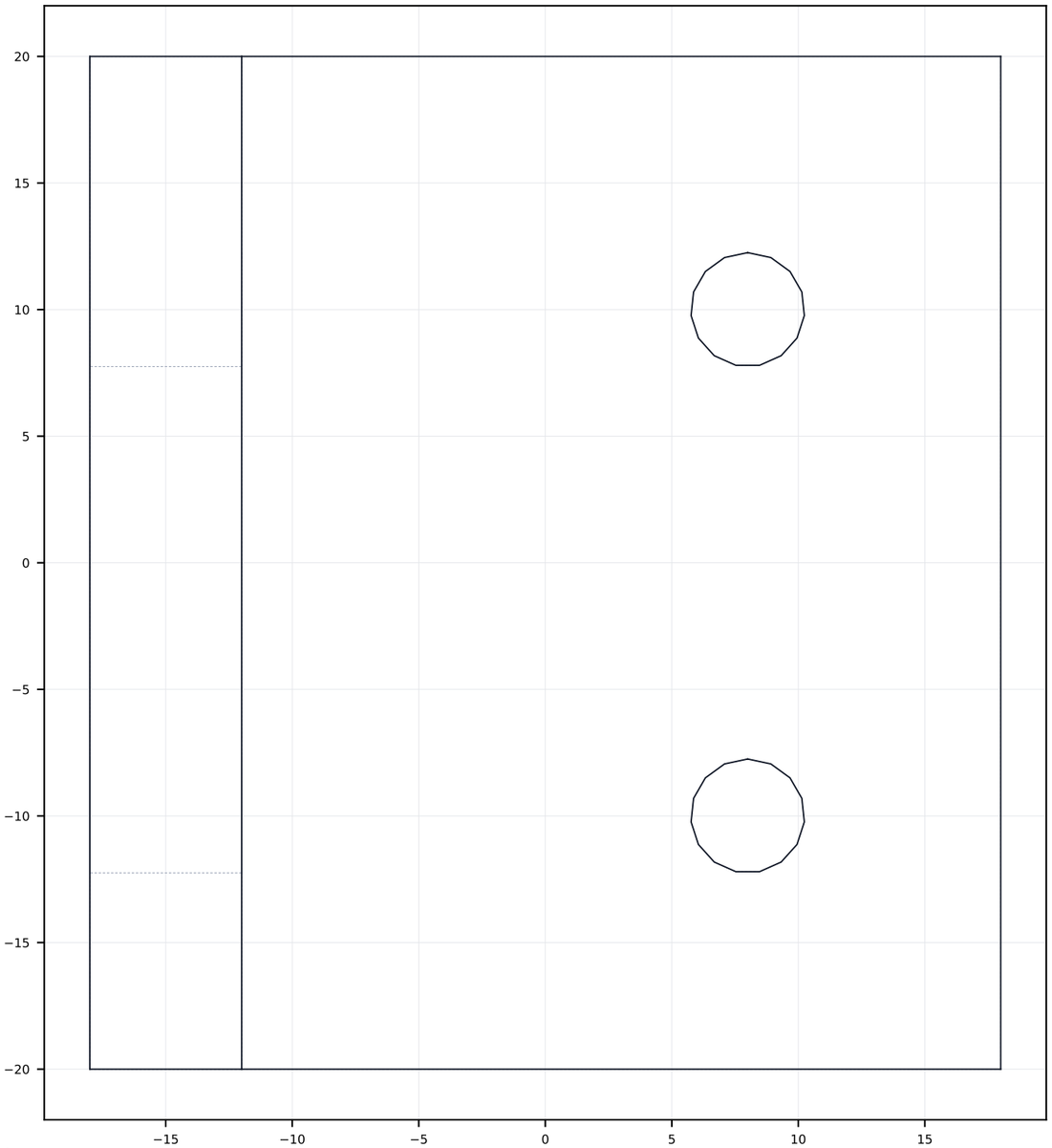


DRAWING arc_plate
qty per robot: 2 (R-hand part = this one mirrored about XZ)
material: 6061-T6 plate 6 mm
process: waterjet + CNC
finish:
mass: 623.8 g overall: 6.0 x 452.7 x 185.2 mm (x, y, z)
note:
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/arc_plate.step
view origin = bounding-box centre = robot (92.00, 0.00, -145.26)

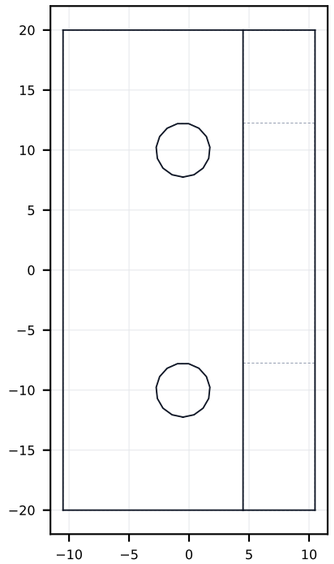
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
M4 tapped 6.0 deep (drill 3.3)	9.00	-16.00	-34.74	-x
M4 tapped 6.0 deep (drill 3.3)	9.00	-16.00	-14.74	-x
M4 tapped 6.0 deep (drill 3.3)	9.00	16.00	-34.74	-x
M4 tapped 6.0 deep (drill 3.3)	9.00	16.00	-14.74	-x
d4.5 through	3.00	-149.91	40.29	-x
d4.5 through	3.00	-149.74	10.43	-x
d4.5 through	3.00	-100.75	-29.25	-x
d4.5 through	3.00	-96.97	-9.94	-x
d4.5 through	3.00	-48.75	-50.26	-x
d4.5 through	3.00	0.00	-37.74	-x
d4.5 through	3.00	48.75	-50.26	-x
d4.5 through	3.00	96.97	-9.94	-x
d4.5 through	3.00	100.75	-29.25	-x
d4.5 through	3.00	149.74	10.43	-x
d4.5 through	3.00	149.91	40.29	-x
d5.5 through	3.00	-179.25	76.45	-x
d5.5 through	3.00	-166.18	81.47	-x
d5.5 through	3.00	166.18	81.47	-x
d5.5 through	3.00	179.25	76.45	-x

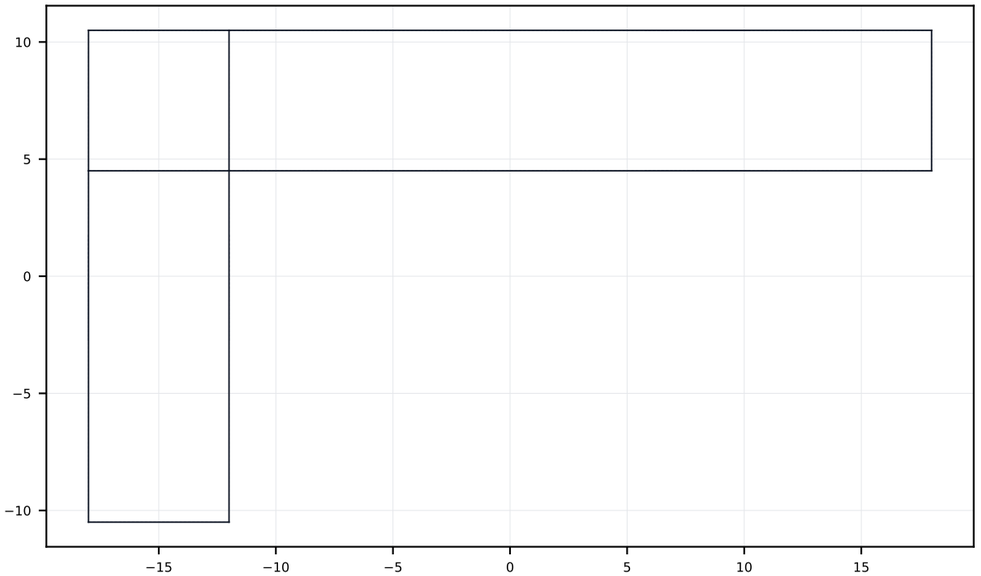
front (-y) (axes in mm from the view origin)



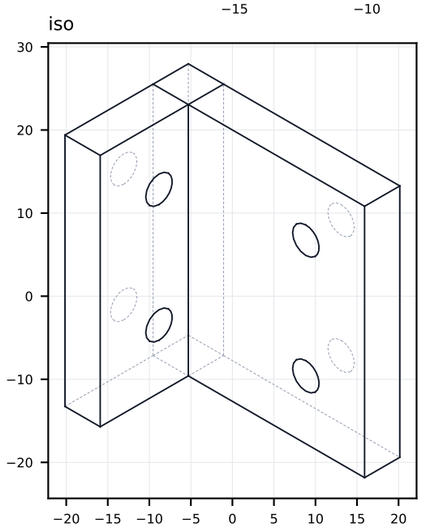
side (+x) (axes in mm from the view origin)



top (+z) (axes in mm from the view origin)



iso

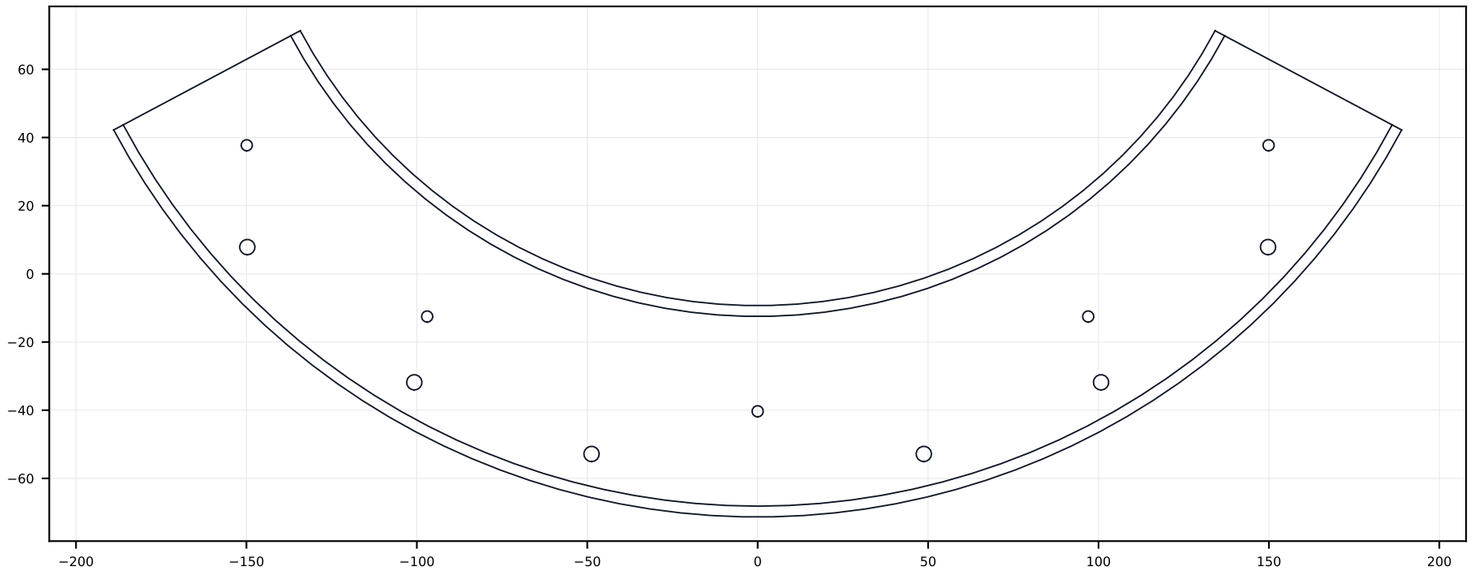


DRAWING arc_bracket
qty per robot: 4 (R-hand part = this one mirrored about XZ)
material: 6061-T6
process: CNC from 40 x 40 x 6 angle
finish:
mass: 32.0 g overall: 36.0 x 21.0 x 40.0 mm (x, y, z)
note:
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/arc_bracket.step
view origin = bounding-box centre = robot (113.00, 16.50, -170.00)

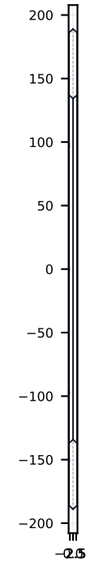
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
d4.5 through	-12.00	-0.50	-10.00	-x
d4.5 through	-12.00	-0.50	10.00	-x
d4.5 through	8.00	4.50	-10.00	+y
d4.5 through	8.00	4.50	10.00	+y

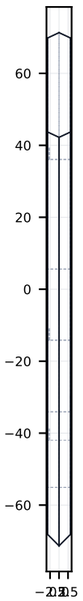
side (+x) (axes in mm from the view origin)



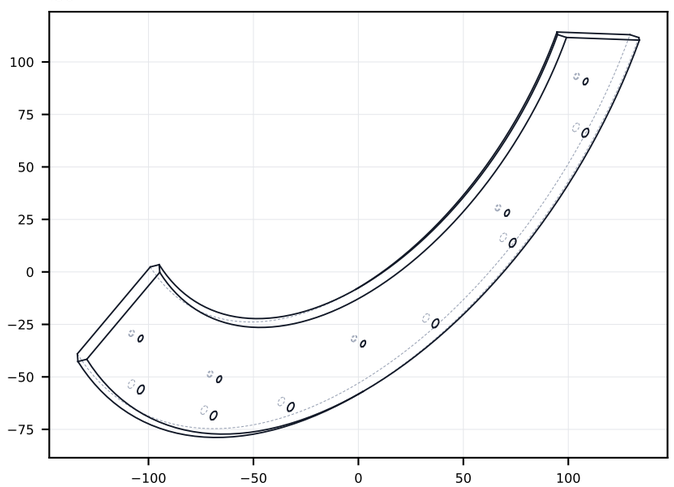
top (+z) (axes in mm from the view origin)



front (-y) (axes in mm from the view origin)



iso

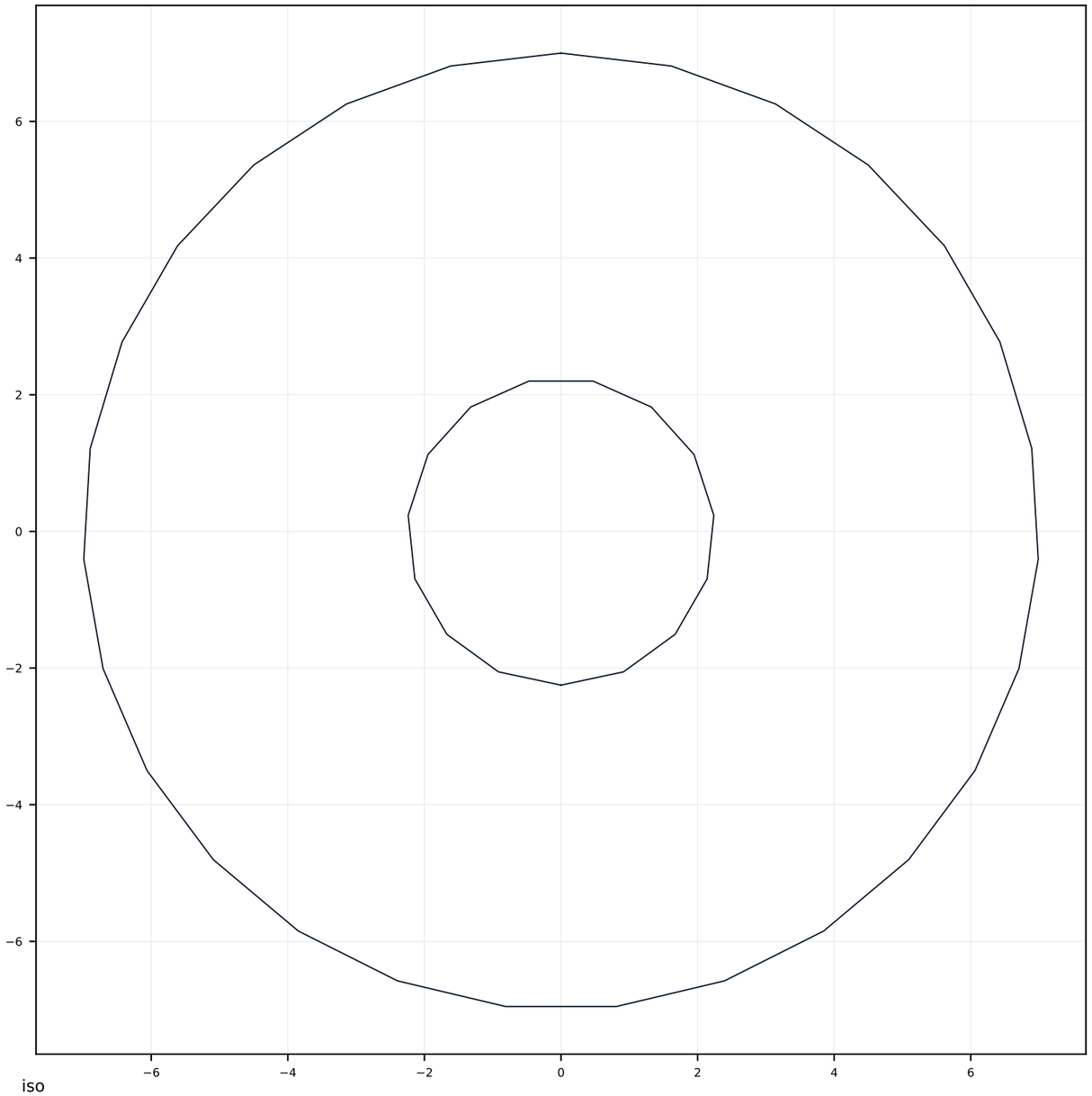


DRAWING track
qty per robot: 2 (R-hand part = this one mirrored about XZ)
material: AISI 1045, edges induction hardened HRC 53 (DualVee track spec)
process: waterjet + CNC V-edges
finish:
mass: 1154.4 g overall: 6.4 x 377.9 x 142.6 mm (x, y, z)
note:
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/track.step
view origin = bounding-box centre = robot (82.00, 0.00, -142.68)

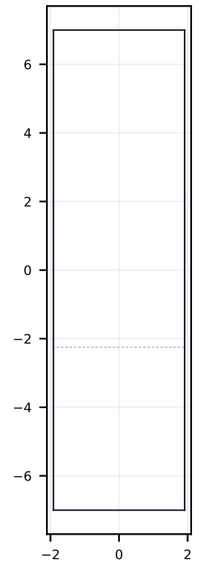
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
M4 tapped 6.3 deep (drill 3.3)	13.00	-149.91	37.72	-x
M4 tapped 6.3 deep (drill 3.3)	13.00	-96.97	-12.51	-x
M4 tapped 6.3 deep (drill 3.3)	13.00	0.00	-40.32	-x
M4 tapped 6.3 deep (drill 3.3)	13.00	96.97	-12.51	-x
M4 tapped 6.3 deep (drill 3.3)	13.00	149.91	37.72	-x
d4.5 through	13.00	-149.74	7.85	-x
d4.5 through	13.00	-100.75	-31.82	-x
d4.5 through	13.00	-48.75	-52.84	-x
d4.5 through	13.00	48.75	-52.84	-x
d4.5 through	13.00	100.75	-31.82	-x
d4.5 through	13.00	149.74	7.85	-x

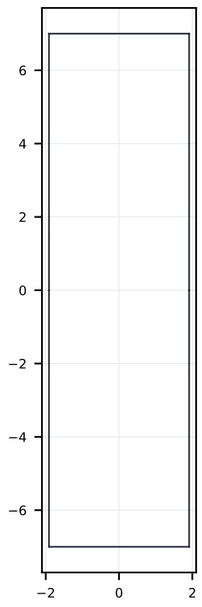
side (+x) (axes in mm from the view origin)



front (-y) (axes in mm from the view origin)



top (+z) (axes in mm from the view origin)



iso

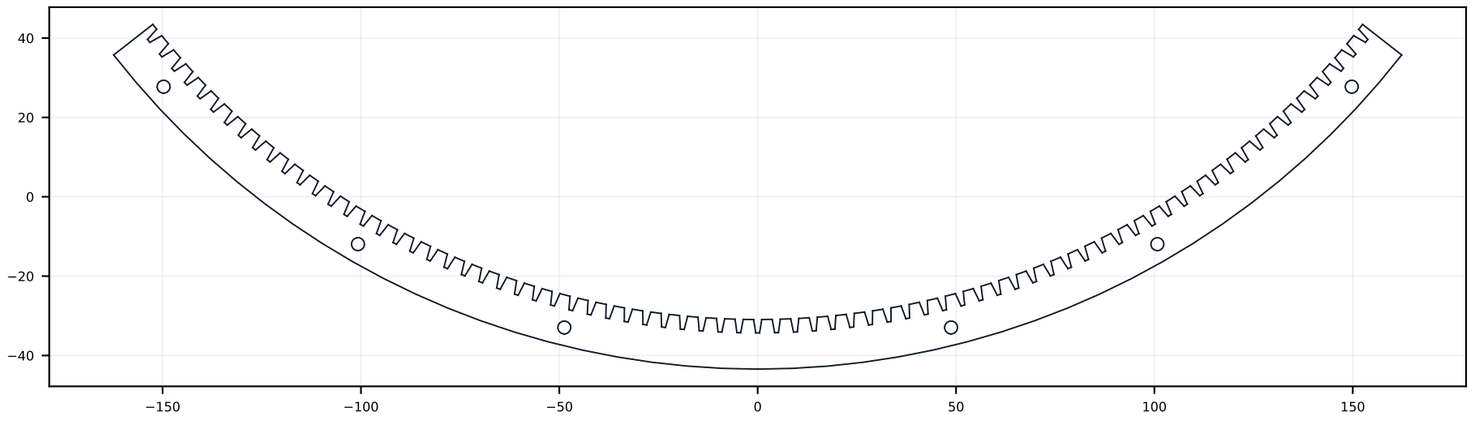


DRAWING track_spacer
qty per robot: 10 (R-hand part = this one mirrored about XZ)
material: 6061
process: turned spacer d14 x 2.8
finish:
mass: 1.4 g overall: 3.8 x 14.0 x 14.0 mm (x, y, z)
note:
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/track_spacer.step
view origin = bounding-box centre = robot (87.09, 0.00, -183.00)

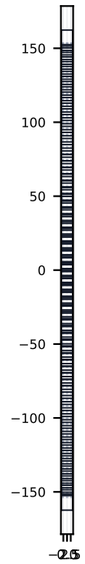
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
d4.5 through	7.91	0.00	0.00	-x

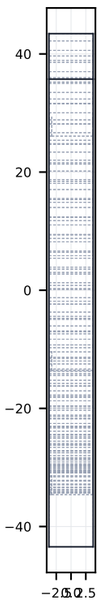
side (+x) (axes in mm from the view origin)



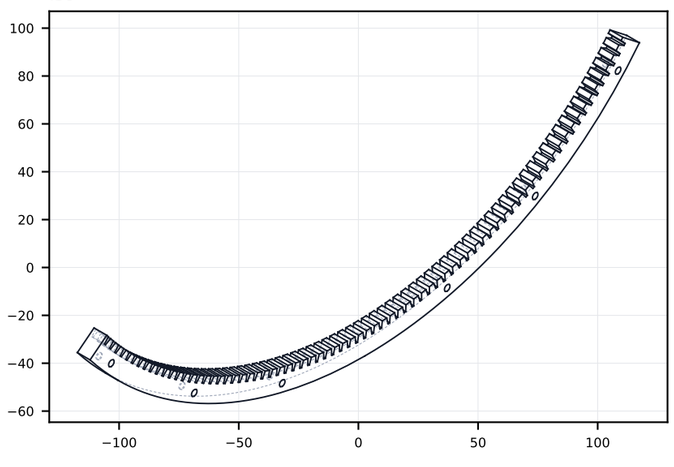
top (+z) (axes in mm from the view origin)



front (-y) (axes in mm from the view origin)



iso

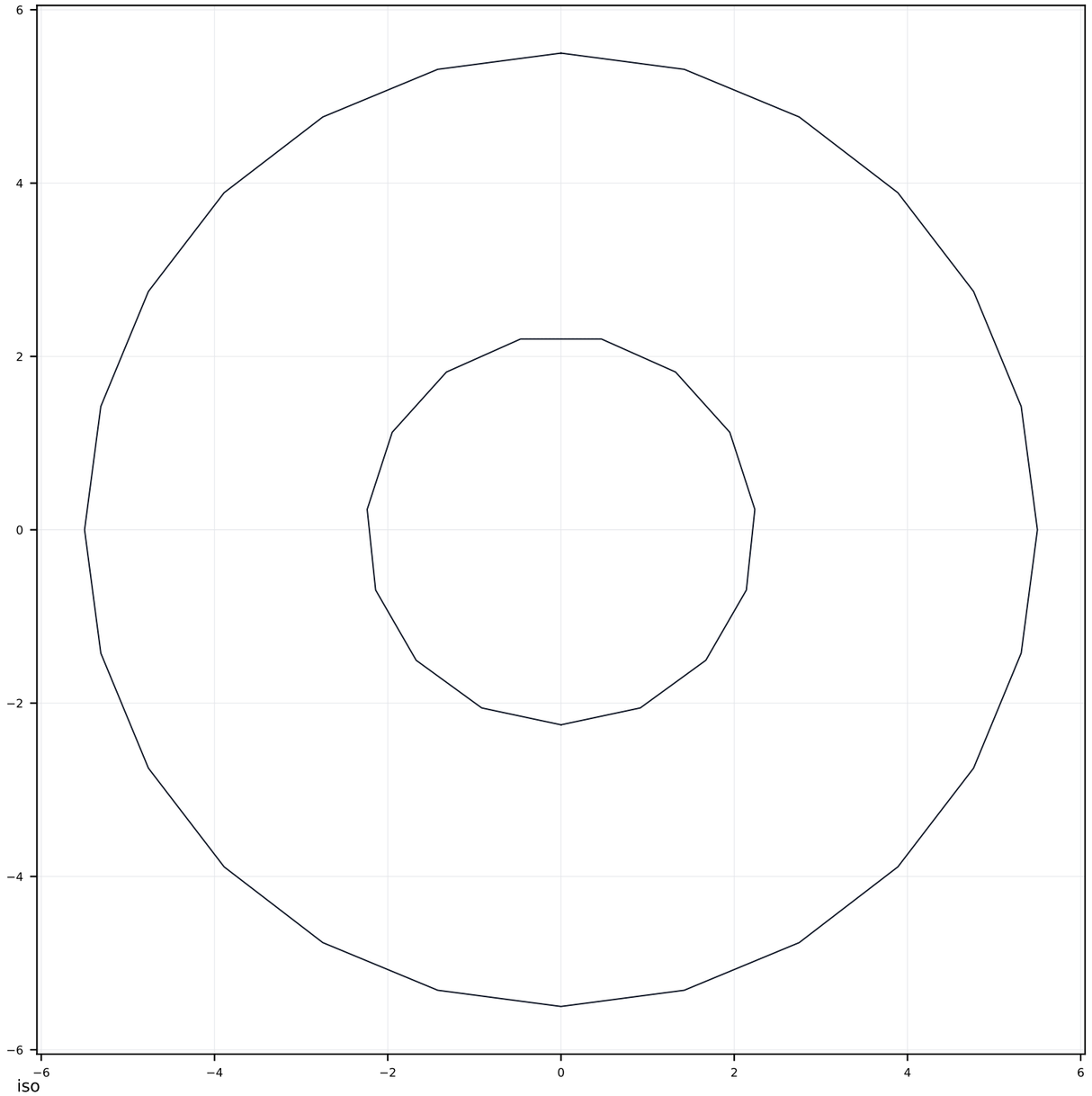


DRAWING lean_rack
qty per robot: 1
material: 7075-T6
process: wire EDM
finish: hard anodise
mass: 86.4 g overall: 7.5 x 324.7 x 86.9 mm (x, y, z)
note: internal m1.5 z260 segment +-52 deg
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/lean_rack.step
view origin = bounding-box centre = robot (72.75, -0.00, -162.56)

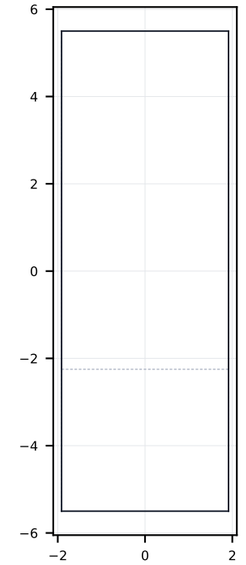
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
M4 tapped 7.5 deep (drill 3.3)	22.25	-149.74	27.73	-x
M4 tapped 7.5 deep (drill 3.3)	22.25	-100.75	-11.94	-x
M4 tapped 7.5 deep (drill 3.3)	22.25	-48.75	-32.95	-x
M4 tapped 7.5 deep (drill 3.3)	22.25	48.75	-32.95	-x
M4 tapped 7.5 deep (drill 3.3)	22.25	100.75	-11.94	-x
M4 tapped 7.5 deep (drill 3.3)	22.25	149.74	27.73	-x

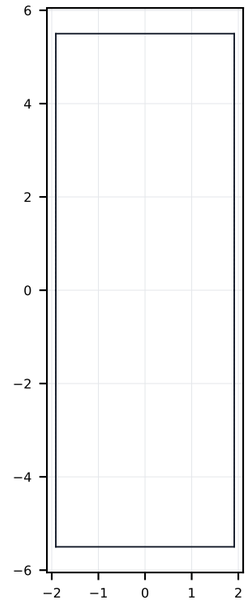
side (+x) (axes in mm from the view origin)



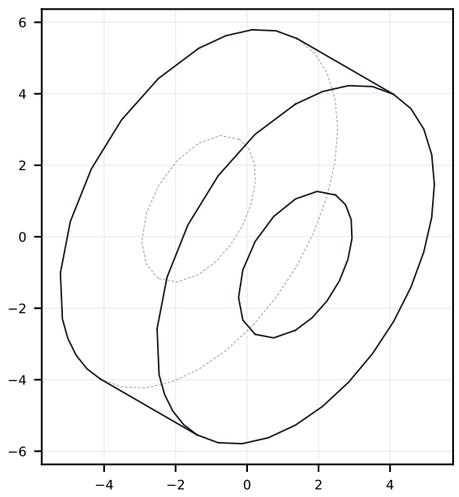
front (-y) (axes in mm from the view origin)



top (+z) (axes in mm from the view origin)



iso

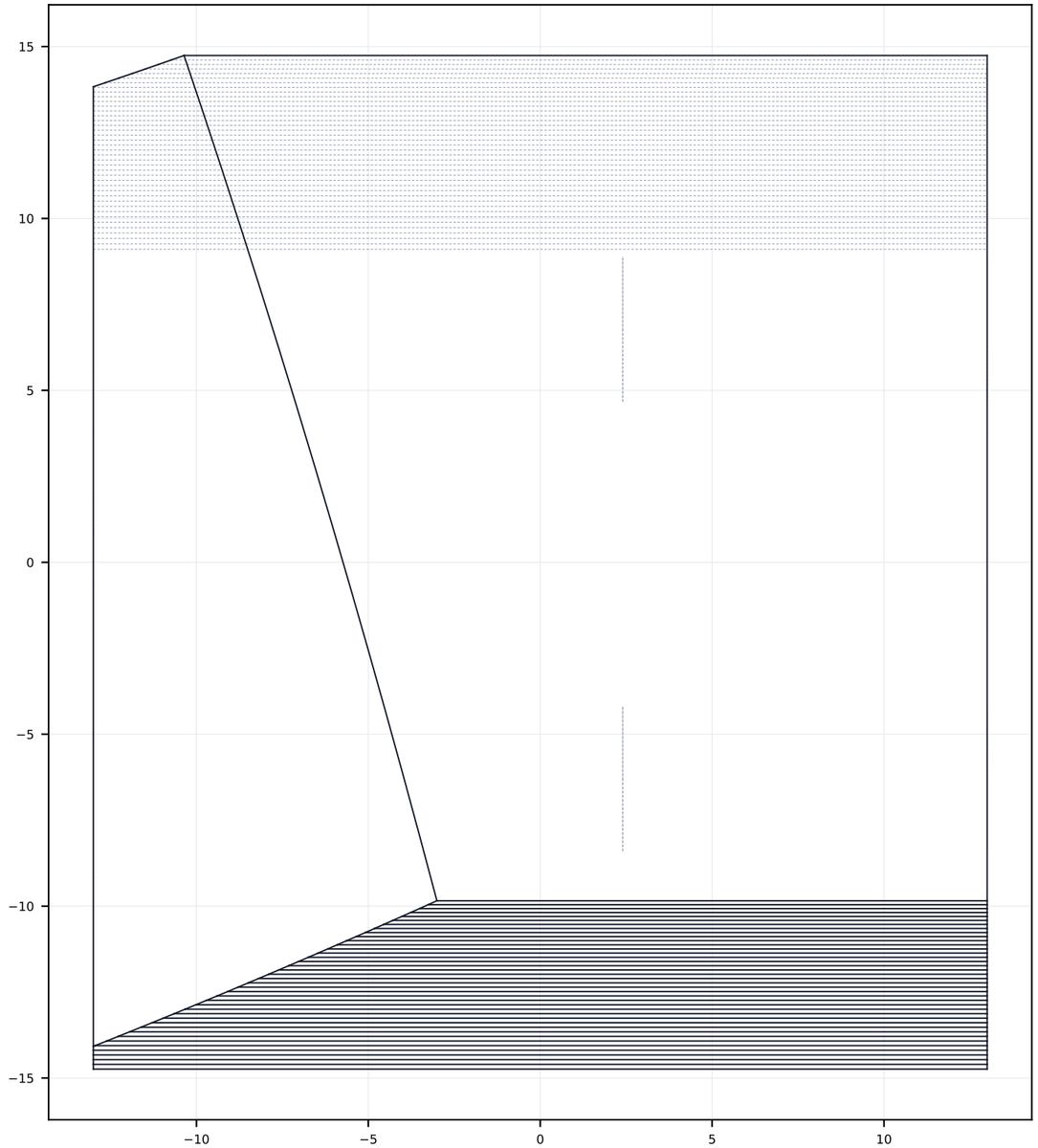


DRAWING rack_spacer
qty per robot: 12 (R-hand part = this one mirrored about XZ)
material: 6061
process: turned spacer d11
finish:
mass: 0.8 g overall: 3.8 x 11.0 x 11.0 mm (x, y, z)
note:
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/rack_spacer.step
view origin = bounding-box centre = robot (87.09, 48.75, -195.51)

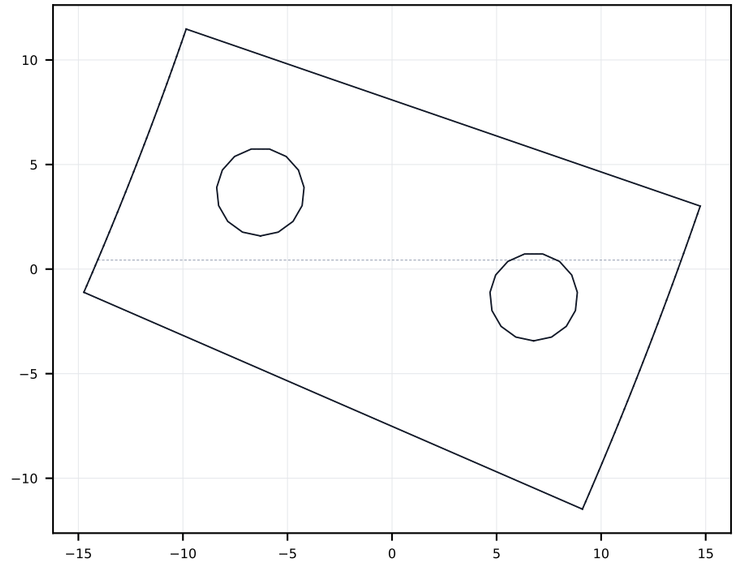
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
d4.5 through	7.91	-0.00	-0.00	-x

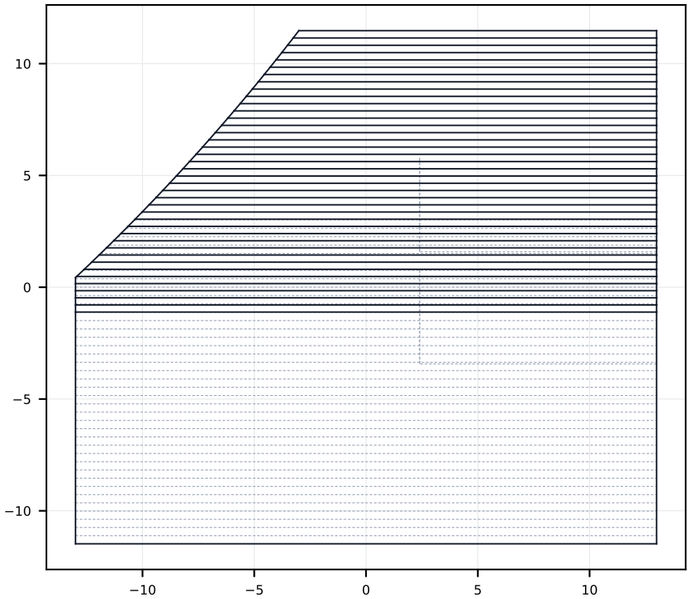
top (+z) (axes in mm from the view origin)



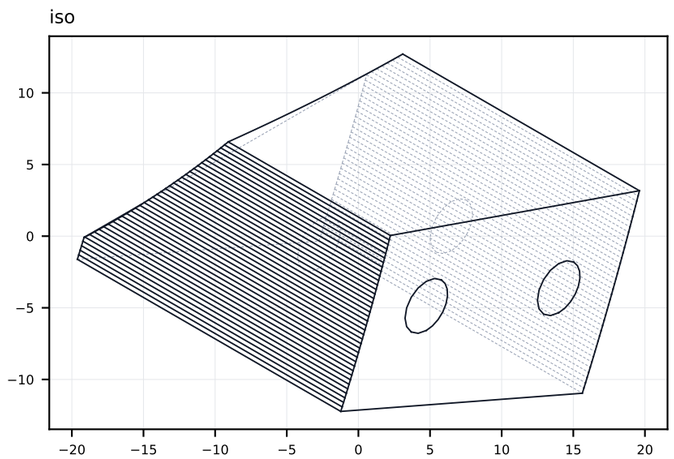
side (+x) (axes in mm from the view origin)



front (-y) (axes in mm from the view origin)



iso

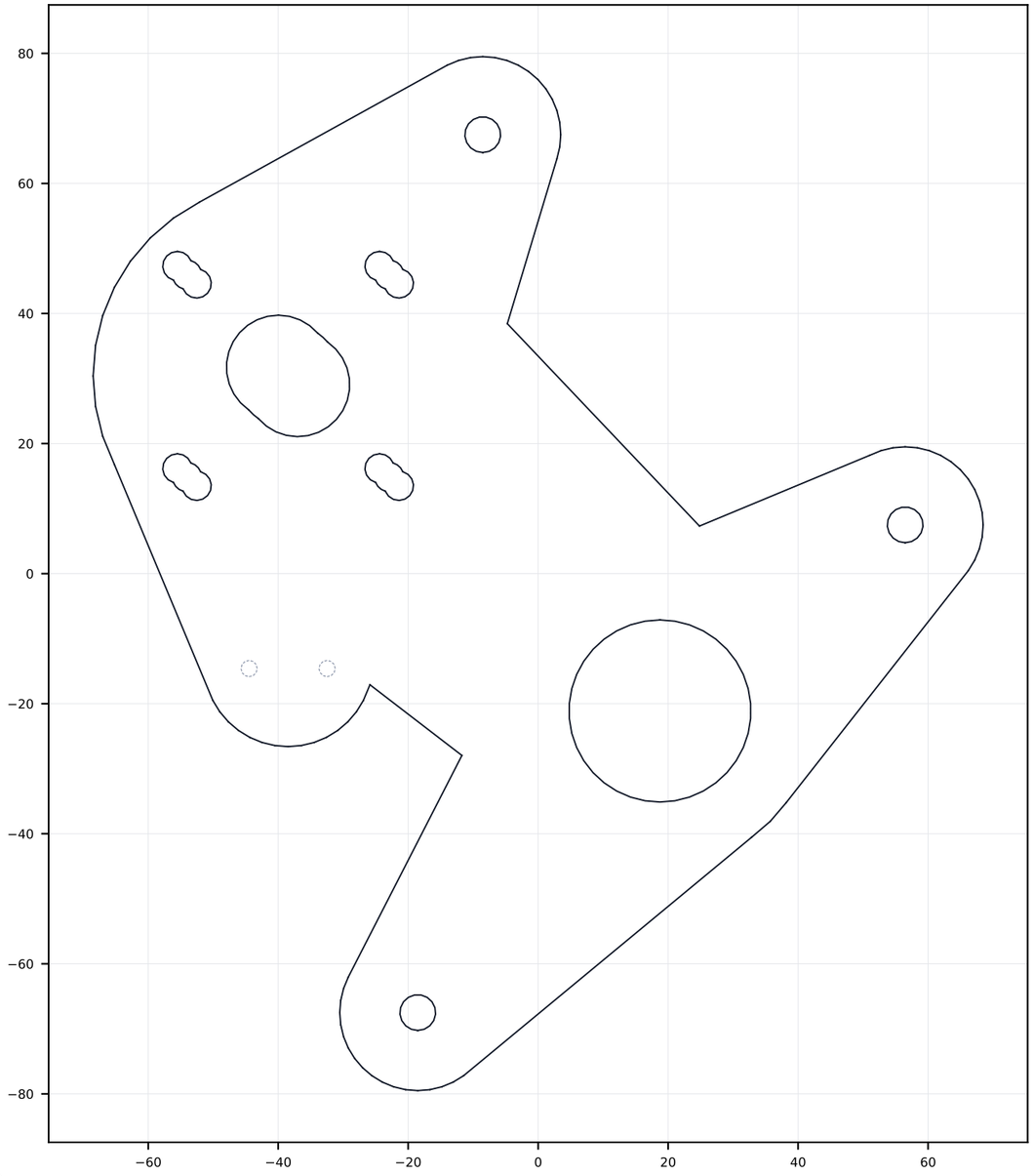


DRAWING lean_stop
qty per robot: 4 (R-hand part = this one mirrored about XZ)
material: 6061-T6
process: CNC
finish:
mass: 23.8 g overall: 26.0 x 29.5 x 23.0 mm (x, y, z)
note: 3 mm PU 90A pad bonded on the face toward the carriage
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/lean_stop.step
view origin = bounding-box centre = robot (76.00, 172.47, -67.48)

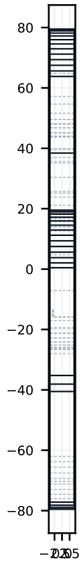
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
M5 tapped 10.6 deep (drill 4.2)	19.00	-6.30	3.69	-x
M5 tapped 10.6 deep (drill 4.2)	19.00	6.77	-1.33	-x

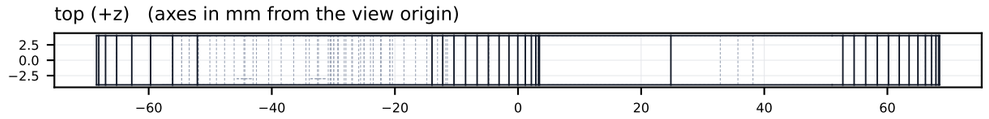
front (-y) (axes in mm from the view origin)



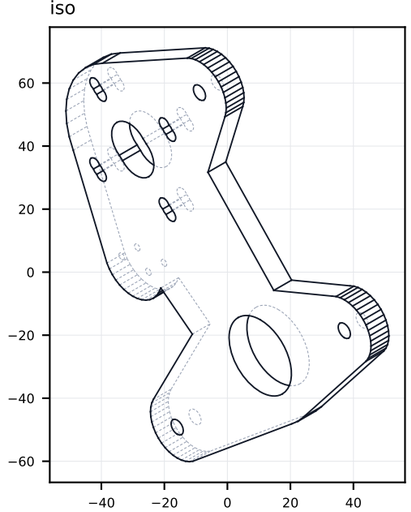
side (+x) (axes in mm from the view origin)



top (+z) (axes in mm from the view origin)



iso

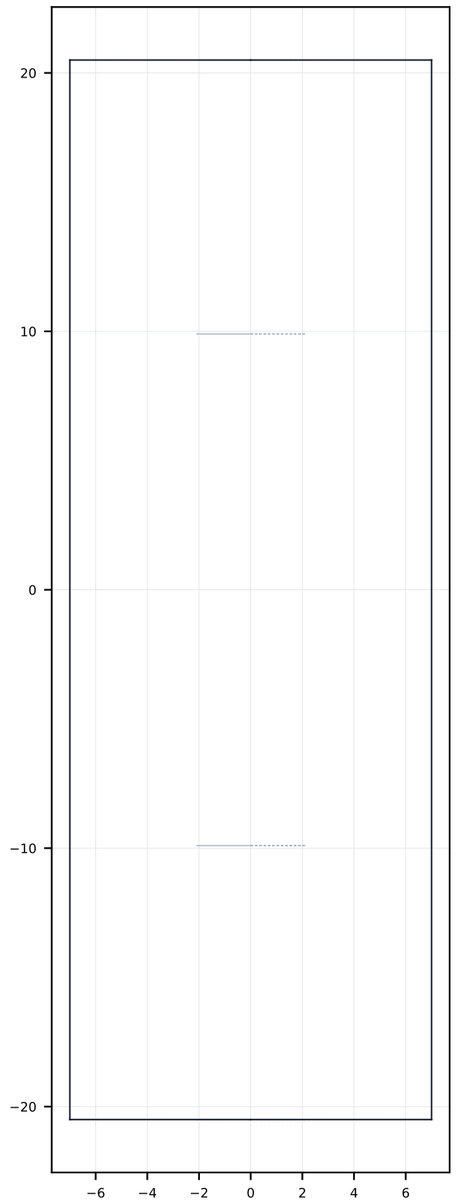


DRAWING outer_plate
qty per robot: 2 (R-hand part = this one mirrored about XZ)
material: 6061-T6 plate 8 mm
process: waterjet + CNC
finish:
mass: 200.7 g overall: 136.9 x 8.0 x 159.0 mm (x, y, z)
note: motor holes slotted +-2 mm for belt tension
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/outer_plate.step
view origin = bounding-box centre = robot (168.53, 80.00, -147.50)

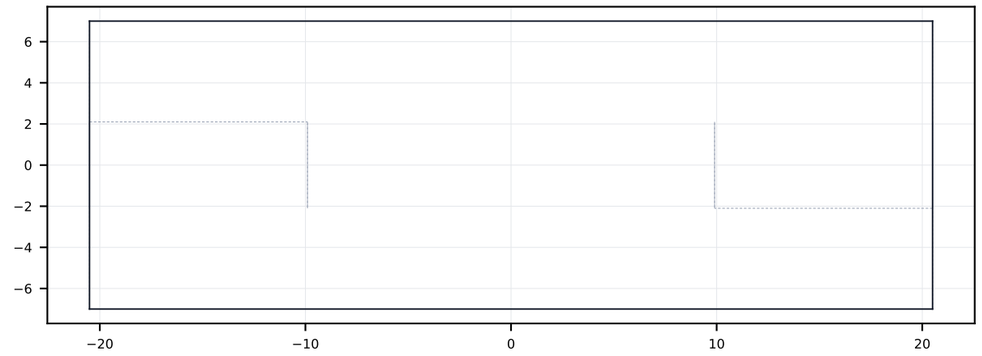
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
M3 tapped 7.0 deep (drill 2.5)	-44.48	7.00	-14.60	-y
M3 tapped 7.0 deep (drill 2.5)	-32.48	7.00	-14.60	-y
d4.5 through	-54.03	-4.00	14.85	+y
d4.5 through	-54.03	-4.00	45.96	+y
d4.5 through	-22.92	-4.00	14.85	+y
d4.5 through	-22.92	-4.00	45.96	+y
d5.5 through	-18.53	4.00	-67.50	-y
d5.5 through	-8.53	4.00	67.50	-y
d5.5 through	56.47	4.00	7.50	-y

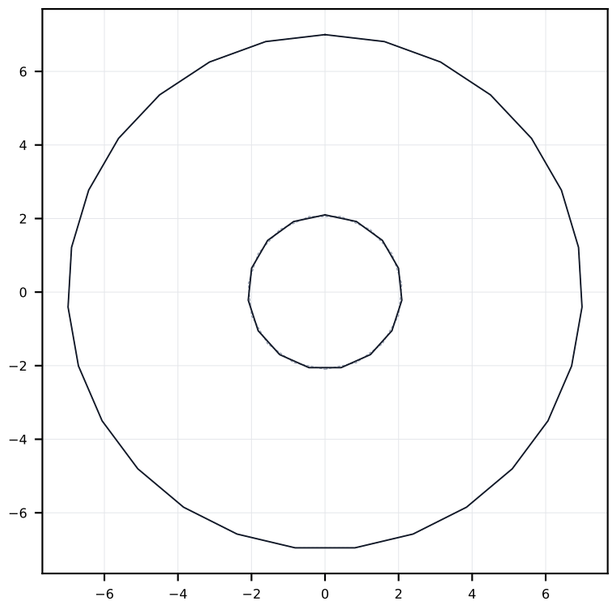
top (+z) (axes in mm from the view origin)



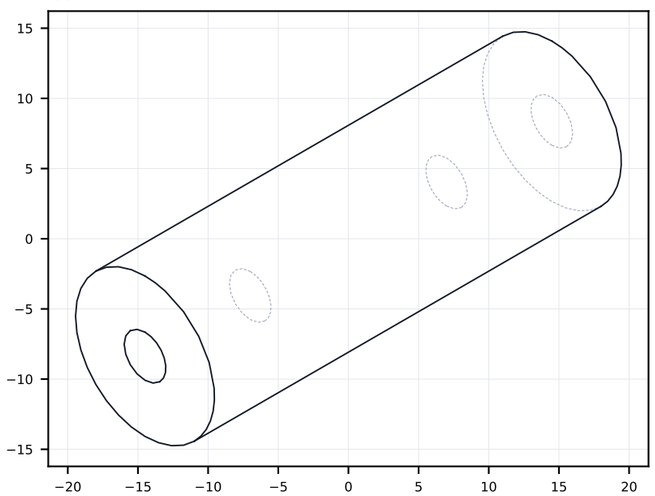
side (+x) (axes in mm from the view origin)



front (-y) (axes in mm from the view origin)



iso

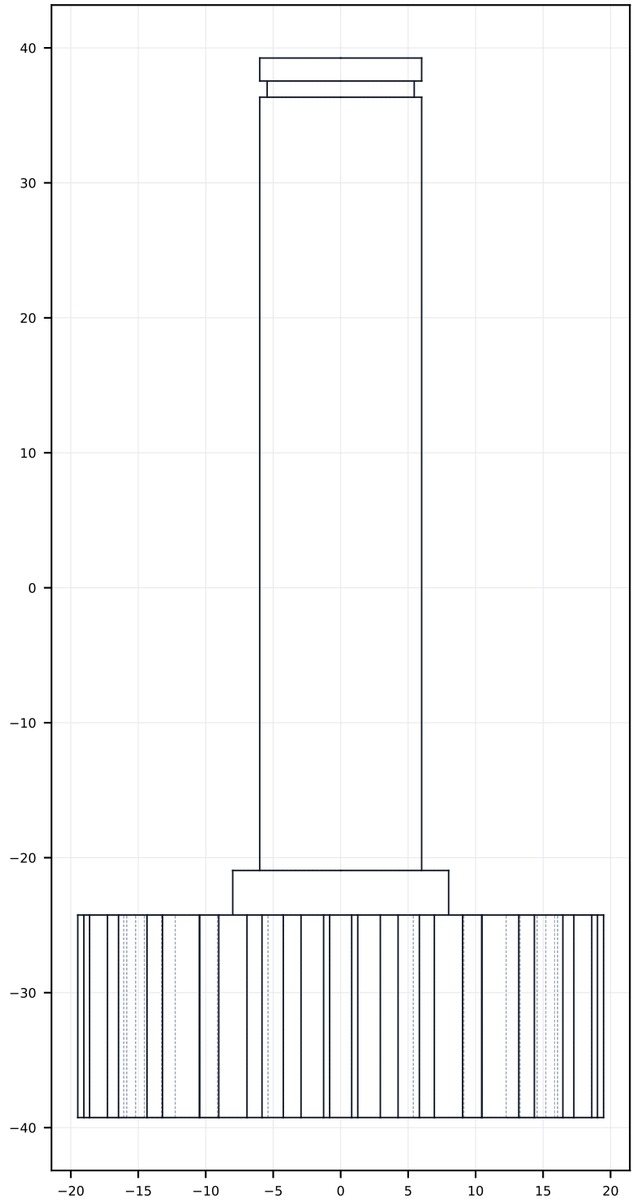


DRAWING cassette_spacer
qty per robot: 6 (R-hand part = this one mirrored about XZ)
material: 6061 d14
process: turned, M5 tapped both ends 12 deep
finish:
mass: 16.2 g overall: 14.0 x 41.0 x 14.0 mm (x, y, z)
note:
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/cassette_spacer.step
view origin = bounding-box centre = robot (160.00, 55.50, -80.00)

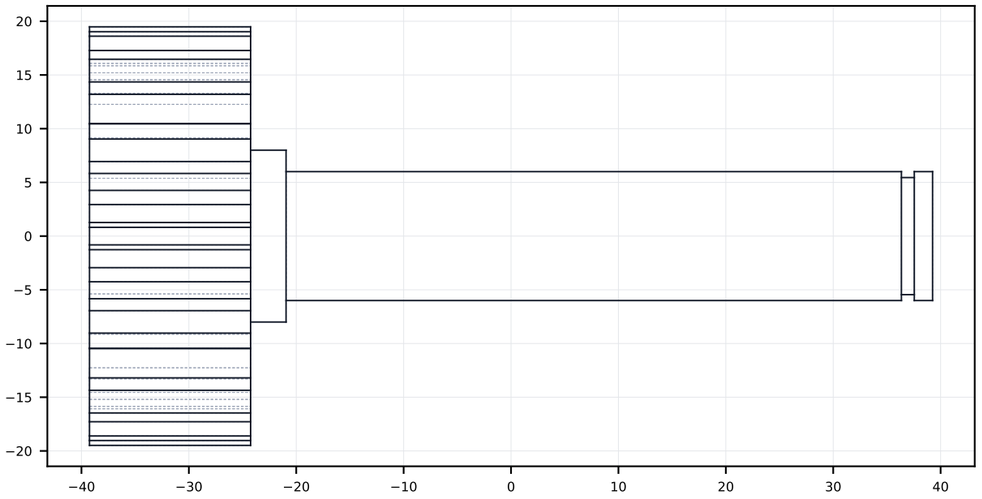
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
M5 tapped 10.6 deep (drill 4.2)	0.00	-28.50	0.00	+y
M5 tapped 10.6 deep (drill 4.2)	0.00	28.50	0.00	-y

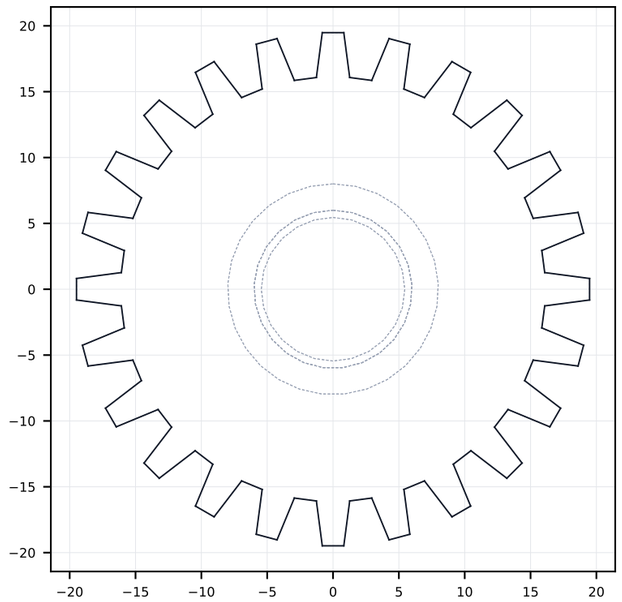
top (+z) (axes in mm from the view origin)



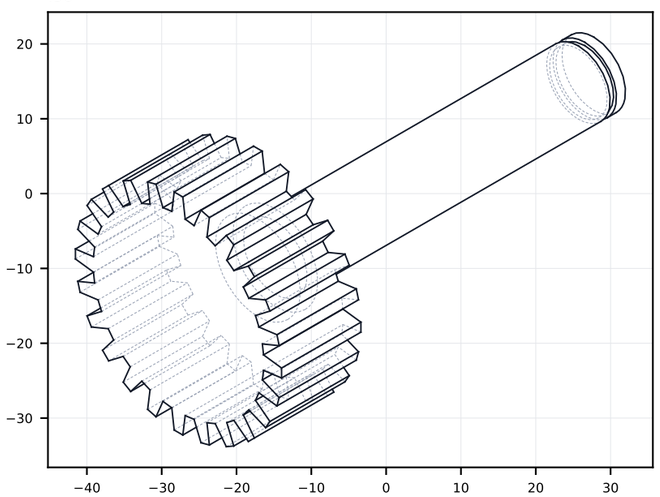
side (+x) (axes in mm from the view origin)



front (-y) (axes in mm from the view origin)

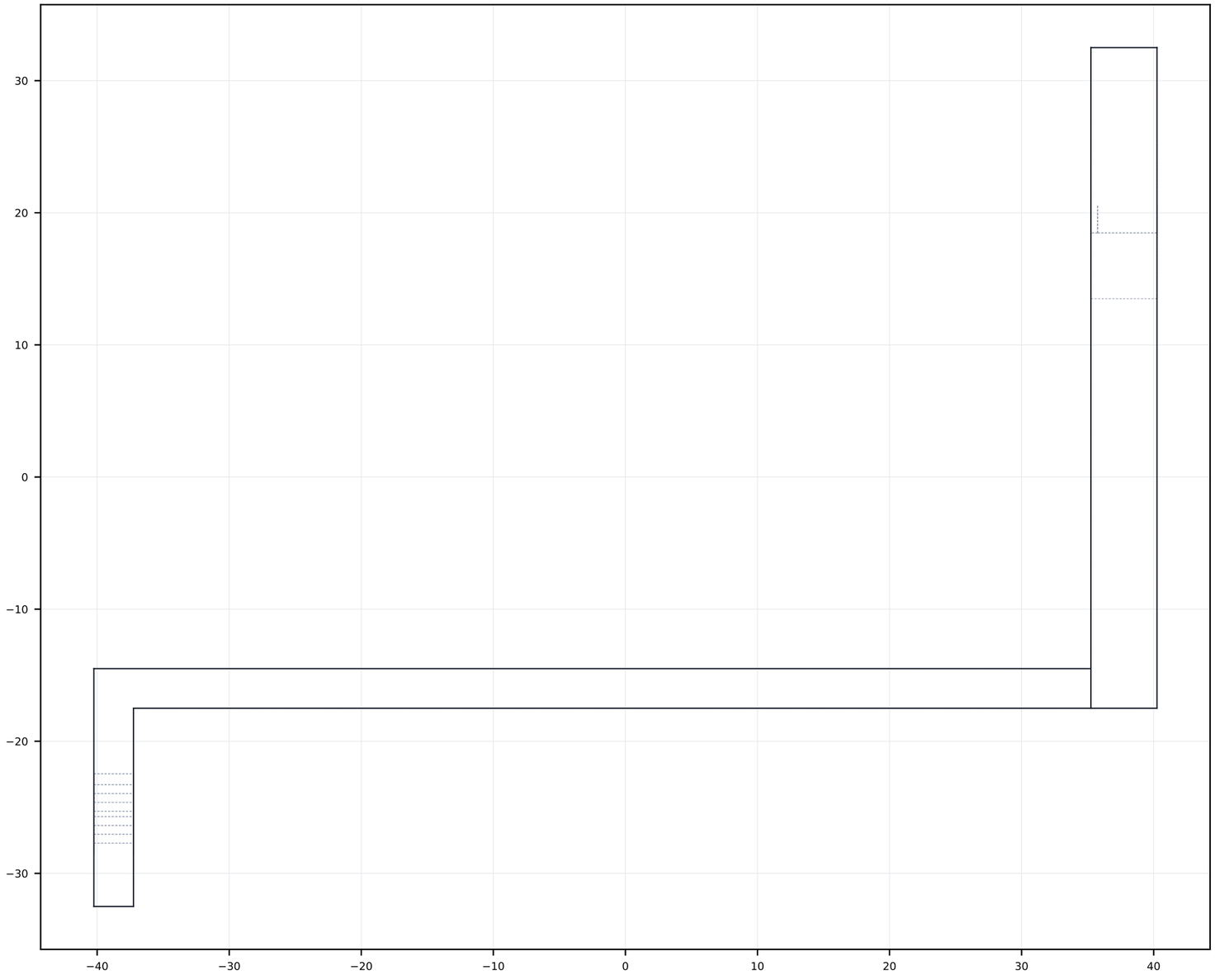


iso

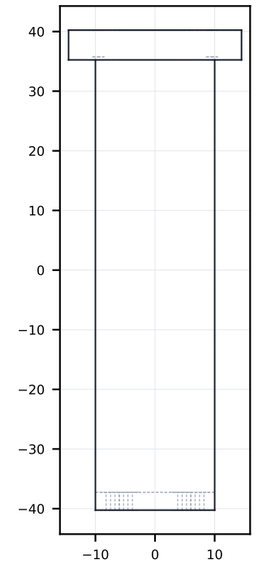


DRAWING pinion_shaft
qty per robot: 2 (R-hand part = this one mirrored about XZ)
material: 42CrMo4 QT + gas nitrided
process: turn, hob 24T m1.5, grind bearing seats
finish:
mass: 174.4 g overall: 39.0 x 78.5 x 39.0 mm (x, y, z)
note: DIN 6885 4x4 keyway for the P32, DIN 471 groove
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/pinion_shaft.step
view origin = bounding-box centre = robot (187.27, 47.75, -168.62)

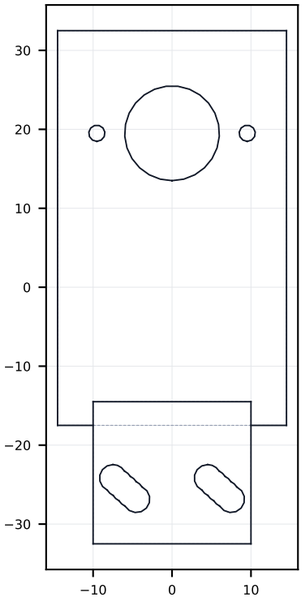
side (+x) (axes in mm from the view origin)



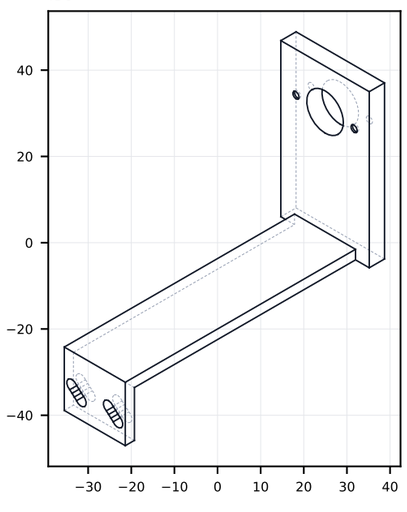
top (+z) (axes in mm from the view origin)



front (-y) (axes in mm from the view origin)



iso

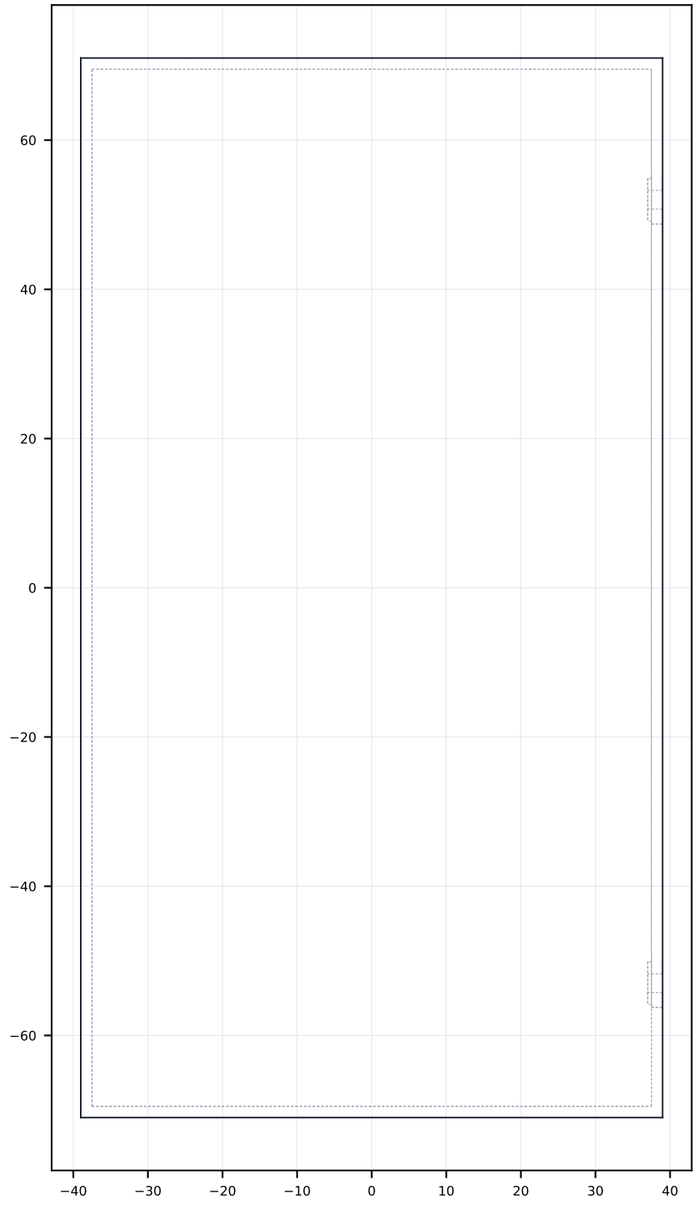


DRAWING drive_enc_bracket
qty per robot: 2 (R-hand part = this one mirrored about XZ)
material: 6061-T6
process: CNC from 6 mm plate (or 3 mm bent L + 5 mm bridge)
finish:
mass: 32.3 g overall: 29.0 x 80.5 x 65.0 mm (x, y, z)
note: M2.5 tapped through in the bridge; foot holes slotted +-2 mm with the motor
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/drive_enc_bracket.step
view origin = bounding-box centre = robot (130.05, 124.25, -136.60)

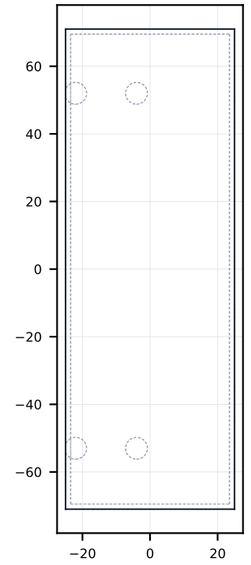
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
M2.5 tapped 5.0 deep (drill 2.05)	-9.53	41.75	19.50	-y
M2.5 tapped 5.0 deep (drill 2.05)	9.52	41.75	19.50	-y
d3.4 through	-6.00	-37.25	-25.50	-y
d3.4 through	6.00	-37.25	-25.50	-y

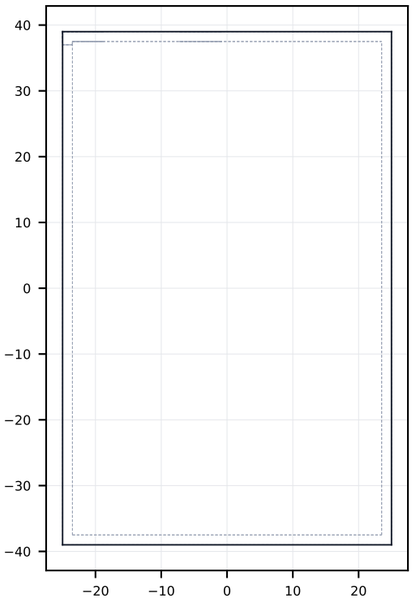
side (+x) (axes in mm from the view origin)



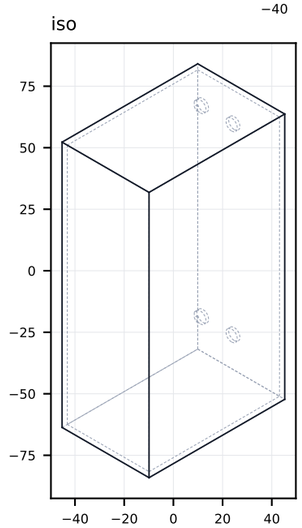
front (-y) (axes in mm from the view origin)



top (+z) (axes in mm from the view origin)



iso

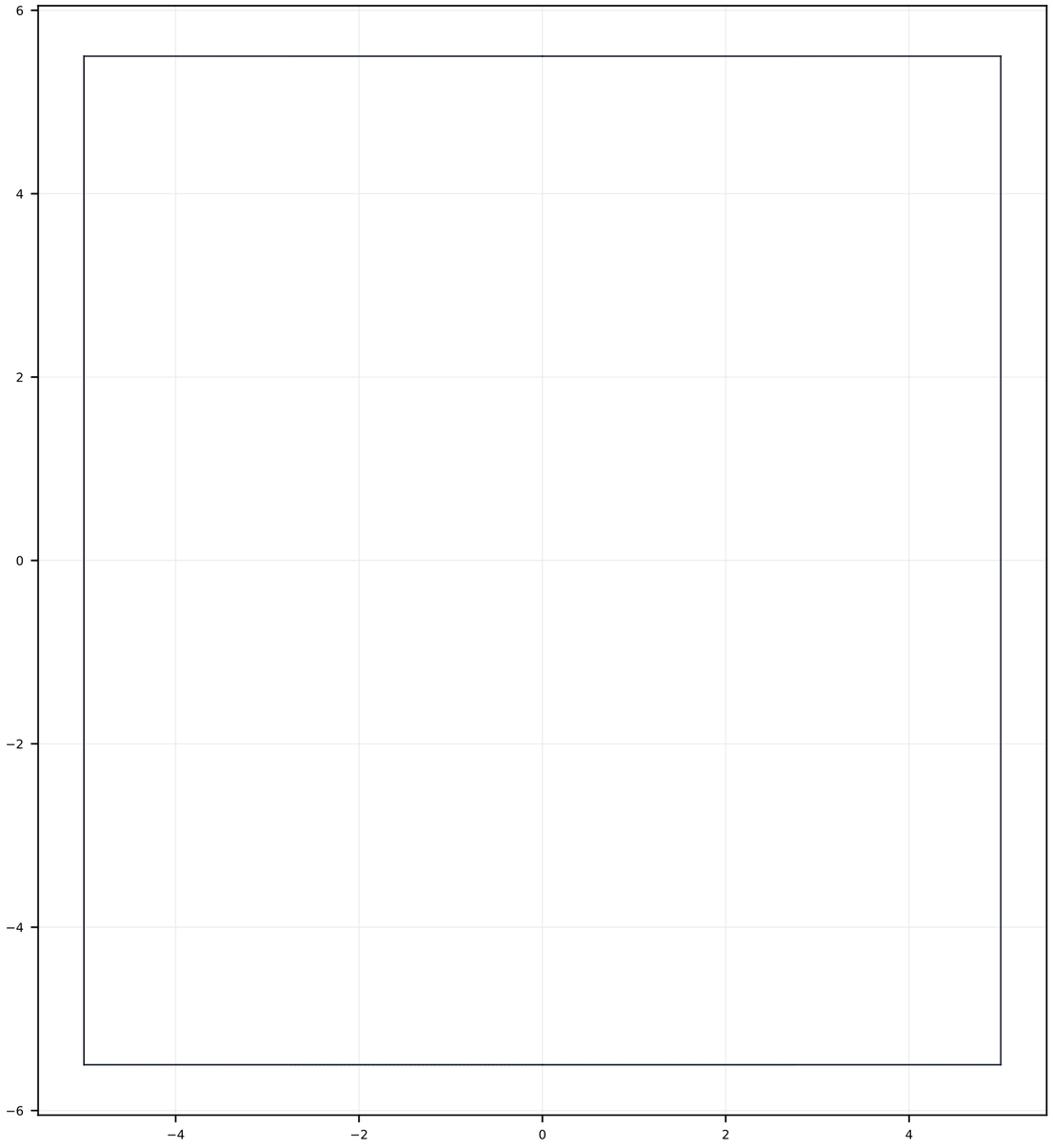


DRAWING battery_case
qty per robot: 2 (R-hand part = this one mirrored about XZ)
material: 5052-H32 1.5 mm
process: laser cut + bent + riveted, PEM nuts on the inboard end
finish:
mass: 171.8 g overall: 50.0 x 78.0 x 142.0 mm (x, y, z)
note: fish-paper lined; IP54 gasketed lid on the outboard face
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/battery_case.step
view origin = bounding-box centre = robot (127.00, -85.00, -147.00)

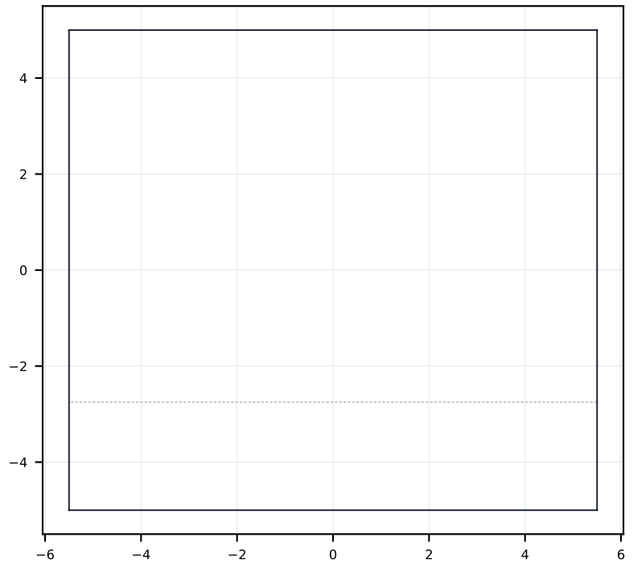
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
d6.5 for M5 PEM nut	-22.00	58.00	-53.00	-y
d6.5 for M5 PEM nut	-22.00	58.00	52.00	-y
d6.5 for M5 PEM nut	-4.00	58.00	-53.00	-y
d6.5 for M5 PEM nut	-4.00	58.00	52.00	-y

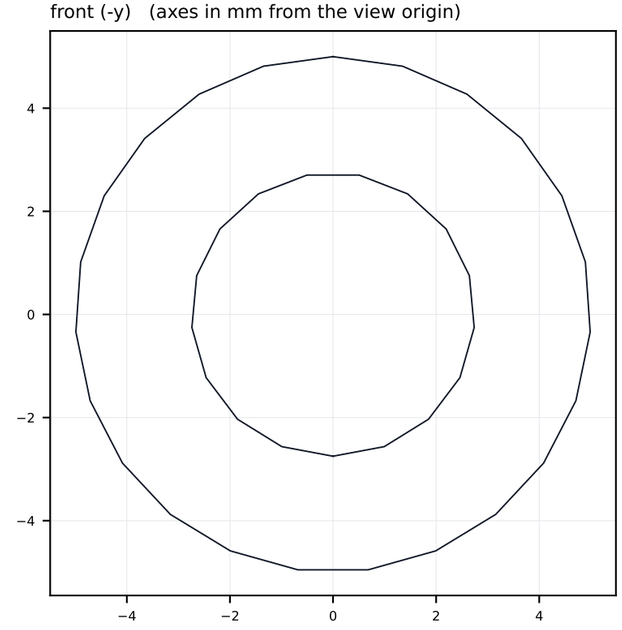
top (+z) (axes in mm from the view origin)



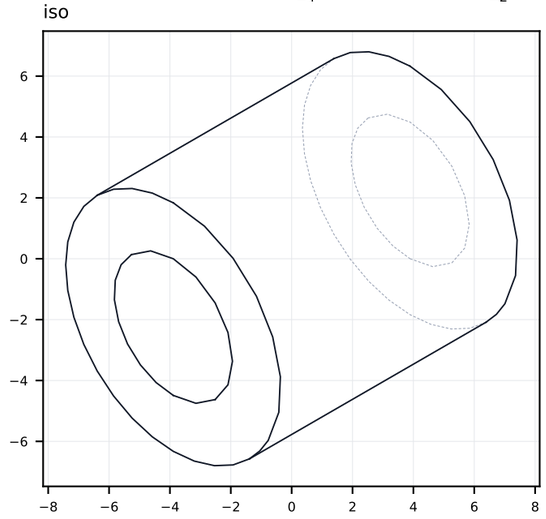
side (+x) (axes in mm from the view origin)



front (-y) (axes in mm from the view origin)



iso

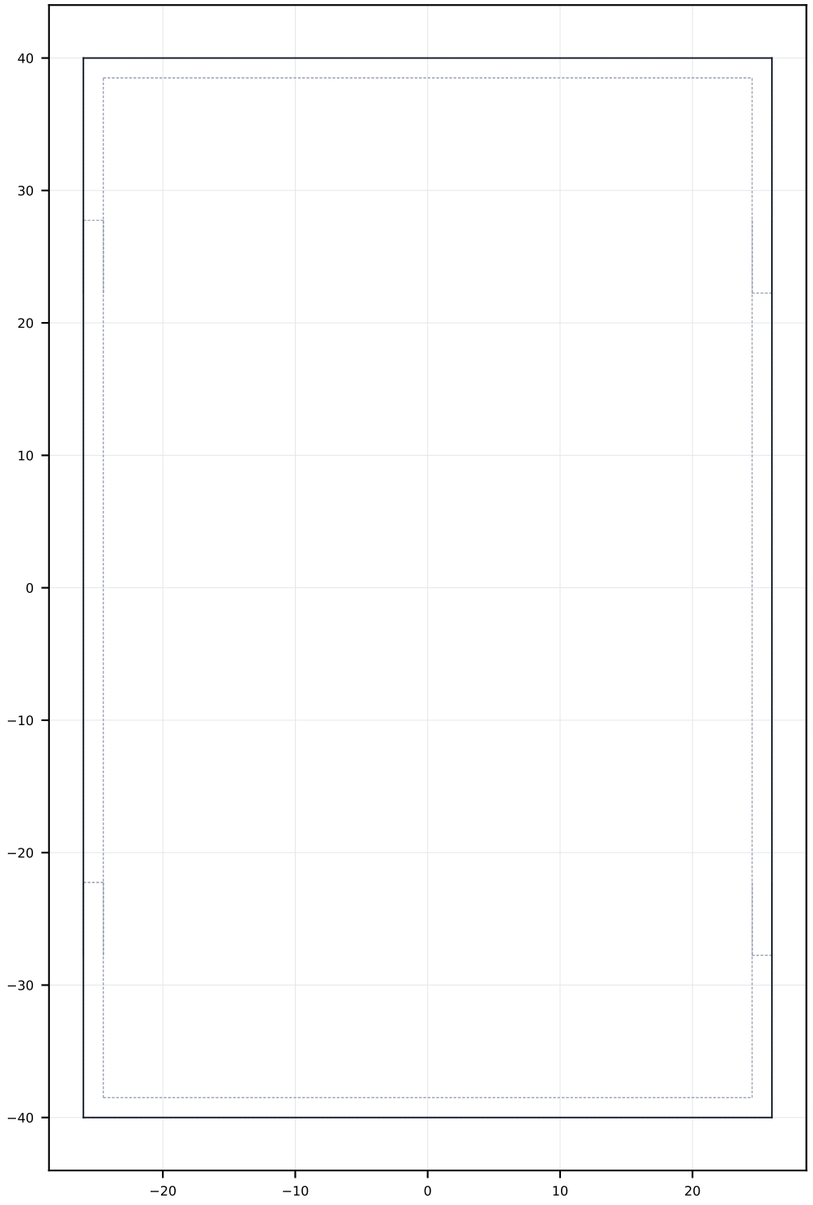


DRAWING battery_spacer
qty per robot: 8 (R-hand part = this one mirrored about XZ)
material: 6061 d10/5.6
process: cut tube spacer, 11 mm
finish:
mass: 1.6 g overall: 10.0 x 11.0 x 10.0 mm (x, y, z)
note:
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/battery_spacer.step
view origin = bounding-box centre = robot (105.00, -40.50, -95.00)

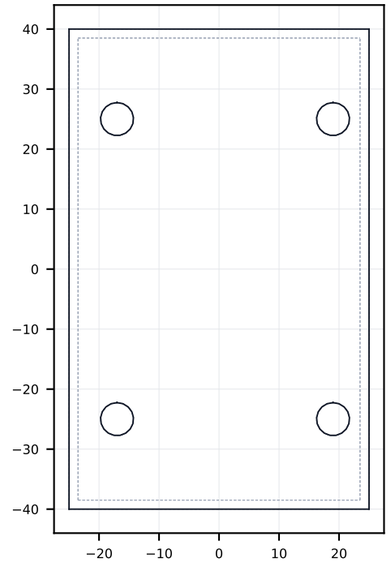
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
d5.5 through	0.00	13.50	0.00	-y

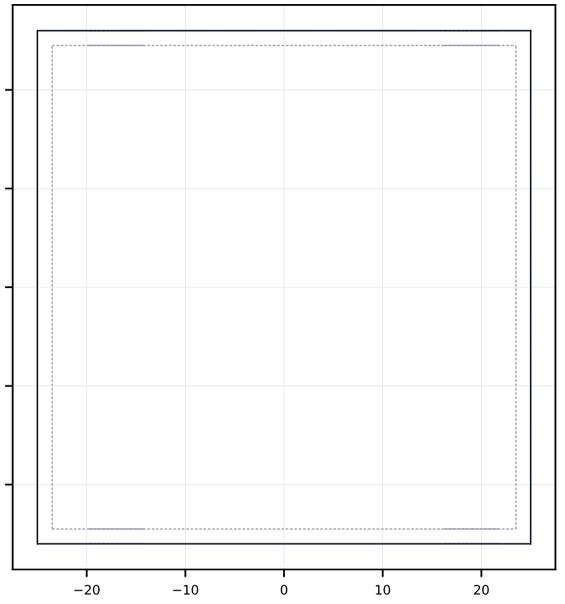
side (+x) (axes in mm from the view origin)



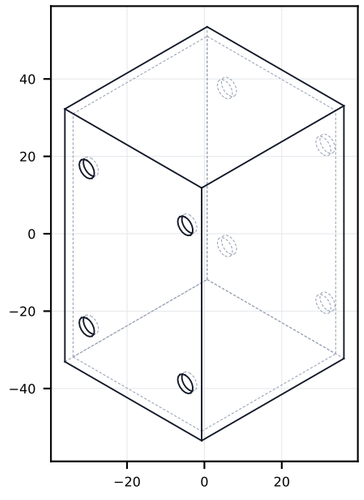
front (-y) (axes in mm from the view origin)



top (+z) (axes in mm from the view origin)



iso

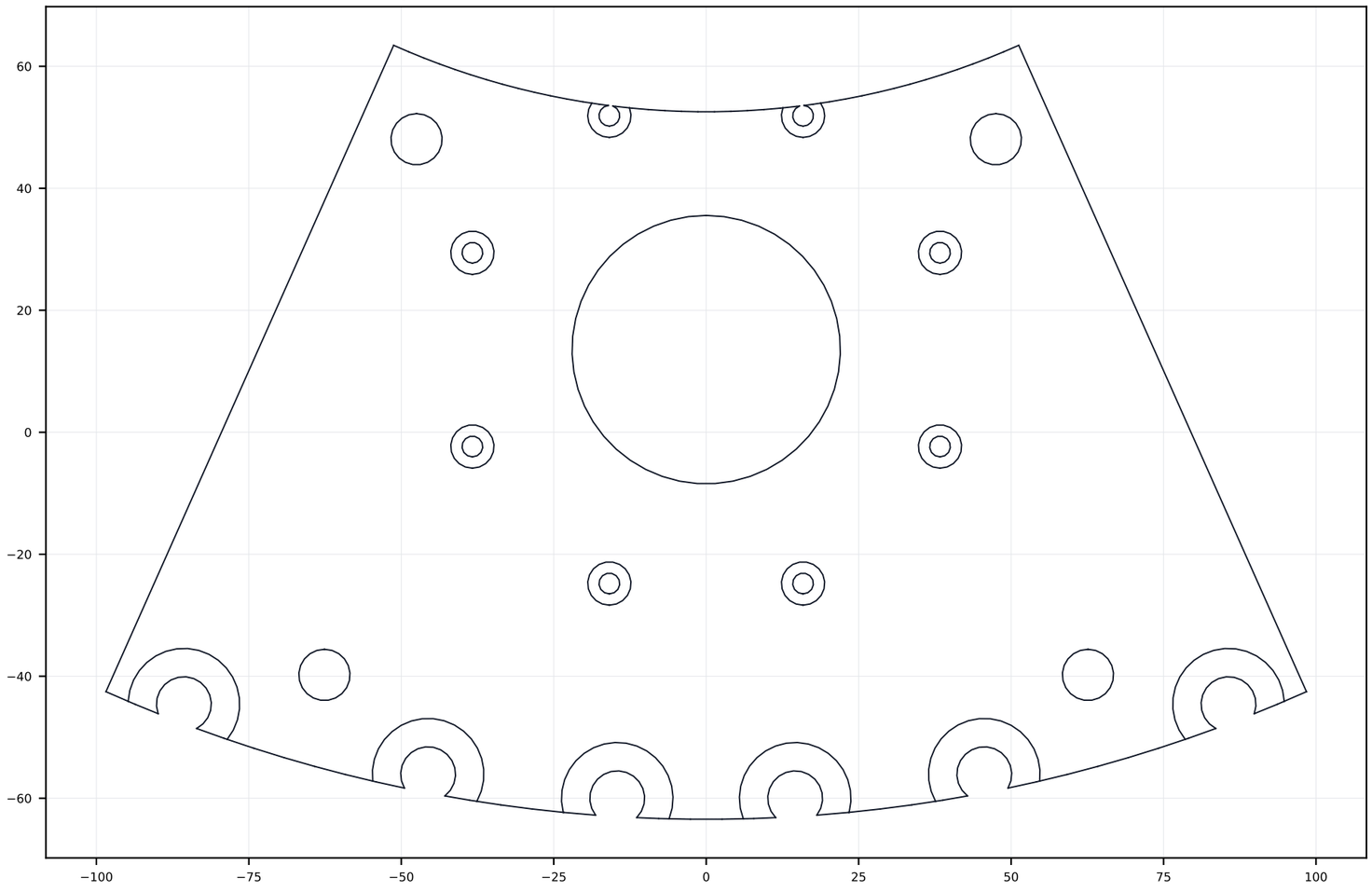


DRAWING power_box
qty per robot: 1
material: 5052-H32 1.5 mm
process: laser cut + bent, PEM nuts
finish:
mass: 82.0 g overall: 50.0 x 52.0 x 80.0 mm (x, y, z)
note: 12S BMS (<= 74 x 44 x 15, on edge), 40 A MIDI fuse, XT90-S anti-spark, CAN hub, e-stop relay
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/power_box.step
view origin = bounding-box centre = robot (-20.00, 0.00, -165.00)

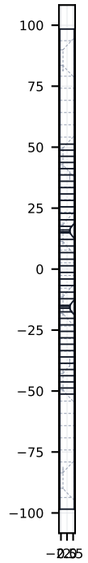
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
d5.5 for M4 PEM nut	-17.00	-35.00	-25.00	+y
d5.5 for M4 PEM nut	-17.00	-35.00	25.00	+y
d5.5 for M4 PEM nut	-17.00	35.00	-25.00	-y
d5.5 for M4 PEM nut	-17.00	35.00	25.00	-y
d5.5 for M4 PEM nut	19.00	-35.00	-25.00	+y
d5.5 for M4 PEM nut	19.00	-35.00	25.00	+y
d5.5 for M4 PEM nut	19.00	35.00	-25.00	-y
d5.5 for M4 PEM nut	19.00	35.00	25.00	-y

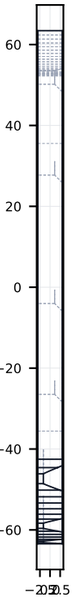
side (+x) (axes in mm from the view origin)



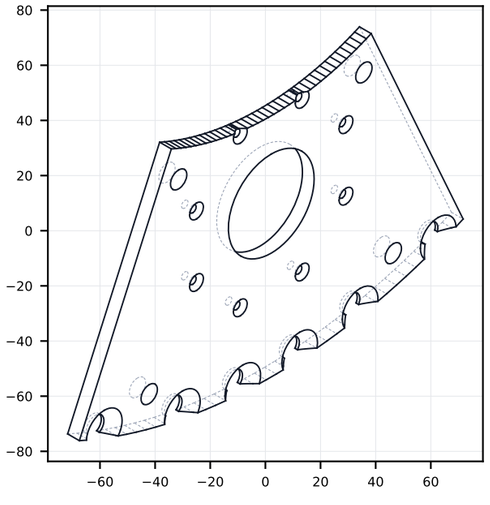
top (+z) (axes in mm from the view origin)



front (-y) (axes in mm from the view origin)



iso

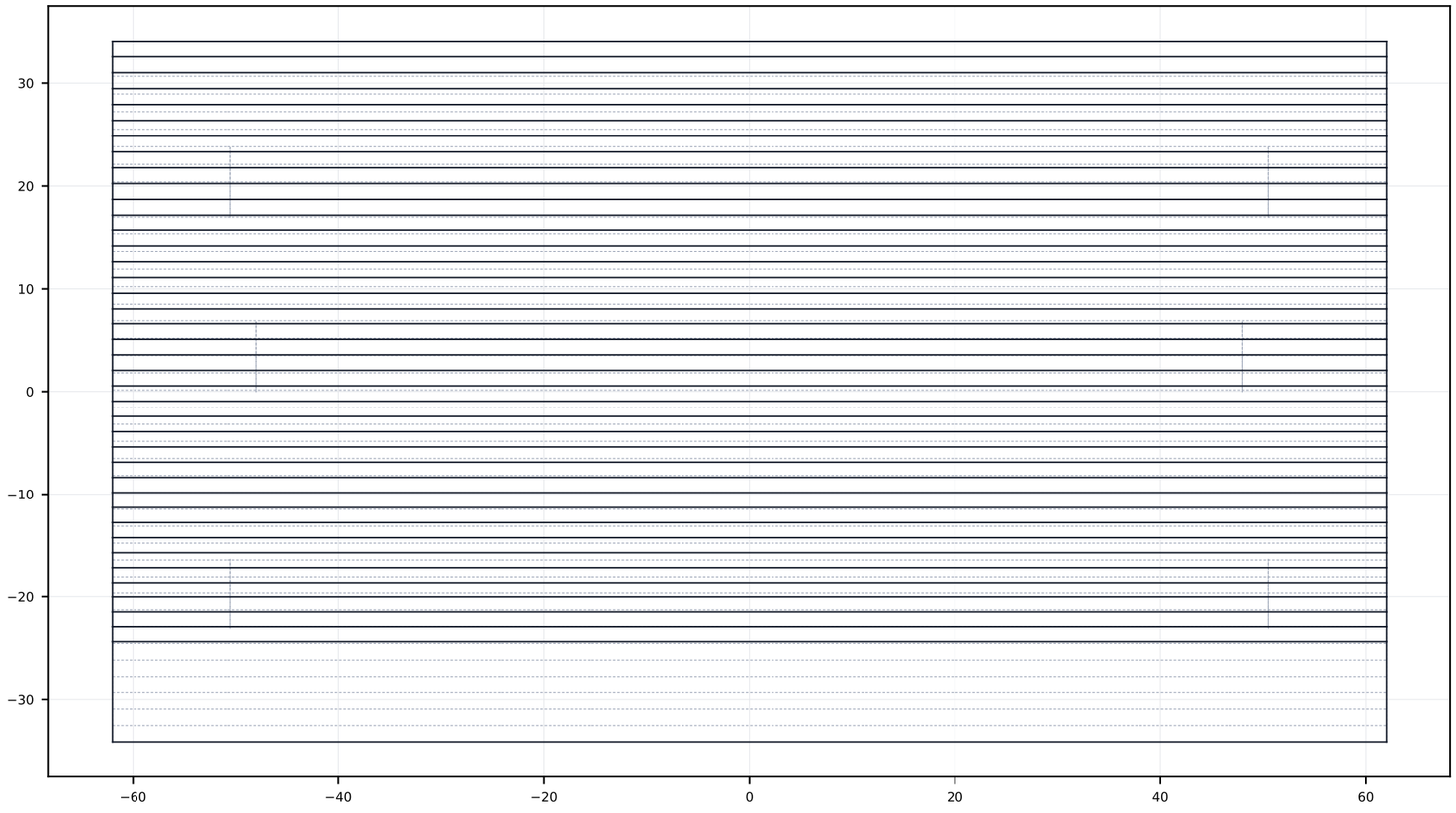


DRAWING carriage
qty per robot: 2 (R-hand part = this one mirrored about XZ)
material: 6061-T6 plate 6 mm
process: waterjet + CNC
finish:
mass: 249.8 g overall: 6.0 x 196.9 x 126.9 mm (x, y, z)
note:
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/carriage.step
view origin = bounding-box centre = robot (65.00, 0.00, -178.55)

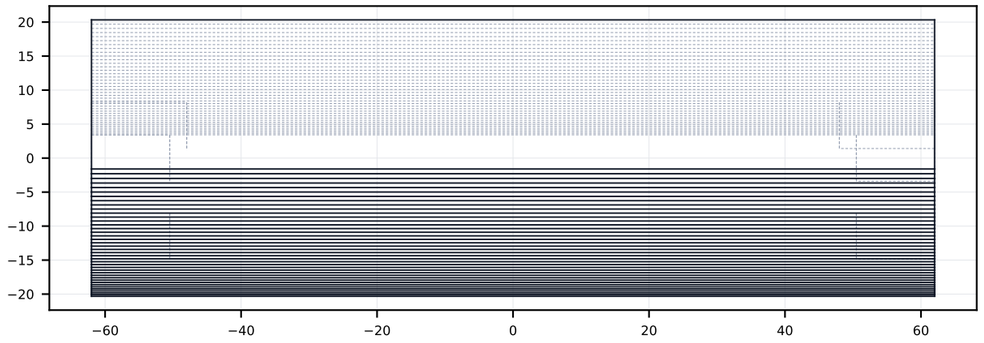
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
d3.4 through, csk 90 deg	3.00	-38.34	-2.33	-x
d3.4 through, csk 90 deg	3.00	-38.34	29.43	-x
d3.4 through, csk 90 deg	3.00	-15.88	-24.79	-x
d3.4 through, csk 90 deg	3.00	-15.88	51.89	-x
d3.4 through, csk 90 deg	3.00	15.88	-24.79	-x
d3.4 through, csk 90 deg	3.00	15.88	51.89	-x
d3.4 through, csk 90 deg	3.00	38.34	-2.33	-x
d3.4 through, csk 90 deg	3.00	38.34	29.43	-x
d9.0 through, csk 90 deg	3.00	-85.65	-44.58	-x
d9.0 through, csk 90 deg	3.00	-45.60	-56.06	-x
d9.0 through, csk 90 deg	3.00	-14.59	-60.01	-x
d9.0 through, csk 90 deg	3.00	14.59	-60.01	-x
d9.0 through, csk 90 deg	3.00	45.60	-56.06	-x
d9.0 through, csk 90 deg	3.00	85.65	-44.58	-x

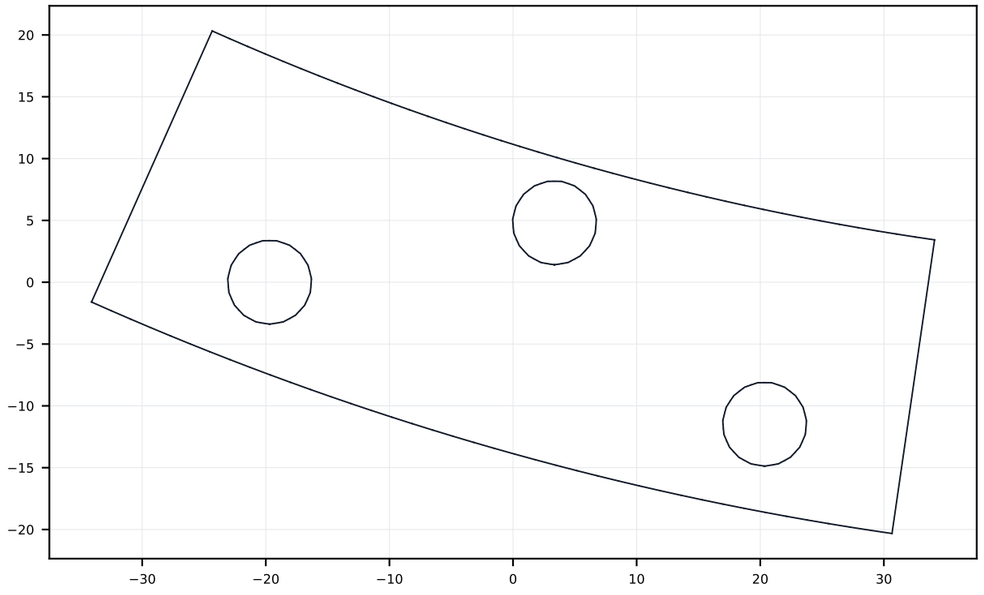
top (+z) (axes in mm from the view origin)



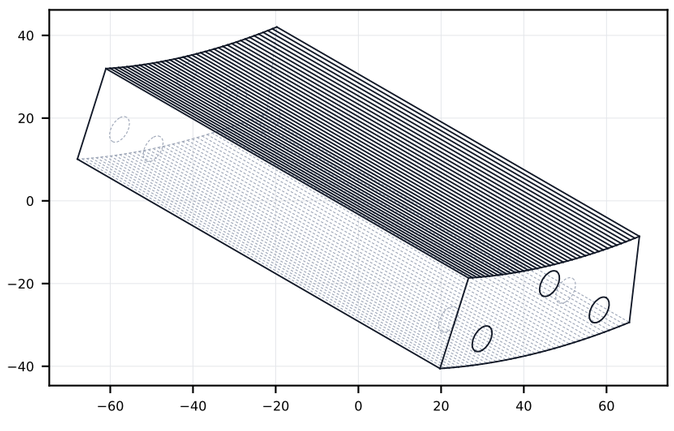
front (-y) (axes in mm from the view origin)



side (+x) (axes in mm from the view origin)



iso

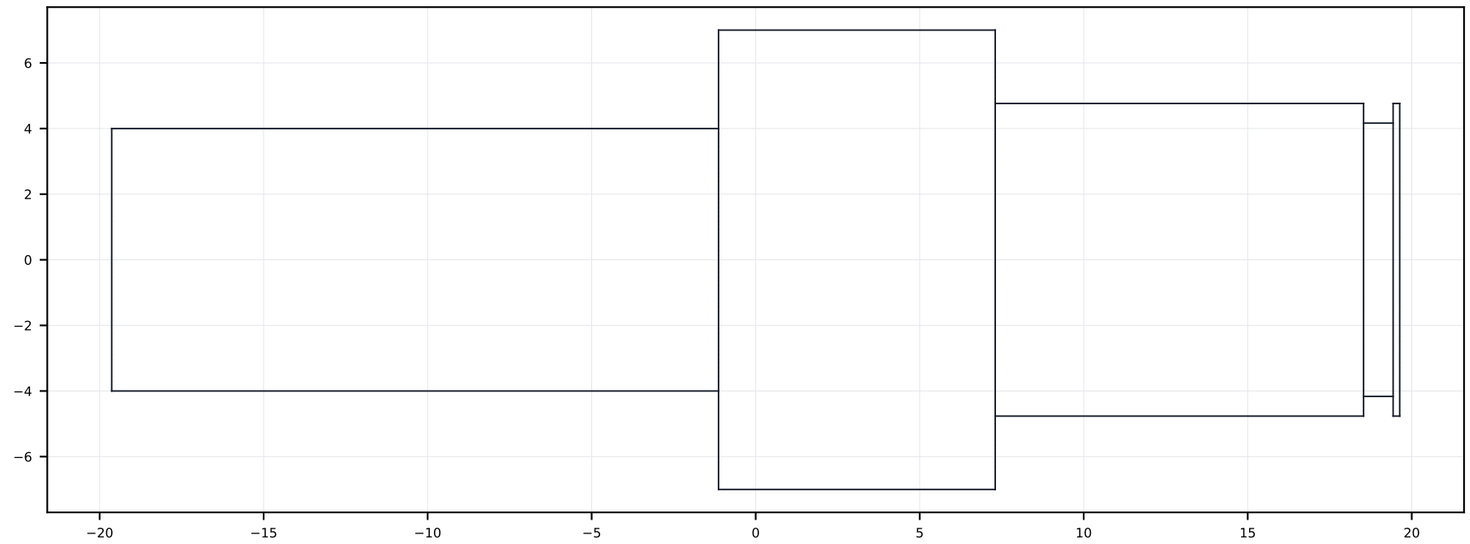


DRAWING tungsten
qty per robot: 3 (R-hand part = this one mirrored about XZ)
material: W-Ni-Fe 95 % heavy alloy, ASTM B777 class 3 (18.0 g/cc)
process: sintered blank, CNC (EDM optional)
finish:
mass: 3397.3 g overall: 124.0 x 68.2 x 40.6 mm (x, y, z)
note: M8 tapped 16 deep in both end faces
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/tungsten.step
view origin = bounding-box centre = robot (0.00, -65.96, -223.13)

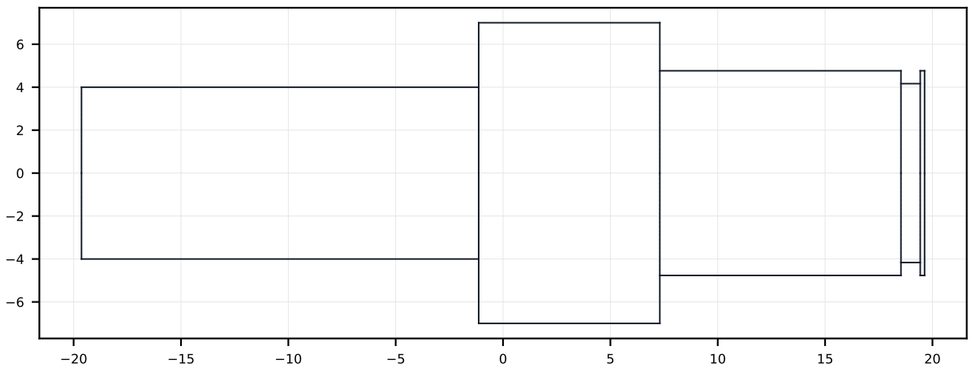
HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
M8 tapped 11.5 deep (drill 6.8)	-68.00	-19.69	0.00	+x
M8 tapped 11.5 deep (drill 6.8)	-68.00	20.35	-11.48	+x
M8 tapped 11.5 deep (drill 6.8)	68.00	-19.69	0.00	-x
M8 tapped 11.5 deep (drill 6.8)	68.00	20.35	-11.48	-x

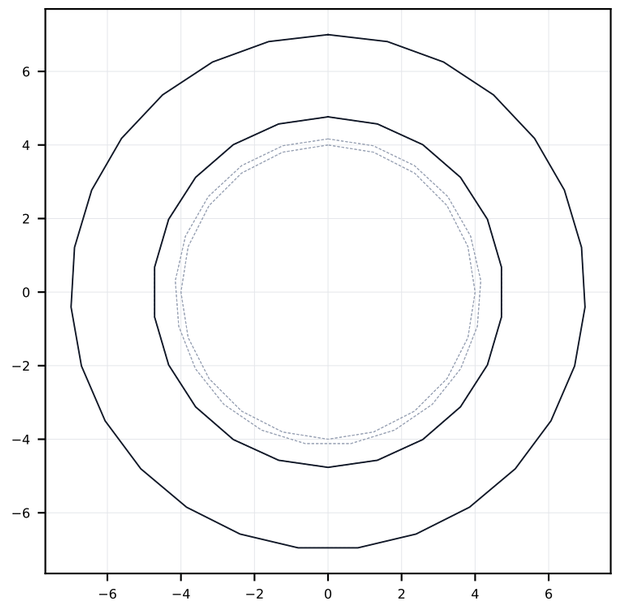
front (-y) (axes in mm from the view origin)



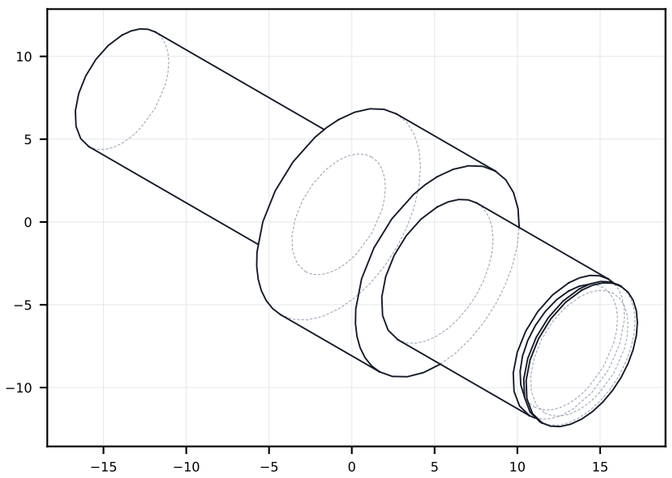
top (+z) (axes in mm from the view origin)



side (+x) (axes in mm from the view origin)

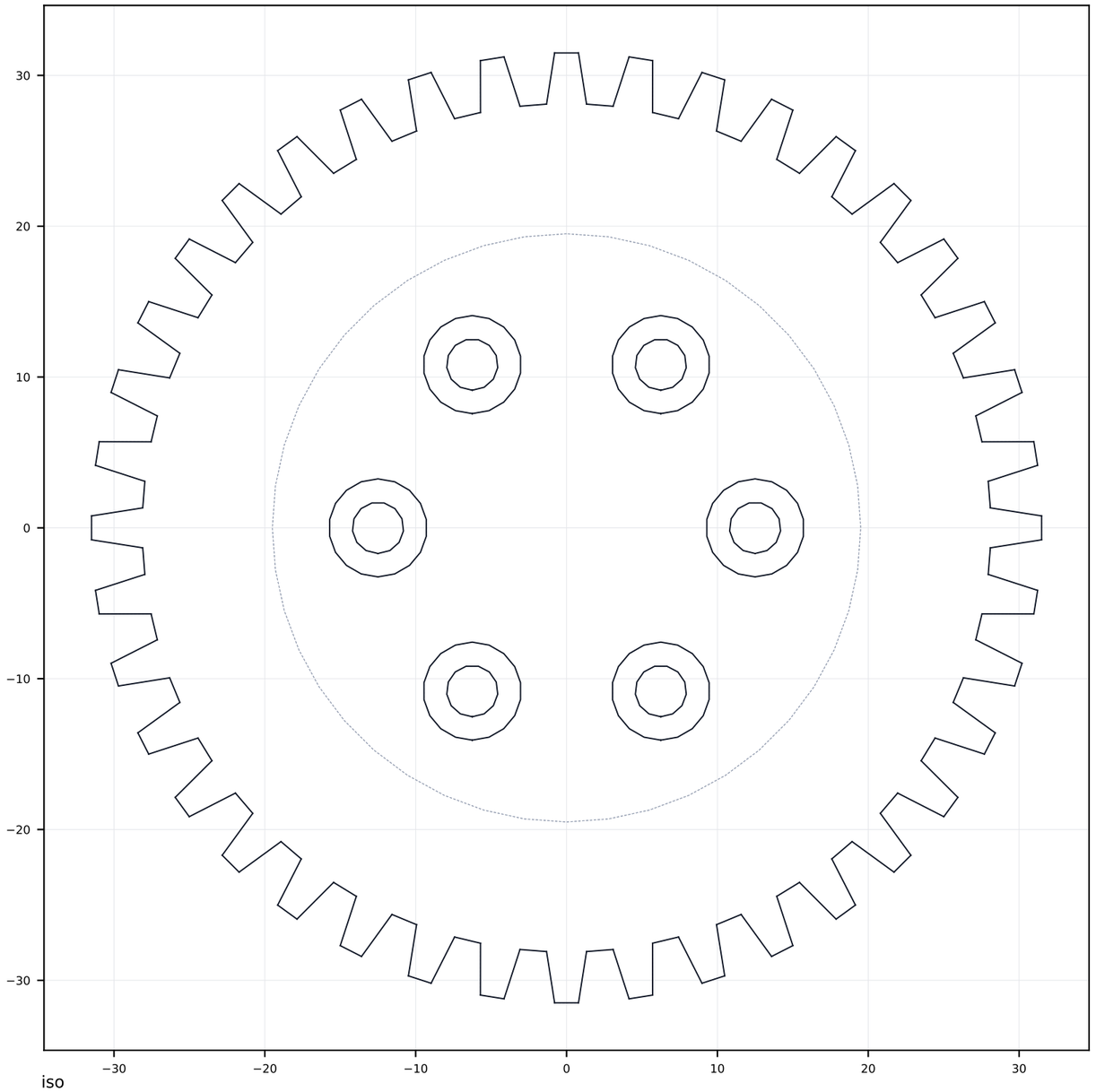


iso

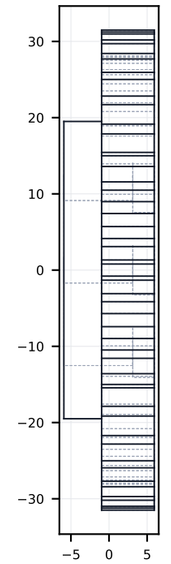


DRAWING bob_stud
qty per robot: 8 (R-hand part = this one mirrored about XZ)
material: 303 stainless
process: CNC turn
finish:
mass: 24.6 g overall: 39.3 x 14.0 x 14.0 mm (x, y, z)
note: seat d9.53 g6, DIN 6799 groove, M8 stem
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/bob_stud.step
view origin = bounding-box centre = robot (69.13, -47.50, -130.50)

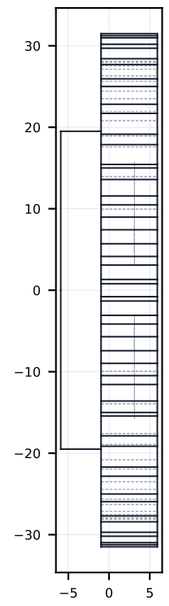
side (+x) (axes in mm from the view origin)



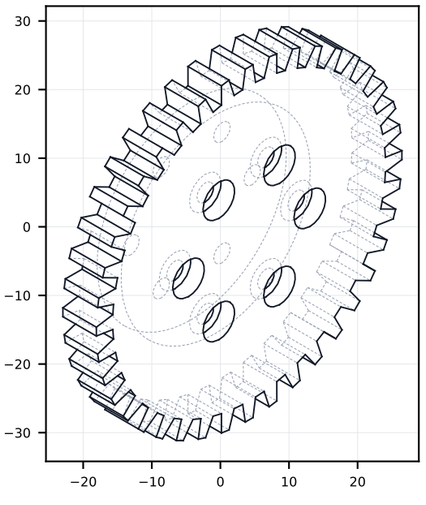
front (-y) (axes in mm from the view origin)



top (+z) (axes in mm from the view origin)



iso



DRAWING lean_pinion
qty per robot: 1
material: 42CrMo4 QT
process: hob + turn
finish:
mass: 188.2 g overall: 11.9 x 63.0 x 63.0 mm (x, y, z)
note: pilot d20 H7 onto the AK70-10 output, 6 x M3
general tolerances ISO 2768-mK, edges broken 0.3 x 45, units mm
geometry master: parts/lean_pinion.step
view origin = bounding-box centre = robot (69.97, 0.00, -165.00)

HOLE TABLE (view coords of the screw seat, mm)

feature	x	y	z	axis
d3.4 through	3.12	-12.50	0.00	-x
d3.4 through	3.12	-6.25	-10.82	-x
d3.4 through	3.12	-6.25	10.82	-x
d3.4 through	3.12	6.25	-10.82	-x
d3.4 through	3.12	6.25	10.82	-x
d3.4 through	3.12	12.50	0.00	-x